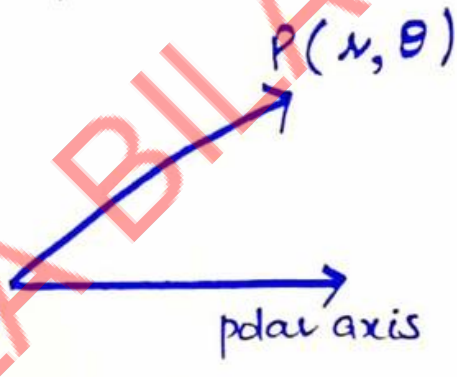
Lecture 23

Polar Coordinate System

Take a point O called origin/pole & draw a ray called polar axis.

Now take a point P with coordinates (r, θ) where "r" is distance of $\overline{OP}$ & θ is angle b/w $\overline{OP}$ & polar axis.



This is how we form a polar coordinate System.

Radial distance:-

The number r is called radial distance of P.

Polar angle:-

θ is called the polar angle of P.

P is written as $P(r, \theta)$ (e.g) $(6, 45^\circ), (4, 330^\circ)$ are plotted in polar coordinate System.

Remark. The points $(3, 225^\circ)$ & $(-3, 45^\circ)$ are same. (-ve sign shows that point is one extension of angle's terminal side rather than on terminal side itself. As $(\theta + 180)$ is extension of θ. So $(-r, \theta)$ can also be written as $(r, \theta + 180)$

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Relation b/w Polar & Rectangular coordinates:-

Polar coordinates $P(r, \theta)$

Rectangular Coordinates $P(x, y)$

$x = r\cos\theta$ & $y = r\sin\theta$

Conversion:-

From polar to Cartesian/Rectangular:-

$x = r\cos\theta$ & $y = r\sin\theta$

From Rectangular to Polar:-

$r^2 = x^2 + y^2$ & $\tan\theta = \frac{y}{x}$

Example:- Find rectangular coordinates of P whose polar coordinates are $(6, 135^\circ)$

$r = 6$, $\theta = 135^\circ$

As $x = r\cos\theta$

$= 6(\cos 135)$

$= 6\left(-\frac{\sqrt{2}}{2}\right)$

$= -3\sqrt{2}$

$y = r\sin\theta$

$y = 6\sin 135^\circ$

$= 6\left(\frac{\sqrt{2}}{2}\right)$

$y = 3\sqrt{2}$

So rectangular coordinates of P are

$(-3\sqrt{2}, 3\sqrt{2})$

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Example:- Find polar coordinates of P whose rectangular coordinates are $(-2, 2\sqrt{3})$.

here $x = -2$ & $y = 2\sqrt{3}$

$$r^2 = x^2 + y^2$$
$$= (-2)^2 + (2\sqrt{3})^2$$
$$= 4 + 4(3)$$
$$= 4 + 12$$
$$r^2 = 16$$
$$r = 4$$

$$\tan\theta = \frac{y}{x}$$
$$\tan\theta = \frac{2\sqrt{3}}{-2}$$
$$\tan\theta = (-\sqrt{3})$$
$$\theta = \tan^{-1}(-\sqrt{3})$$
$$\theta = \frac{2\pi}{3} \text{ or } 120^\circ$$

So the polar coordinates are (r, θ) or $(-r, \theta + 180)$

(i-e) $\left(4, \frac{2\pi}{3}\right), \left(-4, \frac{5\pi}{3}\right)$

Lines in Polar coordinates:-

If we have $x = a$ (i-e) point $(a, 0)$, the equation $x = r\cos\theta$ will be $a = r\cos\theta$.

If we have $y = b$ (i.e) point $(0, b)$, the equation $y = r\sin\theta$ will be $b = r\sin\theta$.

If $\theta = \theta_0$ it means line passes through origin.

As general equation of line is $Ax + By + C = 0$.

Put values:- $A(r\cos\theta) + B(r\sin\theta) + C = 0$

$$r(A\cos\theta + B\sin\theta) + C = 0$$

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Circles in polar coordinates:-

If we have a circle of radius 'a' & its center is (r_0, θ_0), then by applying law of cosine

(general form:- $a^2 = b^2 + c^2 - 2bc\cos\theta$)

$$a^2 = r^2 - 2rr_0\cos(\theta - \theta_0) + r_0^2 \quad \text{---} \quad (1)$$

Special Cases:-

⇒ A circle of radius a & center at origin.

Let $r_0 = 0$ (put in eq 1)

$$a^2 = r^2 - 2r(0)\cos(\theta - \theta_0) + 0^2$$

$$a^2 = r^2$$

This circle consists of all points $P(r, \theta)$ for which $r = a$, regardless of θ.

⇒ If circle of radius a & center on x-axis & passes Through origin, Then polar coordinates of center are either $(a, 0)$ or (a, π) depending on whether center is on right or left of origin.

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Lecture 24 Sketching

graph of the curve having equ $r = \sin\theta$

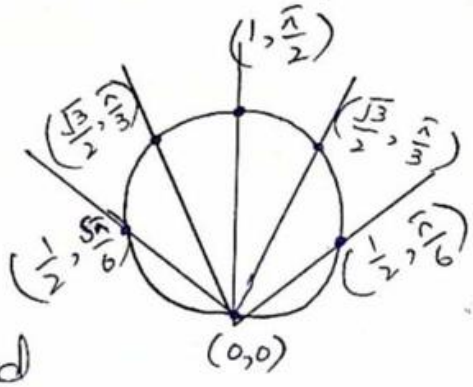
increase θ by 30° or $\frac{\pi}{6}$.

θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
$r = \sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$

θ	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
$r = \sin\theta$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0

There are 13 pairs in Table but only 6 points are plotted bcz others are same as them.

(-ve value of r & θ is increased by 180° or π)



Equation of This circle:- As $r = \sin\theta$

(multiply b/s by r) $\quad r^2 = r\sin\theta$

($r^2 = x^2 + y^2$ & $y = r\sin\theta$) $\quad x^2 + y^2 = y$

$x^2 + y^2 - y = 0$

completing square $\quad x^2 + \left[y^2 - 2y\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2\right] - \left(\frac{1}{2}\right)^2 = 0$

$x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}$

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$$x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4} \quad \text{——— (1)}$$

As general eqn of circle is

$$(x - x_0)^2 + (y - y_0)^2 = r^2 \quad \left[(x_0, y_0) \text{ is center } \& \ r \text{ is radius}\right]$$

In eq (1) $x_0 = 0$ & $y_0 = -\frac{1}{2}$ | $r^2 = \frac{1}{4}$, $r = \frac{1}{2}$

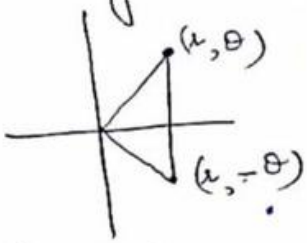
So center (x_0, y_0) is $\boxed{\left(0, -\frac{1}{2}\right)}$ & $\boxed{r = \frac{1}{2}}$

Curves in Polar Coordinates:-

1. Symmetry

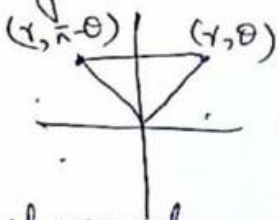
(i) about x-axis/initial line:-

If equation of a line remains unchanged when (r, θ) is replaced by $(r, -\theta)$.



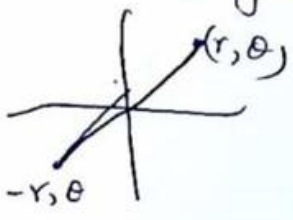
(ii) about y-axis:-

If equation of a line remains unchanged when (r, θ) is replaced by $(r, \pi - \theta)$



(iii) about Pole:-

If the equation of a line remains unchanged when (r, θ) is replaced by $(-r, \theta)$



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2. Position of the pole Relative to curve:-

Put $r=0$ & find value of θ for position.

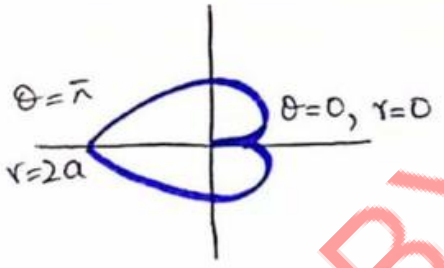
3. Table of values:-

Construct a sufficiently complete table of values.

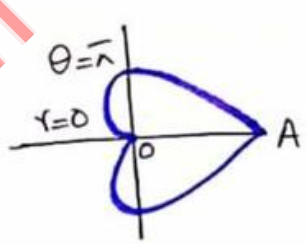
Position of Pole relative to curve:-

When $r=0$ & $\theta=0$ Then the curve passes through the pole.

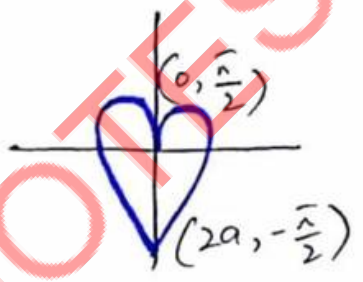
Cardioids:-



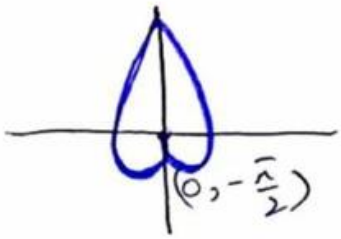
$r = a(1-\cos\theta)$



$r = a(1+\cos\theta)$



$r = a(1-\sin\theta)$



$r = a(1+\sin\theta)$

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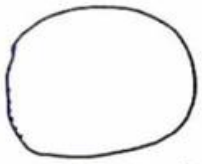
Cardidoids & Limacons:-

$r = a + b\sin\theta$, $r = a - b\sin\theta$

$r = a + b\cos\theta$, $r = a - b\cos\theta$

These equation form polar curves called Limacons. If $a = b$ then above equations become same as for cardidoids. Their position depends upon whether $\sin\theta$ or $\cos\theta$ is +ve or -ve.



$\frac{a}{b} \geq 2$

$\frac{a}{b} < 1$

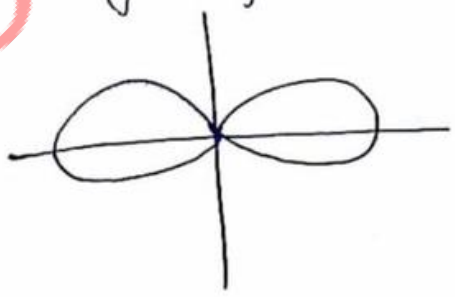
Lemicon with inner loop.

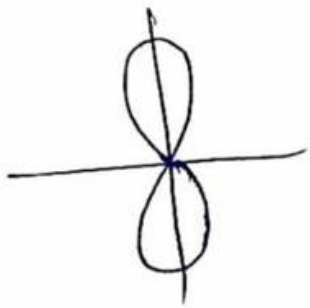
Leminscate:- if $a \neq 0$, then equation of form

$r^2 = a^2\cos 2\theta$, $r^2 = -a^2\cos 2\theta$

$r^2 = a^2\sin 2\theta$, $r^2 = -a^2\sin 2\theta$

represent propeller-shaped curves, called Leminscate. These are centered at origin. But position depends on sign of a^2.





(e.g) $r^2 = 4\cos 2\theta$ is eqn. of a leminscate.

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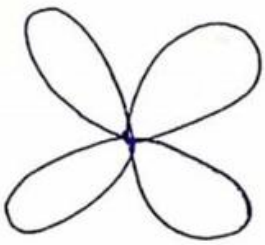
Rose Curves:-

Equations of the form ~~$r = a \sin\theta$~~ $r = a \sin n\theta$ & $r = a \cos n\theta$ represent flower shaped curves called Roses.

The rose has n equally spaced petals or loops if n is odd & $2n$ equally spaced petals if n is even.



($n=2$, four petals)

($n=3$, Three petals)

orientation depends upon sign of a. & if $\sin\theta$ or $\cos\theta$ appeass in equation.

Spiral:-

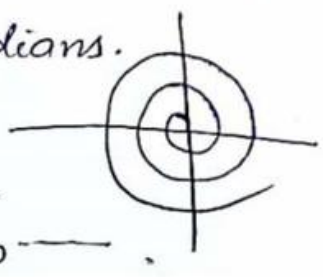
A curve that winds around the origin infinitely many times in such a way that r increases/decreases steadily as θ.

It has equations of form

$r = a\theta$ $(\theta \geq 0)$ or $r = a\theta$ $(\theta \leq 0)$

here a is +ve & θ is in radians.

its intersection with x-axis occurs at $\theta = 0, \pi, 2\pi, 3\pi, \ldots$ & intersection with y-axis occurs at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots$



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Lecture 25

Double integral in Polar Coordinates

Double integral in Polar coordinates is given as

$$\iint_R f(r,\theta)\,dA = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

Example:- Find limits of integration for integrating $f(r,\theta)$ over Region R that lies inside the cardioid $r=1+\cos\theta$ & outside the circle $r=1$.

As

$$\iint_R f(r,\theta)\,dA = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

$r = 1+\cos\theta$

put $r=1$

$1 = 1+\cos\theta$

$1-1 = \cos\theta$

$0 = \cos\theta$

$\cos^{-1}(0) = \theta$

$\pm\frac{\pi}{2} = \theta$

So limits are

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_{1}^{1+\cos\theta} f(r,\theta)\,r\,dr\,d\theta.$$

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Example:- Evaluate $\iint_R \sin\theta \, dA$

where R is region in first quad. That is outside the circle $r=2$ & inside cardioid $r = 2(1+\cos\theta)$

As $r = 2(1+\cos\theta)$

put $r = 2$

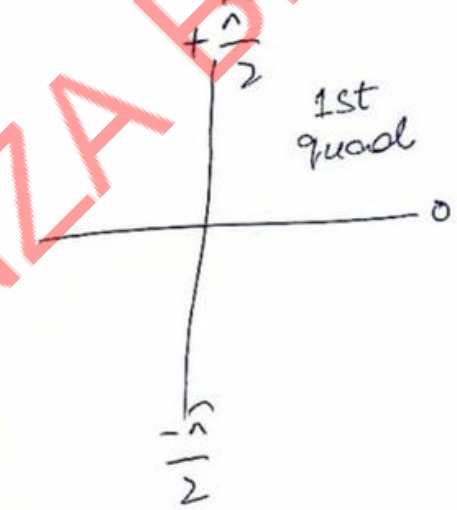
$2 = 2(1+\cos\theta)$

$1 = 1+\cos\theta$

$0 = \cos\theta$

$\cos^{-1}(0) = \theta$

$\pm \frac{\pi}{2} = \theta$



So $\int_0^{\pi/2} \int_2^{2(1+\cos\theta)} \sin\theta \, r \, dr \, d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\sin\theta \, \frac{r^2}{2} \right]_2^{2(1+\cos\theta)} d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\left(\sin\theta \, \frac{(2(1+\cos\theta))^2}{2} \right) - \left(\sin\theta \left(\frac{(2)^2}{2} \right) \right) \right] d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\left(\sin\theta \left(\frac{4(1+\cos\theta)^2}{2} \right) \right) - \left(\sin\theta \, \frac{4}{2} \right) \right] d\theta$

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$$= \int_0^{\frac{\pi}{2}} \left[2\sin\theta(1+\cos\theta)^2 - 2\sin\theta \right] d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \left(-\sin\theta(1+\cos\theta)^2 + \sin\theta \right) d\theta$$

$$= 2\left[-\frac{(1+\cos\theta)^3}{3} + \cos\theta \right]_0^{\frac{\pi}{2}} + C$$

$$= 2\left[\left(-\frac{(1+\cos\frac{\pi}{2})^3}{3} + \cos\frac{\pi}{2} \right) - \left(-\frac{(1+\cos 0^\circ)^3}{3} + \cos 0^\circ \right) \right]$$

$$= 2\left[\left(-\frac{1}{3} + 0 \right) - \left(-\frac{8}{3} + 1 \right) \right]$$

$$= 2\left(-\frac{1}{3} + \frac{8}{3} - \frac{1}{1} \right)$$

$$= 2\left(\frac{-1+8-3}{3} \right) \Rightarrow 2\left(\frac{4}{3} \right)$$

$$= \frac{8}{3} \quad \text{Ans.}$$

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Example:- Use a double polar integral to find area enclosed by Three-petaled rose $r = \sin 3\theta$. We calculate are in 1st quad & multiply by 3.

$$A = 3\iint_R dA$$

$$A = 3\iint_R r\,dr\,d\theta$$

$$= 3\int_0^{\frac{\pi}{3}}\int_0^{\sin 3\theta} r\,dr\,d\theta$$

$$= 3\int_0^{\frac{\pi}{3}}\left[\frac{r^2}{2}\right]_0^{\sin 3\theta} d\theta \Rightarrow \frac{3}{2}\int_0^{\frac{\pi}{3}} r^2\Big|_0^{\sin 3\theta} d\theta$$

$$= \frac{3}{2}\int_0^{\pi/3}\left((\sin^2 3\theta) - 0^2\right) d\theta$$

$$= \frac{3}{2}\int_0^{\frac{\pi}{3}} \frac{1-\cos 6\theta}{2}\, d\theta$$

$$= \frac{3}{2}\int_0^{\frac{\pi}{3}}\left(\frac{1}{2} - \frac{\cos 6\theta}{2}\right) d\theta \Rightarrow \frac{3}{4}\int_0^{\frac{\pi}{3}}(1-\cos 6\theta)\,d\theta$$

$$A = \frac{3}{4}\left(\theta - \frac{\sin 6\theta}{6}\right)\Big|_0^{\frac{\pi}{3}} \Rightarrow \frac{3}{4}\left(\frac{\pi}{3} - \frac{\sin 6\left(\frac{\pi}{3}\right)}{6}\right)$$

$$A = \frac{3}{4}\left(\frac{\pi}{3} - \frac{\sin 2\pi}{6}\right) \Rightarrow \frac{3}{4}\left(\frac{\pi}{3}\right) \Rightarrow \boxed{\frac{\pi}{4}}$$ Ans.

$r = \sin 3\theta$

put $r = 0$

$0 = \sin 3\theta$

$\sin^{-1}(0) = 3\theta$

$\pi = 3\theta$

$\frac{\pi}{3} = \theta$

$\because \quad \sin^2\theta = \frac{1-\cos 2\theta}{2}$

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Find area enclosed by leminscate $r^2 = 4\cos 2\theta$. Total area is four times the first quad portion.

$r = 0$, $r^2 = 4\cos 2\theta$

$r = \sqrt{4\cos 2\theta}$

$0 = 4\cos 2\theta$

$0 = \cos 2\theta$

$\cos^{-1}(0) = 2\theta$

$\frac{\pi}{2} = \theta$

$\frac{\pi}{4} = \theta$, $\theta = 0$

$$A = 4\int_0^{\pi/4}\int_0^{\sqrt{4\cos 2\theta}} r\,dr\,d\theta \Rightarrow 4\int_0^{\frac{\pi}{4}} \frac{r^2}{2}\Big|_0^{\sqrt{4\cos 2\theta}} d\theta$$

$$= \frac{4}{2}\int_0^{\pi/4} \left(\sqrt{4\cos 2\theta}\right)^2 - 0^2 \, d\theta \Rightarrow 2\int_0^{\frac{\pi}{4}} 4\cos 2\theta \, d\theta$$

$$= 8\int_0^{\pi/4} \cos 2\theta \, d\theta \Rightarrow 8\left|\frac{\sin 2\theta}{2}\right|_0^{\frac{\pi}{4}}$$

$$= \frac{8}{2}\left(\sin 2\left(\frac{\pi}{4}\right) - \sin 2(0)\right)$$

$$= 4\left(\sin\frac{\pi}{2} - \sin 0\right)$$

$$= 4(1-0)$$

$$= 4(1)$$

$$\boxed{A = 4}$$ Ans.

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Changing Cartesian integral to polar.

Step 1: put $x = r\cos\theta$ & $y = r\sin\theta$ & replace $dx\,dy$ by $dr\,d\theta$.

Step 2: Supply polar limits of integration for R.

The integral becomes

$$\iint_R f(x,y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta$$

Example:- Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)\,dy\,dx$ by changing into polar coordinates.

$0 \le y \le \sqrt{1-x^2}$ & $0 \le x \le 1$

$y = \sqrt{1-x^2}$ $\qquad \theta = 0$ & $\theta = \frac{\pi}{2}$

$y^2 = (\sqrt{1-x^2})^2$

$y^2 = 1 - x^2$

$x^2 + y^2 = 1$

here $r^2 = 1$

so $r = 1$

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$$\int_0^{\frac{\pi}{2}}\int_0^1 (r^2\cos^2\theta + r^2\sin^2\theta)\, r\, dr\, d\theta$$

$$= \int_0^{\frac{\pi}{2}}\int_0^1 r^2(\cos^2\theta + \sin^2\theta)\, r\, dr\, d\theta$$

$$= \int_0^{\frac{\pi}{2}}\int_0^1 r^2(1)\, r\, dr\, d\theta$$

$$= \int_0^{\pi/2}\int_0^1 r^3\, dr\, d\theta \Rightarrow \int_0^{\frac{\pi}{2}} \frac{r^4}{4}\Big|_0^1 d\theta$$

$$= \frac{1}{4}\int_0^{\pi/2}(1^4 - 0^4)d\theta \Rightarrow \frac{1}{4}\int_0^{\pi/2}(1)\,d\theta$$

$$= \frac{1}{4}|\theta|_0^{\frac{\pi}{2}} \Rightarrow \frac{1}{4}\left(\frac{\pi}{2} - 0\right)$$

$$= \frac{1}{4}\left(\frac{\pi}{2}\right) \Rightarrow \frac{\pi}{8}$$ Ans.

Example:- Evaluate $\iint_D \frac{dx\,dy}{x^2+y^2}$ by changing to polar coordinates where D is region of first quad b/w circles $x^2+y^2 = a^2$ & $x^2+y^2 = b^2$ $(0<a<b)$

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As $x^2+y^2=r^2$ if $a^2 \le r^2 \le b^2$

$a \le r \le b$

As quad is 1st So

$\theta = 0$ & $\theta = \frac{\pi}{2}$

$$= \int_0^{\frac{\pi}{2}} \int_a^b \frac{r\,dr\,d\theta}{r^2\cos^2\theta + r^2\sin^2\theta}$$

$$= \int_0^{\frac{\pi}{2}} \int_a^b \frac{r}{r^2}\,dr\,d\theta \Rightarrow \int_0^{\frac{\pi}{2}} \int_a^b \frac{1}{r}\,dr\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} |\ln r|_a^b\,d\theta \Rightarrow \int_0^{\pi/2} (\ln b - \ln a)\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \ln\left(\frac{b}{a}\right) d\theta \Rightarrow \ln\left(\frac{b}{a}\right) \int_0^{\pi/2} d\theta$$

$$= \ln\left(\frac{b}{a}\right) |\theta|_0^{\frac{\pi}{2}} \Rightarrow \ln\left(\frac{b}{a}\right)\left(\frac{\pi}{2} - 0\right)$$

$$= \frac{\pi}{2} \ln\left(\frac{b}{a}\right) \quad \text{Ans.}$$

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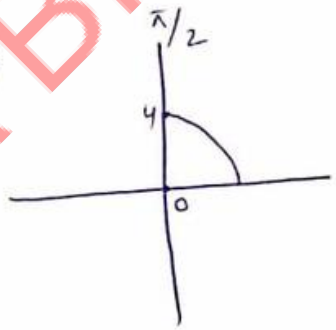
Lecture No. 26

Examples:-

Example:- Evaluate $I = \int_0^4 \int_0^{\sqrt{4y-y^2}} (x^2+y^2)\, dx\, dy$ by changing into polar coordinates.

$0 \le x \le \sqrt{4y-y^2}$ & $0 \le y \le 4$



Take $x = \sqrt{4y-y^2}$

Take square on b/s

$x^2 = 4y - y^2$

$x^2 + y^2 = 4y$ (as general eqn. of circle is $x^2+y^2 = r^2$)

So here

$r^2 = 4y$

$r^2 = 4 r\sin\theta$ $(\because y = r\sin\theta)$

$r = 4\sin\theta$ & $r = 0$

$\theta = 0$ & $\theta = \frac{\pi}{2}$

Now $I = \int_0^{\frac{\pi}{2}} \int_0^{4\sin\theta} r^2 \, r\, dr\, d\theta \Rightarrow \int_0^{\pi/2} \int_0^{4\sin\theta} r^3\, dr\, d\theta$

$= \int_0^{\pi/2} \left| \frac{r^4}{4} \right|_0^{4\sin\theta} d\theta \Rightarrow \int_0^{\pi/2} \left(\frac{(4\sin\theta)^4}{4} - \frac{0^4}{4} \right) d\theta$

$= \int_0^{\pi/2} \frac{4^4 \sin^4\theta}{4}\, d\theta$

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$$= \int_0^{\pi/2} 4^3 \sin^4\theta \, d\theta$$

$$= \int_0^{\pi/2} 64 \sin^4\theta \, d\theta$$

Using Walli's formula

$$= \int_0^{\pi/2} 64 \frac{(4-1)(4-3)}{4(4-2)} d\theta \quad \because \frac{(m-1)(m-3)\cdots}{m(m-2)(m-4)\cdots}$$

$$= \int_0^{\pi/2} 64 \frac{(3)(1)}{4(2)} d\theta$$

$$= \int_0^{\pi/2} 64 \cdot \frac{3}{8} d\theta$$

$$= \int_0^{\pi/2} 24 \, d\theta \Rightarrow 24 \int_0^{\pi/2} d\theta$$

$$= 24 \, |\theta|_0^{\pi/2}$$

$$= 24 \left(\frac{\pi}{2} - 0\right)$$

$$= 24 \left(\frac{\pi}{2}\right)$$

$$= 12\pi \quad \text{Ans.}$$

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Example:- Evaluate $\iint_R e^{x^2+y^2}\,dy\,dx$ where R is semicircular region bounded by x-axis & the curve $y=\sqrt{1-x^2}$

$$y=\sqrt{1-x^2}$$

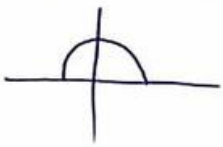
Take square on b/s

$$y^2=1-x^2$$

$$y^2+x^2=1 \quad \text{(comparing with } x^2+y^2=r^2\text{)}$$

here $r^2=1$, $r=1$ & $r=0$ (bcz semi circle

Now $\theta=0$ & $\theta=\pi$ (bcz semi-circle on x-axis)



$$=\frac{1}{2}\int_0^{\pi}\int_0^{1} e^{r^2}\,2r\,dr\,d\theta$$

$$=\frac{1}{2}\int_0^{\pi}\left|e^{r^2}\right|_0^1 d\theta \Rightarrow \frac{1}{2}\int_0^{\pi}(e^1-e^0)\,d\theta$$

$$=\frac{1}{2}\int_0^{\pi}(e-1)\,d\theta \Rightarrow \frac{1}{2}(e-1)\int_0^{\pi} d\theta$$

$$=\frac{1}{2}(e-1)(\theta)\Big|_0^{\pi} \Rightarrow \frac{1}{2}(e-1)(\pi-0)$$

$$=\frac{\pi}{2}(e-1)$$ Ans.

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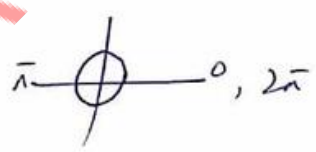
Example:- Let R_a be region bounded by the circle $x^2+y^2=a^2$. Define

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy = \lim_{a\to\infty}\iint_R e^{-(x^2+y^2)}\,dx\,dy$$

$x^2+y^2=a^2$ here $r^2=a^2$

$r=a$ & $r=0$

$\theta=0$ & $\theta=2\pi$ (bcz region is bounded by circle)



$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy = \lim_{a\to\infty}\iint_R e^{-(x^2+y^2)}\,dx\,dy$$

$$= \lim_{a\to\infty} -\frac{1}{2}\int_0^{2\pi}\int_0^{a} e^{-r^2}(-2)r\,dr\,d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}\int_0^{2\pi}\left|e^{-r^2}\right|_0^a d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}\int_0^{2\pi}\left(e^{-a^2}-e^{-0}\right)d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}\int_0^{2\pi}(e^{-a^2}-1)\,d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}(e^{-a^2}-1)\int_0^{2\pi} d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}(e^{-a^2}-1)|\theta|_0^{2\pi}$$

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$$= -\frac{1}{2} \lim_{a \to \infty} \left(e^{-a^2} - 1\right)\left(2\bar{\pi} - 0\right)$$

$$= \lim_{a \to \infty} \left(1 - e^{-a^2}\right)\left(\frac{2\bar{\pi}}{2}\right)$$

$$= \lim_{a \to \infty} \left(1 - \frac{1}{e^{a^2}}\right)\bar{\pi}$$

$$= \left(1 - \frac{1}{e^{\infty}}\right)\bar{\pi} \Rightarrow \left(1 - \frac{1}{\infty}\right)\bar{\pi}$$

$$= (1 - 0)\,\bar{\pi} \Rightarrow \bar{\pi} \quad \underline{\text{Ans.}}$$

Example: Prove That $\int_0^{\infty} e^{-t^2}\,dt = \frac{1}{2}\sqrt{\pi}$

Can't solve directly bcz derivative of $-t^2$ is not given. So

let $I = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2-y^2}\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2}\cdot e^{-y^2}\,dx\,dy$$

$$= \left[\int_{-\infty}^{\infty} e^{-x^2}\,dx\right]\left[\int_{-\infty}^{\infty} e^{-y^2}\,dy\right]$$

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use a dummy variable t.

$$= \left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]\left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]$$

$$= \left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]^2$$

introduce a vawable a & apply limit

$$= \lim_{a\to\infty}\left(\int_{-a}^{a} e^{-t^2}\,dt\right)^2$$

$$= \lim_{a\to\infty} 4\left(\int_{0}^{a} e^{-t^2}\,dt\right)^2 \qquad \left(\int_0^a \text{ shows one quadrant}\right)$$

Here

$$4\lim_{a\to\infty}\left(\int_0^a e^{-t^2}\,dt\right)^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy$$

$$4\lim_{a\to\infty}\left(\int_0^a e^{-t^2}\,dt\right)^2 = \pi \qquad \text{(reference to previous example).}$$

$$\left(\int_0^{\infty} e^{-t^2}\,dt\right)^2 = \frac{\pi}{4}$$

$$\int_0^{\infty} e^{-t^2}\,dt = \frac{\sqrt{\pi}}{2} \text{ or } \frac{1}{2}\sqrt{\pi}$$

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Theorem:- Let G be a rectangular box defined by inequalities

$$a \le x \le b \;,\; c \le y \le d \;,\; k \le z \le l$$

If G is continuous then

$$\iiint_G f(x,y,z)\,dV = \int_a^b \int_c^d \int_k^l f(x,y,z)\,dz\,dy\,dx$$

order of integration doesn't matter.

Example:- Evaluate Tripple integral

$\iiint_G 12xy^2z^3\,dV$ over rectangular box G defined by inequalities

$-1 \le x \le 2 \;,\; 0 \le y \le 3 \;,\; 0 \le z \le 2.$

$$= \int_{-1}^{2}\int_{0}^{3}\int_{0}^{2} 12xy^2z^3\,dz\,dy\,dx$$

$$= \int_{-1}^{2}\int_{0}^{3} 12xy^2 \left|\frac{z^4}{4}\right|_0^2 dy\,dx$$

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$$= \int_{-1}^{2} \int_{0}^{3} \frac{12xy^2}{4} (2^4 - 0^4)\, dy\, dx$$

$$= \int_{-1}^{2} \int_{0}^{3} 3xy^2 (16)\, dy\, dx$$

$$= \int_{-1}^{2} \int_{0}^{3} 48xy^2\, dy\, dx$$

$$= \int_{-1}^{2} 48x \left| \frac{y^3}{3} \right|_0^3 dx$$

$$= \int_{-1}^{2} \frac{48x}{3} (3^3 - 0^3)\, dx$$

$$= \int_{-1}^{2} 16x (27)\, dx$$

$$= \int_{-1}^{2} 432x\, dx$$

$$= 432 \left| \frac{x^2}{2} \right|_{-1}^{2} dx$$

$$= \frac{432}{2} (2^2 - (-1)^2) \Rightarrow 216(4-1)$$

$$\Rightarrow 216(3) \Rightarrow 648 \text{ Ans.}$$

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Example:- $\iiint_R (x-2y+z)\,dx\,dy\,dz$ when

$0 \le x \le 1$, $0 \le y \le x^2$, $0 \le z \le x+y$

$$\int_0^1 \int_0^{x^2} \int_0^{x+y} (x-2y+z)\,dz\,dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left| \frac{(x-2y+z)^2}{2} \right|_0^{x+y} dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left[\frac{(x-2y+x+y)^2}{2} - \frac{(x-2y)^2}{2} \right] dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left(\frac{(2x-y)^2 - (x-2y)^2}{2} \right) dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left(\frac{4x^2+y^2-4xy - x^2-4y^2+4xy}{2} \right) dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{x^2} (3x^2-3y^2)\,dy\,dx$$

$$= \frac{3}{2} \int_0^1 \int_0^{x^2} (x^2-y^2)\,dy\,dx$$

$$= \frac{3}{2} \int_0^1 \left(x^2y - \frac{y^3}{3} \right)\Big|_0^{x^2} dx$$

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$$= \frac{3}{2}\int_0^1 \left(\left[x^2(x^2) - \frac{(x^2)^3}{3} \right] - \left[x^2(0) - \frac{0^3}{3} \right] \right) dx$$

$$= \frac{3}{2}\int_0^1 \left(x^4 - \frac{x^6}{3} \right) dx$$

$$= \frac{3}{2}\left(\frac{x^5}{5} - \frac{x^7}{3 \cdot 7} \right)\Big|_0^1$$

$$= \frac{3}{2}\left(\frac{x^5}{5} - \frac{x^7}{21} \right)\Big|_0^1$$

$$= \frac{3}{2}\left[\left(\frac{(1)^5}{5} - \frac{(1)^7}{21} \right) - \left(\frac{(0)^5}{5} - \frac{(0)^7}{21} \right) \right]$$

$$= \frac{3}{2}\left[\frac{1}{5} - \frac{1}{21} \right]$$

$$= \frac{3}{2}\left[\frac{21 - 5}{105} \right]$$

$$= \frac{3}{2}\left(\frac{16}{105} \right)$$

$$= \frac{8}{35} \quad \text{Ans.}$$

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Example:- Evaluate $\iiint xyz \, dx\,dy\,dz$

where $S = \{(x,y,z) : x^2+y^2+z^2 \le 1,\ x \ge 0, y \ge 0, z \ge 0\}$

$x^2+y^2+z^2=1$

$z^2 = 1-x^2-y^2$

$z = \sqrt{1-x^2-y^2}$

ignore z

$x^2+y^2=1$

$y^2 = 1-x^2$

$y = \sqrt{1-x^2}$

ignore y.

$x^2 = 1$

$x = 1$

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dz\,dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \left| \frac{z^2}{2} \right|_0^{\sqrt{1-x^2-y^2}} dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \frac{\left(\sqrt{1-x^2-y^2}\right)^2}{2} dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} xy(1-x^2-y^2) \, dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} x\left(y - x^2y - y^3\right) dy\,dx$$

$$= \frac{1}{2} \int_0^1 \left[x \left| \frac{y^2}{2} - \frac{x^2y^2}{2} - \frac{y^4}{4} \right|_0^{\sqrt{1-x^2}} \right] dx$$

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$$= \frac{1}{2}\int_0^1 x\left(\frac{\left(\sqrt{1-x^2}\right)^2}{2} - \frac{x^2\left(\sqrt{1-x^2}\right)^2}{2} - \frac{\left(\sqrt{1-x^2}\right)^4}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-x^2}{2} - \frac{x^2(1-x^2)}{2} - \frac{(1-x^2)^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-x^2}{2} - \frac{x^2-x^4}{2} - \frac{1+x^4-2x^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{2-2x^2-2x^2+2x^4-1-x^4+2x^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-2x^2+x^4}{4}\right) dx$$

$$= \frac{1}{8}\int_0^1 \frac{x-2x^3+x^5}{} dx$$

$$= \frac{1}{8}\int_0^1 x^5 - 2x^3 + x \quad dx$$

$$= \frac{1}{8}\left[\frac{x^6}{6} - \frac{2x^4}{4} + \frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{8}\left[\frac{(1)^6}{6} - \frac{2(1)^4}{4} + \frac{(1)^2}{2}\right] - 0$$

$$= \frac{1}{8}\left(\frac{1}{6} - \frac{2}{4} + \frac{1}{2}\right) \Rightarrow \frac{1}{8}\left(\frac{1-3+3}{6}\right) \Rightarrow \frac{1}{12}$$ Ans.

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Lecture 27

Vector Valued Functions

Those functions for which the domain consists of real numbers & the range consists of vectors in 2d-space & or 3d-space are called Vector valued functions.

Such functions can be expressed in

2d-space like $r(t) = (x(t), y(t))$

$= x(t)i + y(t)j$

Such functions can be expressed in

3d-space like $r(t) = (x(t), y(t), z(t))$

$= x(t)i + y(t)j + z(t)k$

here $x(t)$, $y(t)$ & $z(t)$ are called real valued functions of the real variable t. These functions are called component functions or components of r.

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Representation:-

Vector valued functions are represented by bold letters $\mathbf{f(t)}$, $\mathbf{g(t)}$ etc

Real valued functions are represented by italic lightface type. $f(t)$, $g(t)$ etc.

Parametric Equations in vector Form:-

Vector valued functions can be used to express parametric equations in 2-space/3-space in a compact Form.

For example consider parametric equations

$$x = x(t) \quad \& \quad y = y(t).$$

Two vectors are equivalent iff their corresponding components are equal, so this pair of equations can be replaced by single vector equation.

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$x = x(t)$, $y = y(t)$

$x\underline{i} + y\underline{j} = x(t)\underline{i} + y(t)\underline{j}$

Similarly for Three dimensional space These equations

$x = x(t)$, $y = y(t)$, $z = z(t)$

can be replaced by single vector equation

$$x\underline{i} + y\underline{j} + z\underline{k} = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$$

In 2-d

$r = x\underline{i} + y\underline{j}$ & $r(t) = x(t)\underline{i} + y(t)\underline{j}$ are same.

In 3-d

$r = x\underline{i} + y\underline{j} + z\underline{k}$ & $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$

are same.

So we can write $r = r(t)$.

Example:- Express given parametric equ. in single vector equ.

(a) $x = t^2$ & $y = 3t$

$r = x\underline{i} + y\underline{j} = t^2\underline{i} + 3t\underline{j}$

(b) $x = \cos t$, $y = \sin t$, $z = t$

$r = x\underline{i} + y\underline{j} + z\underline{k} = \cos t\,\underline{i} + \sin t\,\underline{j} + t\,\underline{k}$

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Example:- Find parametric equ. That correspond to the vector equation.

$$x\underline{i} + y\underline{j} + z\underline{k} = (t^3+1)\underline{i} + 3\underline{j} + e^t\underline{k}$$

Comparing

$$x = t^3+1 \;,\quad y = 3 \;,\quad z = e^t$$

Example:- Sketch graph of vector-valued function $r(t) = \cos t\,\underline{i} + \sin t\,\underline{j}$ $\quad 0 \le t \le 2\pi$

As $r(t) = x\underline{i} + y\underline{j}$

$\cos t\,\underline{i} + \sin t\,\underline{j} = x\underline{i} + y\underline{j}$

Comparing

$$x = \cos t \;\&\; y = \sin t \quad (0 \le t \le 2\pi)$$

Take square on b/s

$$x^2 = \cos^2 t \;\&\; y^2 = \sin^2 t$$

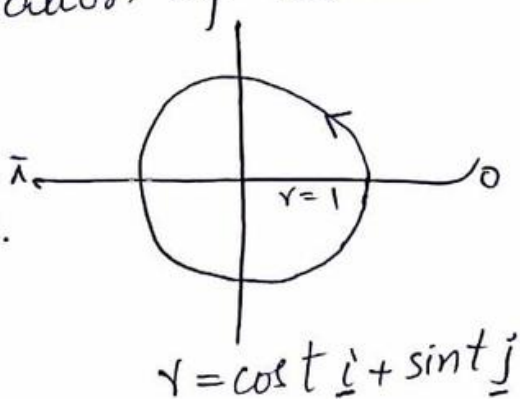
Now adding both equations

$$x^2 + y^2 = \cos^2 t + \sin^2 t$$

$$x^2 + y^2 = 1$$

So we got an equation of circle whose radius is 1.

in anti-clockwise direction.



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Example:- Sketch the graph of vector-valued function $r(t) = \cos t\, i + \sin t\, j + 2k \quad 0 \le t \le 2\pi$

As $r = r(t)$

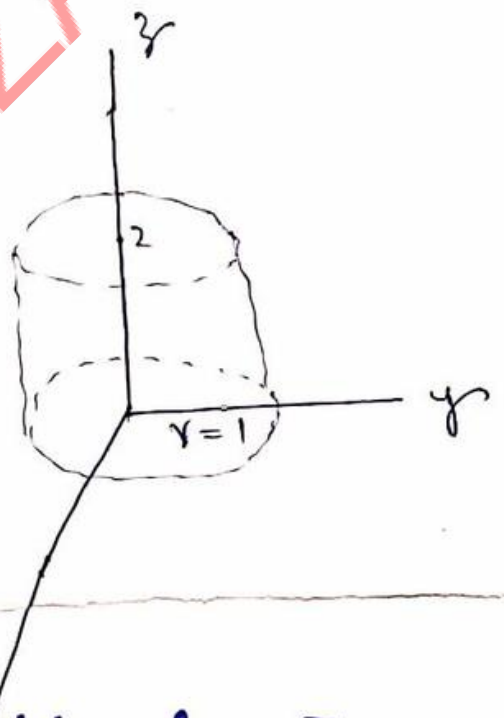
$r = xi + yj + zk$

$\cos t\, i + \sin t\, j + 2k = xi + yj + zk$

Comparing $x = \cos t$, $y = \sin t$, $z = 2$

~~or~~ squaring & adding first two equations

$x^2 + y^2 = 1$



stretch along z-axis.

$r = \cos t\, i + \sin t\, j + 2k$

Example:- Describe the graph of vector equ.

$r = (-2+t)i + 3tj + (5-4t)k$

Comparing $x = -2+t$, $y = 3t$, $z = 5-4t$

Graph is a line passing through $(-2, 0, 5)$ & parallel to $i + 3j - 4k$

(ignoring constants & t).

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Example:- The graph of vector-valued func $r(t) = t\underline{i} + t^2\underline{j} + t^3\underline{k}$ is called twisted cubic. Show That This curve ~~lies~~ Lies on parabolic cyllinder $y = x^2$ & sketch graph for $t \geq 0$. $r(t) = t\underline{i} + t^2\underline{j} + t^3\underline{k}$

Corresponding equations are:-

$x = t$, $y = t^2$, $z = t^3$

If we put $x = t$ in $y = t^2$ it becomes $y = x^2$ which is equation of parabolic cyllinder. So This curve Lies on parabolic cyllinder.

If we start t from origion ($t = 0$) by increasing "t" values of x, y, z also increases.

Graphs of constant vector-valued func.

If c is a constant vector in The sense That it does not depend on parameter, Then The graph of $r = c$ is a single point since The radius vector remains fixed with its tip at c.

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In 2d-space If $c = x_0 \underline{i} + y_0 \underline{j}$, then graph is the point (x_0, y_0) & if $c = x_0 \underline{i} + y_0 \underline{j} + z_0 k$ (3d-space) then graph is point (x_0, y_0, z_0).

x_0, y_0, z_0 are constant components.

Example:- The graph of $r(t) = \cos t \underline{i} + \sin t \underline{j} + 2k$

As $x = \cos t$ & $y = \sin t$ & $z = 2$

They form equ. of circle with radius 1 & lying in plane $z = 2$.

$$|r(t)| = \sqrt{x^2 + y^2 + z^2} = \sqrt{\cos^2 t + \sin^2 t + (2)^2}$$

$$= \sqrt{1 + 4} = \sqrt{5}$$

So the circle lies on surface of a sphere of radius $\sqrt{5}$.

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Lecture 28

Limits of vector valued functions

Limit of vector valued function is defined to be vector that results by taking limit of each component.

In 2d-Space $r(t) = x(t)\underline{i} + y(t)\underline{j}$ is defined as

$$\lim_{t\to\alpha} r(t) = \lim_{t\to\alpha} x(t)\underline{i} + \lim_{t\to\alpha} y(t)\underline{j}$$

In 3d-space $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$ is defined as

$$\lim_{t\to\alpha} r(t) = \lim_{t\to\alpha} x(t)\underline{i} + \lim_{t\to\alpha} y(t)\underline{j} + \lim_{t\to\alpha} z(t)\underline{k}$$

If lim of any component does not exist Then limit of $r(t)$ does not exist.

Same goes for $\lim_{t\to\alpha^+}$, $\lim_{t\to\alpha^-}$, $\lim_{t\to\infty^+}$, $\lim_{t\to\infty^-}$

We can write $\lim_{t\to\alpha} r(t) = L$

length & direction of $r(t)$ approaches to length & direction of L as $t\to\alpha$.

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Continuity of vector valued functions

r is continuous at t_0 if

$r(t_0)$ is defined.

$\lim_{t \to t_0} r(t)$ exists.

$\lim_{t \to t_0} r(t) = r(t_0)$

The graph of a continuous vector-valued function is unbroken curve.

Derivatives of vector valued functions:-

The vector-valued function's derivative is

$$r'(t) = \lim_{h \to 0} \frac{r(t+h) - r(t)}{h}$$

$\frac{f(x+h) - f(x)}{h}$

Theorem:- If $r(t) = x(t)\underline{i} + y(t)\underline{j}$ in 2d.

Then $r'(t) = x'(t)\underline{i} + y'(t)\underline{j}$

Similarly if $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$ in 3d Then $r'(t) = x'(t)\underline{i} + y'(t)\underline{j} + z'(t)k$

Proof:-

As $r'(t) = \lim_{h \to 0} \frac{r(t+h) - r(t)}{h}$

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$$r'(t) = \lim_{h\to 0} \frac{[x(t+h)-x(t)]}{h}\underline{i} + \lim_{h\to 0}\frac{[y(t+h)-y(t)]}{h}\underline{j}$$

$x(t)\underline{i} + y(t)\underline{j}$

$$= x'(t)\underline{i} + y'(t)\underline{j}$$

Similarly for 3d-space.

Notations:-

$$\frac{d}{dt}r(t), \quad \frac{dr}{dt}, \quad r'(t), \quad \dot{r}$$

Example:- $r(t) = t^2\underline{i} + t^3\underline{j}$, $r'(t) = ?$

$r'(1) = ?$

$$r'(t) = \frac{d}{dt}(t^2)\underline{i} + \frac{d}{dt}(t^3)\underline{j}$$

$$r'(t) = 2t\underline{i} + 3t^2\underline{j}$$

now put $t = 1$

$$r'(1) = 2(1)\underline{i} + 3(1)^2\underline{j}$$

$$r'(1) = 2\underline{i} + 3\underline{j}$$

Tangent vectors & Tangent lines

geometrical interpretation of derivative

Let P be a point on graph of vector-valued func $r(t)$ & let $r(t_0)$ be radius vector from origin to P. If $r'(t_0)$ exists & $r'(t_0) \neq 0$ then we call

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$r'(t_0)$ The tangent vector to the graph of r at $r(t_0)$

Equation:- Vector equation of Tangent line is

$$r = r(t_0) + t\, r'(t_0)$$

Example:- Find parametric equ. of Tangent line to circular helix $x=\cos t$, $y=\sin t$, $z=t$ at point where $t_0 = \frac{\pi}{6}$.

Equating $r(t) = x\underline{i} + y\underline{j} + z\underline{k}$

$$r(t) = (\cos t)\underline{i} + (\sin t)\underline{j} + (t)\underline{k} \longrightarrow (1)$$

$$r'(t) = (-\sin t)\underline{i} + (\cos t)\underline{j} + \underline{k} \longrightarrow (2)$$

Put $t = \frac{\pi}{6}$ in (1) & (2) respectively

$$r\left(\frac{\pi}{6}\right) = \left(\cos\frac{\pi}{6}\right)\underline{i} + \left(\sin\frac{\pi}{6}\right)\underline{j} + \left(\frac{\pi}{6}\right)\underline{k}$$

$$r\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{\pi}{6}\underline{k} \quad \&$$

$$r'\left(\frac{\pi}{6}\right) = -\frac{1}{2}\underline{i} + \frac{\sqrt{3}}{2}\underline{j} + \underline{k}$$

Now equ. of Tangent

$$r = r(t_0) + t\, r'(t_0)$$

$$= \left(\frac{\sqrt{3}}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{\pi}{6}\underline{k}\right) + t\left(\frac{-1}{2}\underline{i} + \frac{\sqrt{3}}{2}\underline{j} + \underline{k}\right)$$

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$$= \left(\frac{\sqrt{3}}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{\pi}{6}\underline{k}\right) + -\frac{1}{2}t\,\underline{i} + \frac{\sqrt{3}}{2}t\,\underline{j} + tk$$

$$= \left(\frac{\sqrt{3}}{2} - \frac{1}{2}t\right)\underline{i} + \left(\frac{1}{2} + \frac{\sqrt{3}}{2}t\right)\underline{j} + \left(\frac{\pi}{6} + t\right)\underline{k}$$

Comparing with $r = x\underline{i} + y\underline{j} + z\underline{k}$

$$x = \frac{\sqrt{3}}{2} - \frac{1}{2}t \;,\quad y = \frac{1}{2} + \frac{\sqrt{3}}{2}t \;,\quad z = \frac{\pi}{6} + t$$

Example:- The graph of $r(t) = t^2\underline{i} + t^3\underline{j}$ is called a semi-cubical parabola. Find a vector equation of tangent line to graph of $r(t)$ at

(a) The point $(0,0)$

(b) The point $(1,1)$

$$r(t) = t^2\underline{i} + t^3\underline{j}$$

$$r'(t) = 2t\underline{i} + 3t^2\underline{j}$$

(a) here $t=0$ so $r'(0) = 0$. There is no Tangent vector at the point & Tangent Line does not exist.

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(b) here $t=1$ So vector equation will be

$$r = r(t_0) + t\,r'(t_0)$$

$$r = r(1) + t\,r'(1) \quad \text{—— (1)}$$

$$r(t) = t^2\underline{i} + t^3\underline{j}$$

So $r(1) = \underline{i} + \underline{j}$

& $r'(t) = 2t\underline{i} + 3t\underline{j}$

$r'(1) = 2\underline{i} + 3\underline{j}$

putting values in (1)

$$r = r(1) + t\,r'(1)$$

$$= (\underline{i} + \underline{j}) + t(2\underline{i} + 3\underline{j})$$

$$= \underline{i} + \underline{j} + 2t\underline{i} + 3t\underline{j}$$

$$= (1+2t)\underline{i} + (1+3t)\underline{j}$$

Smooth Function:-

$r(t)$ is called smooth function if components of r have continuous derivatives w.r.t t.

& $r'(t) \neq 0$ for any value of t.

Same for 2d & 3d- space.

A parametric eq will be called smooth if has a Tangent line at every point on its graph.

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Properties of derivatives:-

- $\frac{d}{dt}(c) = 0$
- $\frac{d}{dt}[k\, r(t)] = k \frac{d}{dt}[r(t)]$
- $\frac{d}{dt}[r_1(t) + r_2(t)] = \frac{d}{dt}[r_1(t)] + \frac{d}{dt}[r_2(t)]$
- $\frac{d}{dt}[r_1(t) - r_2(t)] = \frac{d}{dt}[r_1(t)] - \frac{d}{dt}[r_2(t)]$
- $\frac{d}{dt}[r_1(t) \cdot r_2(t)] = [r_1(t)]\frac{d}{dx}[r_2(t)] + [r_2(t)]\frac{d}{dx}[r_1(t)]$

Dot Product in 2d.

- $\frac{d}{dt}[r_1(t) \cdot r_2(t)] = r_1 \cdot \frac{dr_2}{dt} + \frac{dr_1}{dt} \cdot r_2$

Cross product in 3d.

- $\frac{d}{dt}[r_1(t) \times r_2(t)] = r_1 \times \frac{dr_2}{dt} + \frac{dr_1}{dt} \times r_2$

Order doesn't matter in 2d (Dot Product)
But matters in 3d (Cross Product)

Theorem:- If $r(t)$ is a vector-valued function in 2 space or 3 space & $|r(t)|$ is constant for all t, then $r(t) \cdot r'(t) = 0$ (orthogonal vectors)

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keeping $r_1(t) = r_2(t) = r(t)$ derivative of Dot product becomes

$$\frac{d}{dt}\left[r(t).r(t)\right] = r(t).\frac{dr}{dt} + \frac{dr}{dt}.r(t)$$

or

$$\frac{d}{dt}\left[|r(t)|\right]^2 = 2r(t).\frac{dr}{dt}$$

But $\left[|r(t)|\right]^2$ is constant, so its derivative is zero. So

$$2r(t)\frac{dr}{dt} = 0$$

$$r(t).\frac{dr}{dt} = 0$$

(i-e) $r(t)$ is perpendicular to $\frac{dr}{dt}$.

Integrals of vector valued functions:-

(a) if $r(t) = x(t)\underline{i} + y(t)\underline{j}$, then

$$\int r(t)\,dt = \left(\int x(t)\,dt\right)\underline{i} + \left(\int y(t)\,dt\right)\underline{j}$$

$$\int_a^b r(t)\,dt = \left(\int_a^b x(t)\,dt\right)\underline{i} + \left(\int_a^b y(t)\,dt\right)\underline{j}$$

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If $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$

Then

$$\int r(t)\,dt = \left(\int x(t)\,dt\right)\underline{i} + \left(\int y(t)\,dt\right)\underline{j} + \left(\int z(t)\,dt\right)\underline{k}$$

$$\int_a^b r(t)\,dt = \left(\int_a^b x(t)\,dt\right)\underline{i} + \left(\int_a^b y(t)\,dt\right)\underline{j} + \left(\int_a^b z(t)\,dt\right)\underline{k}$$

Example: $r(t) = 2t\,\underline{i} + 3t^2\,\underline{j}$

(a) $\int r(t)\,dt$

$= \int (2t\,\underline{i} + 3t^2\,\underline{j})\,dt$

$= \left(\int 2t\,dt\right)\underline{i} + \left(\int 3t^2\,dt\right)\underline{j}$

$= \left(\frac{2t^2}{2} + C_1\right)\underline{i} + \left(\frac{3t^3}{3} + C_2\right)\underline{j}$

$= t^2\underline{i} + C_1\underline{i} + t^3\underline{j} + C_2\underline{j}$

$= t^2\underline{i} + t^3\underline{j} + C$ where $C_1\underline{i} + C_2\underline{j} = C$

(b) $\int_0^2 r(t)\,dt$

$= \left[t^2\right]_0^2\underline{i} + \left[t^3\right]_0^2\underline{j}$

$= (2^2 - 0^2)\underline{i} + (2^3 - 0^3)\underline{j}$

$= 4\underline{i} + 8\underline{j}$

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Properties of integral:-

$\frac{d}{dx}\left(\int x^2 dx\right)$

$$\int c\, r(t)\, dt = c \int r(t)\, dt$$

$$\int [r_1(t) + r_2(t)]\, dt = \int r_1(t)\, dt + \int r_2(t)\, dt$$

$$\int [r_1(t) - r_2(t)]\, dt = \int r_1(t)\, dt - \int r_2(t)\, dt$$

These properties also hold for definite integral.

In addition $$\frac{d}{dt}\left(\int r(t)\, dt\right) = r(t)$$

This shows indefinite integral of $r(t)$'s set of antiderivatives of $r(t)$ just as for real valued function.

$$\int r(t)\, dt = R(t) + C$$

$$\int_a^b r(t)\, dt = R(t)\Big]_a^b = R(b) - R(a).$$

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Lecture 29.

Change of Parameter

It is possible for different valued functions to have same graph. For example

$$r = (3\cos t)\hat{i} + 3(\sin t)\hat{j} \qquad 0 \le t \le 2\pi$$

&

$$r = \left[3\cos\left(\frac{s}{3}\right)\right]\hat{i} + \left[3\sin\left(\frac{s}{3}\right)\right]\hat{j} \qquad 0 \le s \le 6\pi$$

both the function have same graph ↑ circle of radius centered at origin. (here we made substitution $t = \frac{s}{3}$. real valued function). We can say that $t = g(u)$.

New vector valued function of u is smooth if original vector valued function of t is smooth.

If $\Rightarrow g$ is differentiable.

$\Rightarrow g'$ is continuous.

$\Rightarrow g'(u) \neq 0$ for any u in domain of g.

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Arc length:-

⇒ If $x = x(t)$ & $y(t)$ in 2-space

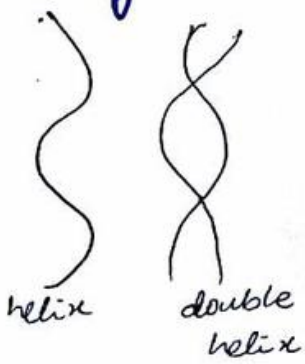
Then arc length $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

⇒ If $x = x(t)$, $y = y(t)$ & $z = z(t)$ is 3-space

arc length $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$

Example:- Find arc length of that portion of the circular helix $x = \cos t$, $y = \sin t$, $z = t$

Take $t = 0$ & $t = \pi$, $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, $\frac{dz}{dt} = 1$



$$L = \int_0^{\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2}\, dt$$

$$= \int_0^{\pi} \sqrt{\sin^2 t + \cos^2 t + 1}\, dt$$

$$= \int_0^{\pi} \sqrt{2}\, dt$$

$$= \sqrt{2} \int_0^{\pi} 1\, dt$$

$$= \sqrt{2}\,(t)\Big]_0^{\pi} \Rightarrow \sqrt{2}\,(\pi - 0)$$

$$L = \sqrt{2}\,\pi \quad \text{Ans.}$$

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Arc length as a parameter:-

The best parameter to use for representing a curve in 2-space or 3-space parametrically is length of arc measured along the curve from some reference point.

Steps:-

⇒ Select an arbitrary point on curve as reference point. (P)

⇒ Starting from reference point, choose one direction along curve ~~as~~ to be +ve & other to be -ve.

⇒ Let s be an arc Then s is +ve if point is +ve & ~~conversely~~ vice versa.

Theorem:- (a) Let C be a curve in 2-space parametrically by $x = x(t)$ & $y = y(t)$ where $x'(t)$ & $y'(t)$ are continuous functions. if S is arc length Then

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

(b) Let C be a curve in 3-space parametrically by $x = x(t)$, $y = y(t)$ & $z = z(t)$ where $x'(t)$, $y'(t)$ & $z'(t)$ are continuous, Then arc length

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$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}$$

Proof:- if $t > t_0$

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, dy\right.$$

$$= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

Similarly for 3-d.

$\frac{ds}{dt}$ does not depend on t_0, its value is independent.

Example:- $x = 2t + 1$, $y = 3t - 2$

here reference point $(1, -2)$

replace t by u

$$x = 2u + 1, \quad y = 3u - 2$$

$$\frac{dx}{du} = 2, \quad \frac{dy}{du} = 3$$

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$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{(2)^2 + (3)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{13}\, du$$

$$= \sqrt{13}\int_{0}^{t} 1\, du \Rightarrow \sqrt{13}\,[u]_0^t$$

$$= \sqrt{13}(t-0) \Rightarrow \sqrt{13}\,t$$

So $s = \sqrt{13}\,t$

$$\frac{s}{\sqrt{13}} = t$$

put in

$$x = 2t + 1 \Rightarrow 2\left(\frac{s}{\sqrt{13}}\right) + 1 \Rightarrow \frac{2}{\sqrt{13}}s + 1$$

$$y = 3(t) - 2 \Rightarrow 3\left(\frac{s}{\sqrt{13}}\right) - 2 \Rightarrow \frac{3}{\sqrt{13}}s - 2$$

So These are parametric equations.

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Example:- Find parametric equations for the circle $x = a\cos t$, $y = a\sin t$ using arc length s as parameter, with reference point for s being $(a, 0)$, $a > 0$. $\quad 0 \le t \le 2\pi$

Replace t by u.

$$x = a\cos u, \quad y = a\sin u$$

$$\frac{dx}{du} = -a\sin u, \quad \frac{dy}{du} = a\cos u$$

Now

$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{(-a\sin u)^2 + (a\cos u)^2}\, du$$

$$= \int_{0}^{t} \sqrt{a^2\sin^2 u + a^2\cos^2 u}\, du$$

$$= \int_{0}^{t} \sqrt{a^2(\sin^2 u + \cos^2 u)}\, du$$

$$= \int_{0}^{t} \sqrt{a^2}\, du \Rightarrow \int_{0}^{t} a\, du \Rightarrow a\int_{0}^{t} du$$

$$= a[u]_0^t \Rightarrow a(t - 0) \Rightarrow at$$ Ans.

$$s = at, \text{ & } t = \frac{s}{a}, \text{ So}$$

$$x = a\cos\left(\frac{s}{a}\right) \text{ & } y = a\sin\left(\frac{s}{a}\right)$$

$0 \le t \le 2\pi$

$0 \le \frac{s}{a} \le 2\pi$

$0 \le s \le 2\pi a$

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Example:- Find arc length of curve

$$r(t) = t^3 \underline{i} + t \underline{j} + \frac{1}{2}\sqrt{6}\, t^2 \underline{k} \qquad 1 \le t \le 3$$

$$x = t^3, \quad y = t, \quad z = \frac{1}{2}\sqrt{6}\, t^2$$

$$\frac{dx}{dt} = 3t^2, \quad \frac{dy}{dt} = 1, \quad \frac{dz}{dt} = \frac{1}{2}\sqrt{6}(2t) = \sqrt{6}\, t$$

$$\text{Arc length} = \int_1^3 \sqrt{(3t^2)^2 + (1)^2 + (\sqrt{6}\, t)^2}\, dt$$

$$= \int_1^3 \sqrt{9t^4 + 1 + 6t^2}\, dt$$

$$= \int_1^3 \sqrt{(3t^2)^2 + (1)^2 + 2(3t^2)(1)}\, dt$$

$$= \int_1^3 \sqrt{(3t^2+1)^2}\, dt$$

$$= \int_1^3 (3t^2+1)\, dt$$

$$= \left[\frac{3t^3}{3} + t\right]_1^3 \Rightarrow \left[t^3 + t\right]_1^3$$

$$= (3^3 + 3) - (1^3 + 1) \Rightarrow (27 + 3) - (1+1)$$

$$= 30 - 2 \Rightarrow 28 \quad \underline{\text{Ans}}.$$

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Lecture 30

Exact Differential

If $z = f(x,y)$, Then $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

If $z = f(x,y,w)$, Then $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy + \frac{\partial z}{\partial w}dw$

Exercise

1. $z = x^2 + y^2$, $dz = 2x\,dx + 2y\,dy \Rightarrow 2(x\,dx + y\,dy)$

2. $z = x^3 \sin 2y$, $dz = 3x^2 \cdot \sin 2y\,dx + x^3 \cos 2y \cdot 2\,dy$

$= x^2(3\sin 2y\,dx + 2x\cos 2y\,dy)$

3. $z = (2x-1)e^{3y}$, $dz = 2e^{3y}dx + (2x-1)e^{3y}\cdot 3dy$

$= e^{3y}(2dx + 3(2x-1)dy)$

4. $z = x^2 + 2y^2 + 3w^2$, $dz = 2x\,dx + 4y\,dy + 6w\,dw$

$= 2(x\,dx + 2y\,dy + 3w\,dw)$

5. $z = x^3y^2w$, $dz = 3x^2 \cdot y^2w\,dx + x^3(2y)w\,dy + x^3y^2\,dw$

$= x^2y(3ywdx + 2xw\,dy + xy\,dw)$

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Exact differential:-

As in $z = f(x,y)$, $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

Work in reverse now

Any expression like $dz = Pdx + Qdy$ can be written as $P = \frac{\partial z}{\partial x}$ & $Q = \frac{\partial z}{\partial y}$

Now $\frac{\partial P}{\partial y} = \frac{\partial^2 z}{\partial y \partial x}$ & $\frac{\partial Q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y}$

Also $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

So dz is exact differential if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

Example:- $dz = (3x^2+4y^2)dx + 8xy\,dy$

Comparing with $Pdx + Qdy$

$P = 3x^2+4y^2$ & $Q = 8xy$

$\frac{\partial P}{\partial y} = 8y$ $\qquad$ $\frac{\partial Q}{\partial x} = 8y$

So dz is an exact differential.

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Example:- $dz = (1+8xy)\,dx + 5x^2\,dy$

here $P = 1+8xy$ & $Q = 5x^2$

$\frac{\partial P}{\partial y} = 8x$ $\qquad$ $\frac{\partial Q}{\partial x} = 10x$

Both are not equal so dz is not an exat differential.

Exercise:-

1. $dz = 4x^3y^3\,dx + 3x^4y^2\,dy$

$P = 4x^3y^3$ | $Q = 3x^4y^2$

$\frac{\partial P}{\partial y} = 4x^3(3y^2)$ | $\frac{\partial Q}{\partial x} = 3(4x^3)y^2$

$= 12x^3y^2$ | $= 12x^3y^2$

dz is exact differential.

2. $dz = (4x^3y + 2xy^3)\,dx + (x^4 + 3x^2y^2)\,dy$

$P = 4x^3y + 2xy^3$ | $Q = x^4 + 3x^2y^2$

$\frac{\partial P}{\partial y} = 4x^3(1) + 2x(3y^2)$ | $\frac{\partial Q}{\partial x} = 4x^3 + 3(2x)y^2$

$= 4x^3 + 6xy^2$ | $= 4x^3 + 6xy^2$

YES

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3. $dz = (15y^2e^{3x} + 2xy^2)dx + (10ye^{3x} + x^2y)dy$

$P = 15y^2e^{3x} + 2xy^2$

$\frac{\partial P}{\partial y} = 15(2y)e^{3x} + 2x(2y)$

$= \underline{30ye^{3x} + 4xy}$

$Q = 10ye^{3x} + x^2y$

$\frac{\partial Q}{\partial x} = 10ye^{3x} \cdot 3 + 2xy$

$= \underline{30ye^{3x} + 2xy}$

dz is not exact differential.

4. $dz = (3x^2e^{2y} - 2y^2e^{3x})dx + (2x^3e^{2y} - 2ye^{3x})dy$

$P = 3x^2e^{2y} - 2y^2e^{3x}$

$\frac{\partial P}{\partial y} = 3x^2e^{2y} \cdot 2 - 4y^2e^{3x}$

$= \underline{6x^2e^{2y} - 4y^2e^{3x}}$

NO

$Q = 2x^3e^{2y} - 2ye^{3x}$

$\frac{\partial Q}{\partial x} = 6xe^{2y} - 2ye^{3x} \cdot 3$

$= \underline{6x^2e^{2y} - 6ye^{3x}}$

5. $dz = (4y^3\cos 4x + 3x^2\cos 2y)dx + (3y^2\sin 4x - 2x^3\sin 2y)dy$

$P = 4y^3\cos 4x + 3x^2\cos 2y$

$\frac{\partial P}{\partial y} = 4(3y^2)\cos 4x + 3x^2(-\sin 2y) \cdot 2$

$= \underline{12y^2\cos 4x - 6x^2\sin 2y}$

$Q = 3y^2\sin 4x - 2x^3\sin 2y$

$\frac{\partial Q}{\partial x} = 3y^2(\cos 4x) \cdot 4 - 2(3x^2)\sin 2y$

$= \underline{12y^2\cos 4x - 6x^2\sin 2y}$

YES

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Integration of Exact differentials

$dz = P\,dx + Q\,dy$ where $P = \frac{\partial z}{\partial x}$ & $Q = \frac{\partial z}{\partial y}$

$\therefore z = \int P\,dx$ & $z = \int Q\,dy$

Example:- $dz = (2xy+6x)\,dx + (x^2+2y^3)\,dy$

$P = 2xy+6x$, $Q = x^2+2y^3$

$z = \int P\,dx$

$= \int (2xy+6x)\,dx$

$= 2y\cdot\frac{x^2}{2} + 6\frac{x^2}{2} + F(y)$

$z = x^2y + 3x^2 + F(y)$

$z = \int Q\,dy$

$= \int (x^2+2y^3)\,dy$

$= x^2y + 2\frac{y^4}{4} + F(x)$

$z = x^2y + \frac{y^4}{2} + F(x)$

To make them equal $F(y)$ should be $\frac{y^4}{2}$ & $F(x)$ should be $3x^2$

So $z = x^2y + 3x^2 + \frac{y^4}{2}$

Example: Integrate $dz = (8e^{4x} + 2xy^2)\,dx + (4\cos 4y + 2x^2y)\,dy$. to make

$z = 2e^{4x} + x^2y^2 + \sin 4y$

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$$dz = (8e^{4x} + 2xy^2)dx + (4\cos 4y + 2x^2y)dy$$

here

$$P = 8e^{4x} + 2xy^2$$

$$z = \int P\,dx$$

$$= \int (8e^{4x} + 2xy^2)\,dx$$

$$= \frac{8e^{4x}}{4} + 2\frac{x^2}{2}\cdot y^2 + F(y)$$

$$= 2e^{4x} + x^2y^2 + F(y)$$

$$Q = 4\cos 4y + 2x^2y$$

$$z = \int Q\,dy$$

$$z = \int (4\cos 4y + 2x^2y)\,dy$$

$$= \frac{4(\sin 4y)}{4} + 2x^2\frac{y^2}{2} + f(x)$$

$$= \sin 4y + x^2y^2 + F(x)$$

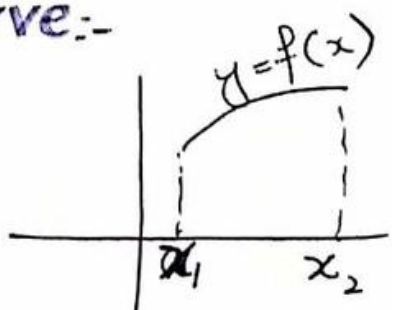
here $F(y) = \sin 4y$ & $F(x) = 2e^{4x}$

So

$$z = 2e^{4x} + x^2y^2 + \sin 4y.$$

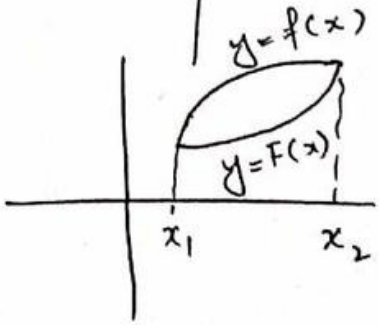
Area enclosed by closed curve:-

Remember we used integration to find area of a plane.



$$A_1 = \int_{x_1}^{x_2} y\,dx = \int_{x_1}^{x_2} f(x)\,dx$$

If there are two curves



$$A = A_1 - A_2$$

$$A = \int_{x_1}^{x_2} f(x)\,dx - \int_{x_1}^{x_2} F(x)\,dx$$

$$= -\left\{ \int_{x_1}^{x_2} f(x)\,dx + \int_{x_1}^{x_2} F(x)\,dx \right.$$

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Final result can also be written as

$$A = -\oint y\,dy$$

$\oint$ indicates that integral is to be ~~written~~ evaluated round closed boundry in the +ve.

Example:- Determine area enclosed by

The graph of $y = x^3$ & $y = 4x$ for $x \geq 0$.

$$Y = x^3 \text{ \& } y = 4x$$

To find limits

$$x^3 = 4x$$
$$x^3 - 4x = 0$$
$$x(x^2 - 4) = 0$$

$x = 0$

$x^2 - 4 = 0$
$x^2 = 4$
$\sqrt{x^2} = \sqrt{(2)^2}$
$x = \pm 2$
$x = 2 \quad \because x \geq 0$

in anticlockwise direction

C_1 : $y = x^3$, limit $x = 0$ & $x = 2$

C_2 : $y = 4x$, limit $x = 2$ & $x = 0$

$$A = -\oint y\,dy \Rightarrow -\left\{ \int_0^2 x^3\,dx + \int_2^0 4x\,dx \right\}$$

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$$= -\left\{ \left[\frac{x^4}{4}\right]_0^2 + \left[4\frac{x^2}{2}\right]_2^0 \right\}$$

$$= -\left(\left[\frac{(2)^4}{4} - \frac{(0)^4}{2}\right] + \left[2(0)^2 - 2(2)^2\right] \right)$$

$$= -\left(\frac{16}{4} - 8\right)$$

$$= -(4-8) \Rightarrow -(-4) \Rightarrow 4 \text{ Ans.}$$

Example:- Find area of Triangle with vertices $O(0,0)$ $A(5,3)$ & $B(2,6)$.

m = Slope of $OA = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-0}{5-0} = \frac{3}{5}$

m = Slope of $BA = \frac{6-3}{2-5} = \frac{3}{-3} = -1$

m = Slope of $OB = \frac{6-0}{2-0} = \frac{6}{2} = 3$

eq. of $OA \Rightarrow y - y_1 = m(x - x_1)$

$$y - 0 = \frac{3}{5}(x - 0)$$

$$y = \frac{3}{5}x$$

eq. of $BA \Rightarrow y - 5 = -1(x - 3)$

$$y = -x + 3 + 5$$

$$y = -x + 8$$

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eq. of OB ⇒ $y-0 = 3(x-0)$

$y = 3x$

Now

$$A = -\oint y\,dy$$

$$= -\left(\int_0^5 \frac{3}{5}x\,dx + \int_5^2 (-x+8)\,dy + \int_2^0 3x\,dx \right)$$

$$= -\left(\frac{3}{5}\left[\frac{x^2}{2}\right]_0^5 + \left[\frac{-x^2}{2} + 8x\right]_5^2 + \left[\frac{3x^2}{2}\right]_2^0 \right)$$

$$= -\left(\frac{3}{5}\left[\frac{5^2-0^2}{2}\right] + \left[\left(\frac{-(2)^2}{2} + 8(2)\right) - \left(\frac{(5)^2}{2} + 8(5)\right)\right] + \left[\left[\frac{3(0)^2}{2} - \frac{3(2)^2}{2}\right] \right.$$

$$= -\left[\frac{3}{5}\left(\frac{25}{2}\right) + \left(14 - \frac{55}{2}\right) + (0-6) \right]$$

$$= -\left(\frac{15}{2} + 14 - \frac{55}{2} - 6 \right)$$

$$= -\left(\frac{15+28-55-12}{2} \right) \Rightarrow -\left(\frac{-24}{2}\right)$$

$\Rightarrow$ 12 Ans.

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Line integral:-

If a field exists in xy-plane, producing a Force F on a particle at K. Then F can be divided into two components.

F_1 along Tangent &

F_a along normal.

Approximate workdone in moving particle is $F_t \delta s$ if δs is the distance covered.

$$W = \sum F_t \delta s$$

$$= \lim_{\delta s \to 0} \sum F_t \delta s = \int F_t \, ds$$

or $$\int_{AB} F_t \, ds = \int_C F_t \, ds$$

where C is curve $y = f(x)$ between $A(x_1, y_1)$ & $B(x_2, y_2)$.

Alternative form:-

If F_t has a component P in x-direction & Q in y-direction then workdone is

$$P\delta x + Q\delta y$$

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So $\int_{AB} F_t \, ds = \int_{AB} (P\,dx + Q\,dy)$

P & Q are functions of x & y.

In general.

$$\int_C F_t \, ds = \int_C (P\,dx + Q\,dy) \quad \left(\text{where } c \text{ is the curve}\right)$$

Lecture 31 Line Integral.

Example:- Evaluate $\int_C (x+3y)\,dx$ from $A(0,1)$ to $B(2,5)$ along the curve $y = 1+x^2$

In $\int_C (x+3y)\,dx$, $P = x+3y$ & $Q = 0$ & $c = 1+x^2$

$$\int_C (x+3(1+x^2))\,dx = \int_0^2 (x+3+3x^2)\,dx$$

$$= \left[\frac{x^2}{2} + 3x + \frac{3x^3}{3}\right]_0^2 = \left[\frac{2^2}{2} + 3(2) + (2)^3\right] - 0$$

$$= (2+6+8) \Rightarrow 16 \quad \text{Ans.}$$

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Example 2:- Evaluate $I = \int_C (x^2+y)dx + (x-y^2)dy$ from $A(0,2)$ to $B(2,5)$ along curve $y = 2+x$

$y = 2+x$

$dy = dx$

$P = x^2+y \Rightarrow x^2+2+x$

$Q = x-y^2 \Rightarrow x-(2+x)^2 \Rightarrow x-4-x^2-4x$

$= -3x-4-x^2 \Rightarrow -(3x+4+x^2)$

So

$$\int_C (P\,dx + Q\,dy)$$

$$= \int_0^2 \left[(x^2+x+2)\frac{dx}{} + (-(3x+4+x^2))\,dx \right]$$

$$= \int_0^2 \left[x^2+x+2-3x-4-x^2 \right] dx$$

$$= \int_0^2 (-2x-2)\,dx \Rightarrow \left[-\frac{2x^2}{2} - 2x \right]_0^2$$

$$= \left(-(2)^2 - 2(2) \right) - \left(-0^2 - 2(0) \right)$$

$$= (-4-4) - 0$$

$$= -8 \quad \text{Ans.}$$

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Example Evaluate $I = \int_C (x^2+2y)dx + xy\,dy$ from $(0,0)$ To $B(1,4)$ along curve $y = 4x^2$

$y = 4x^2$

$dy = 8x\,dx$

$P = x^2+2y \Rightarrow x^2+2(4x^2)$

$= x^2+8x^2 \Rightarrow 9x^2$

$Q = xy \Rightarrow x(4x^2) \Rightarrow 4x^3$

So

$$\int_C P\,dx + Q\,dy$$

$$= \int_0^1 9x^2\,dx + 4x^3(8x\,dx)$$

$$= \int_0^1 (9x^2 + 32x^4)\,dx \Rightarrow \left[\frac{9x^3}{3} + 32\frac{x^5}{5}\right]_0^1$$

$$\Rightarrow \left(\frac{9(1)^3}{3} + 32\frac{(1)^5}{5}\right) - \left(\frac{9(0)^3}{3} + 32\frac{(0)^5}{5}\right)$$

$$= \left(\frac{9}{3} + \frac{32}{5}\right) - 0 \Rightarrow 3 + \frac{32}{5} \Rightarrow \frac{15+32}{5}$$

$$= \frac{47}{5} \Rightarrow 9.4 \quad \text{Ans.}$$

Example:- Evaluate $\int_C (x^2+2y)dx + xy\,dy$ from $O(0,0)$ to $A(1,0)$ along line $y=0$ & Then from $A(1,0)$ To $B(1,4)$ along line $x=1$.

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(i) for line $y=0$ & $O(0,0)$, $A(1,0)$

$P = x^2+2y \Rightarrow x^2+2(0) = x^2$

$Q = xy \Rightarrow x(0) \Rightarrow 0$

$y = 0$

$dy = 0$

So

$$I_{OA} = \int_C P\,dx + Q\,dy$$

$$= \int_0^1 x^2\,dx + 0 \Rightarrow \int_0^1 x^2\,dx \Rightarrow \left[\frac{x^3}{3}\right]_0^1$$

$$= \left(\frac{(1)^3}{3} - \frac{0^3}{3}\right) \Rightarrow \frac{1}{3}$$

(ii) for line $x=1$ & $A(1,0)$ To $B(1,4)$

$P = x^2+2y \Rightarrow 1^2+2y \Rightarrow 1+2y$

$Q = xy = (1)y \Rightarrow y$

$x = 1$

$dx = 0$

So

$$I_{AB} = \int_C P\,dx + Q\,dy \Rightarrow \int_0^4 (1+2y)(0) + (y)\,dy$$

$$= \int_0^4 y\,dy \Rightarrow \left[\frac{y^2}{2}\right]_0^4 \Rightarrow \frac{(4)^2}{2} \Rightarrow \frac{16}{2} \Rightarrow 8$$

Then $I = I_{OA} + I_{AB}$

$$= \frac{1}{3} + 8 \Rightarrow \frac{1+24}{3} \Rightarrow \frac{25}{3} \text{ or } 8\frac{1}{3}$$ Ans.

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Properties of line integral:-

$\Rightarrow \int_C F\,ds = \int_c (P\,dx + Q\,dy)$

$\Rightarrow \int_{AB} F\,ds = -\int_{BA} F\,ds$

or

$\int_{AB} (P\,dx + Q\,dy) = -\int_{BA} (P\,dx + Q\,dy).$

Sign is reversed when direction is reversed.

$\Rightarrow$ For a path parallel to x-axis i.e $x = k$, $dx = 0$

So $\int_C P\,dx = 0 \quad \therefore \int_C Q\,dy$

$\Rightarrow$ For a path parallel to y-axis i.e $y = k$, $dy = 0$

$\int_C Q\,dy = 0 \quad \therefore \int_C P\,dx$

$\Rightarrow$ If path is not single valued, then it is divided into two sections. $y = f_1(x)$ A to K & $y = f_2(x)$ from K to B.

$\Rightarrow$ If path AB is divided into two parts AK & BK then $I_c = I_{AB} = I_{AK} + I_{KB}$

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$\Rightarrow$ In all cases actual path must be continuous & single valued.

Example:- Evaluate $I = \int_C (x+y)\,dx$ from $A(0,1)$ to $B(0,-1)$ along semi-circle $x^2+y^2=1$ for $x \geq 0$

here path is not single valued

$$x^2+y^2=1, \quad y^2 = 1-x^2$$

$$y = \pm\sqrt{1-x^2}$$

So (i) $y = \sqrt{1-x^2}$ from A to K

(ii) $y = -\sqrt{1-x^2}$ from K to B.

As $I = \int_C (x+y)\,dx$ here $Q = 0$

$P = x+y \Rightarrow x+\sqrt{1-x^2}$

So $I = \int_C (P\,dx + Q\,dy)$

will be $I = \int_C P\,dx$

$$I_{AB} = I_{AK} + I_{KB}$$

$$= \int_0^1 (x+\sqrt{1-x^2})\,dx + \int_1^0 (x-\sqrt{1-x^2})\,dx$$

$$= \int_0^1 (x+\sqrt{1-x^2})\,dx - \int_0^1 (x-\sqrt{1-x^2})\,dx$$

$$= \int_0^1 (x+\sqrt{1-x^2} - x + \sqrt{1-x^2})\,dx$$

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$$= \int_0^1 2\sqrt{1-x^2}\,dx$$

$$= 2\int_0^1 \sqrt{1-x^2}\,dx$$

If we put $x = \sin\theta$

$dx = \cos\theta\, d\theta$

put $x = 0$ in $x = \sin\theta$

$0 = \sin\theta$

$\theta = 0$

put $x = 1$

$1 = \sin\theta$

$\sin^{-1}(1) = \theta$

$\frac{\pi}{2} = \theta$

$\because \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}$

$$= 2\int_0^{\pi/2} \sqrt{1-\sin^2\theta}\,\cos\theta\, d\theta$$

$$= 2\int_0^{\pi/2} \cos\theta\cdot\cos\theta\, d\theta$$

$$= 2\int_0^{\pi/2} \cos^2\theta\, d\theta$$

$$= 2\int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= \int_0^{\pi/2} (1+\cos 2\theta)\, d\theta \Rightarrow$$

$$= \left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}}$$

$$= \left(\frac{\pi}{2} + \frac{\sin\pi}{2}\right) - \left(0 - \frac{\sin 0}{2}\right)$$

$$= \left(\frac{\pi}{2} + 0\right) - 0 \Rightarrow \frac{\pi}{2} \quad \text{Ans.}$$

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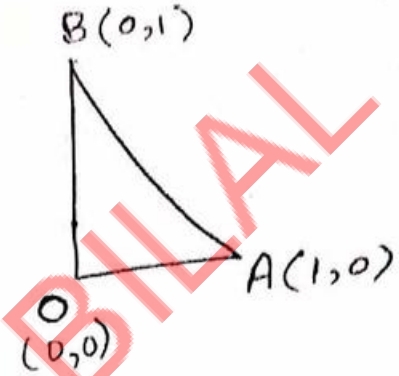
Example:- Evaluate $\oint (x^2 dx - 2xy\,dy)$ where c comprises three sides of triangle joining O(0,0) A(1,0) & B(0,1).



For OA, $y = 0$

$dy = 0$

$P = x^2$, $Q = 2xy \Rightarrow 2x(0) \Rightarrow 0$

So

$$I_1 = \int_{OA} x^2 dx \Rightarrow \int_0^1 x^2 dx \Rightarrow \left| \frac{x^3}{3} \right|_0^1$$

$$I_1 = \frac{(1)^3}{3} - \frac{(0)^3}{3} \Rightarrow \frac{1}{3} - 0 \Rightarrow \boxed{\frac{1}{3}}$$

For AB, $y = 1 - x$

$dy = -1\,dx$

$P = x^2$, $Q = -2xy$

$P = x^2$, $Q = -2x(1-x) \Rightarrow -2x + 2x^2$

$$y - y_0 = m(x - x_0)$$
$$y - 0 = -1(x - 1)$$
$$y = -x + 1$$
$$y = 1 - x$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-1}{1-0} = -1$$

So

$$I_2 = \int_{AB} x^2 dx + (-2x + 2x^2)\,dx$$

$$\int_1^0 (x^2 + 2x - 2x^2)\,dx$$

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$$I_2 = \int_1^0 (-x^2 + 2x)\,dx$$

$$= \left[-\frac{x^3}{3} + \frac{2x^2}{2} \right]_1^0 \Rightarrow 0 - \left(-\frac{1^3}{3} + (1)^2 \right)$$

$$I_2 = -\left(-\frac{1}{3} + 1 \right) \Rightarrow -\left(\frac{-1+3}{3} \right) \Rightarrow \boxed{-\frac{2}{3}}$$

For BO, line $x = 0$

$dx = 0$

$P = x^2$, $Q = -2xy$

$P = 0$, $Q = 0$

$$I_3 \int_{BO} 0\,dy \Rightarrow 0$$

Finally $I = I_1 + I_2 + I_3$

$$= \frac{1}{3} - \frac{2}{3} + 0$$

$$= \frac{1-2}{3} \Rightarrow \boxed{\frac{-1}{3}} \text{ Ans.}$$

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Example:- Evaluate $\oint_c y\,dx$ when c is the circle $x^2+y^2=4$.

$$y^2 = 4 - x^2$$

$$y = \pm\sqrt{4-x^2}$$

$r^2 = 4$

$x = r = \pm 2$

y is not single valued so

$$I = \int_2^{-2} \sqrt{4-x^2}\,dx + \int_{-2}^{2} -\sqrt{4-x^2}\,dx$$

$$= \int_{+2}^{-2} \sqrt{4-x^2}\,dx + \int_{2}^{-2} +\sqrt{4-x^2}\,dx$$

$$= \int_{2}^{-2} 2\sqrt{4-x^2}\,dx$$

$$= 2\int_{-2}^{2} \sqrt{4-x^2}\,dx$$

$$= 2\times 2\int_{0}^{2} \sqrt{4-x^2}\,dx$$

$$= 4\int_{0}^{2} \sqrt{4-x^2}\,dx$$

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put $x = 2\sin\theta$,

$dx = 2\cos\theta\, d\theta$

Limits

$x = 0$	$x = 2$
$x = 2\sin\theta$	$2 = 2\sin\theta$
$0 = 2\sin\theta$	$1 = \sin\theta$
$0 = \sin\theta$	$\sin^{-1}(1) = \theta$
$\sin^{-1}(0) = \theta$	$\frac{\pi}{2} = \theta$
$0 = \theta$	

NOW

$$-4\int_0^{\pi/2} \sqrt{4-(2\sin\theta)^2}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} \sqrt{4-4\sin^2\theta}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} \sqrt{4(1-\sin^2\theta)}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} 2\cos\theta\, 2\cos\theta\, d\theta$$

$$= -4\times 2\int_0^{\pi/2} 2\cos^2\theta\, d\theta \qquad \because 2\cos^2\frac{\theta}{2} = 1+\cos\theta$$

$$= -16\int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= -8\int_0^{\pi/2} (1+\cos 2\theta)\, d\theta \Rightarrow -8\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}}$$

$$= -8\left[\frac{\pi}{2} + \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right] \Rightarrow -8\left[\frac{\pi}{2}+0\right] \Rightarrow -8\left(\frac{\pi}{2}\right)$$

$$= -4\pi \quad \text{Ans.}$$

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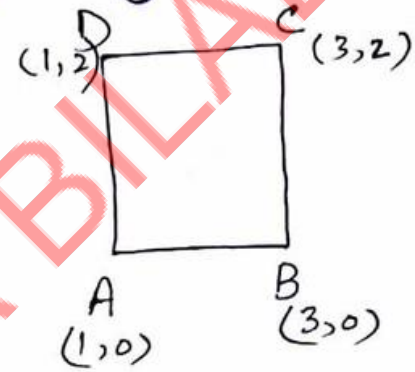
Lecture 32. Examples

Evaluate $I = \oint \{xy\,dx + (1+y^2)\,dy\}$ where c is boundry of rectangle joining A(1,0), B(3,0), C(3,2), D(1,2).



I_1 → line $y=0$, $dy=0$

$I_{AB} = I_1 = 0$

$I_2 = I_{BC}$ line $x=3$, $dx=0$

$$I_2 = \int_0^2 (1+y^2)\,dy \Rightarrow \left[y + \frac{y^3}{3}\right]_0^2 \Rightarrow \left(2 + \frac{2^3}{3}\right) - 0$$

$$= 2 + \frac{8}{3} \Rightarrow \frac{6+8}{3} = \frac{14}{3}$$

$I_3 = I_{CD}$ $y=2$, $dy=0$

$$I_3 = \int_3^1 x(2)\,dx \Rightarrow \left[2\,\frac{x^2}{2}\right]_3^1 = \left[x^2\right]_3^1$$

$$= (1)^2 - (3)^2 \Rightarrow 1 - 9 \Rightarrow -8$$

$I_4 = I_{DA}$ line $x=1$, $dx=0$

$$I_4 = \int_2^0 (1+y^2)\,dy \Rightarrow \left[y + \frac{y^3}{3}\right]_2^0 \Rightarrow 0 - \left(2 + \frac{2^3}{3}\right)$$

$$\Rightarrow \frac{-14}{3}$$ Ans

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Now $I = I_1 + I_2 + I_3 + I_4$

$= 0 + \frac{14}{3} - 8 - \frac{14}{3} \Rightarrow -8$ Ans.

Line integral w.r.t arc length

As $I = \int_{AB} F_t \, ds = \int_{AB} (P\,dx + Q\,dy)$

if $I = \int_c f(x,y)\,ds = \int_c f(x,y) \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

where $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

Example:- Evaluate $I = \int_c (4x + 3xy)\,ds$ where c is straight line joining O(0,0) to A(1,2). c is line $y = 2x$

$dy = 2$

$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$= \sqrt{1 + (2)^2} \Rightarrow \sqrt{5}$

$y - y_0 = m(x - x_0)$

$y - 0 = 2(x - 0)$

$y = 2x$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-0}{1-0} = 2$

Now

$\int_0^1 (4x + 3xy)\sqrt{5}\,dx$

$\int_0^1 (4x + 3x(2x))\sqrt{5}\,dx$

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$$\int_0^1 (4x + 6x^2)\sqrt{5}\, dx$$

$$= \sqrt{5}\left[\frac{4x^2}{2} + \frac{6x^3}{3}\right]_0^1$$

$$= \sqrt{5}\left[2x^2 + 2x^3\right]_0^1$$

$$= \sqrt{5} \times 2\left[(1^2 + 1^3) - (0)\right]$$

$$= 2\sqrt{5}\,(2) \Rightarrow 4\sqrt{5} \text{ ans.}$$

Parametric Equations:-

When x & y are expressed in parametric form $x = y(t)$ & $y = g(t)$, Then

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$I = \int_c f(x,y)\, ds = \int_{t_1}^{t_2} f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \frac{dy}{(dt)^2}}\, dt$$

Example:- Evaluate $I = \oint 4xy\, ds$, where c is defined as curve $x = \sin t$, $y = \cos t$. between $t = 0$ & $t = \frac{\pi}{4}$.

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$x = \sin t \;,\; y = \cos t$

$\frac{dx}{dt} = \cos t \;,\; \frac{dy}{dt} = -\sin t$

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \Rightarrow \sqrt{\cos^2 t + (-\sin t)^2}$$

$$= \sqrt{1} = 1$$

$$I = \int_{t_1}^{t_2} f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^{\pi/4} 4xy\,(1)\, dt \Rightarrow \int_0^{\pi/4} 4 \sin t \cos t\, dt$$

$$= 2\int_0^{\pi/4} 2 \sin t \cos t\, dt \Rightarrow 2\int_0^{\pi/4} \sin 2t\, dt$$

$$= +2\left[-\frac{\cos 2t}{2}\right]_0^{\frac{\pi}{4}} \Rightarrow -\left[\cos 2t\right]_0^{\frac{\pi}{4}}$$

$$= -\left[\cos 2\left(\frac{\pi}{4}\right) - \cos 2(0)\right]$$

$$= -\left(\cos\frac{\pi}{2} - \cos 0\right)$$

$$= -(0 - 1) \Rightarrow -(-1) \Rightarrow 1 \quad \underline{\text{Ans.}}$$

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Dependence of linear integral on path of integration.

Example:- Evaluate $I = \oint \{3x^2y^2dx + 2x^3y\,dy\}$

b/w $O(0,0)$ & $A(2,4)$

(a) along C_1 (i-e) $y = x^2$

(b) along C_2 (ie) $y = 2x$

(c) along C_3 (i-e) $x = 0$ from $(0,0)$ to $(0,4)$

& $y = 4$ from $(0,4)$ to $(2,4)$

(a)

$$I_1 = \int_C (3x^2y^2dx + 2x^3y\,dy) \quad \text{line} \quad y = x^2, \; dy = 2x\,dx$$

$$= \int_0^2 \left(3x^6dx + 2x^3(x^2)(2x)\,dx\right)$$

$$= \int_0^2 (3x^6 + 4x^6)\,dx \Rightarrow \int_0^2 7x^6\,dx$$

$$= 7\left[\frac{x^7}{7}\right]_0^2 \Rightarrow [x^7]_0^2 \Rightarrow (2)^7 - (0)^7 \Rightarrow 128$$

$$\boxed{I_1 = 128}$$

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(b) line $y = 2x$

$dy = 2dx$

$$I_2 = \int_C (3x^2y^2dx + 2x^3y\,dy)$$

$$= \int_0^2 \left[3x^2(2x)^2dx + 2x^3(2x)(2dx) \right]$$

$$= \int_0^2 \left[12x^4 + 8x^4 \right]dx \Rightarrow \int_0^2 20x^4\,dx$$

$$= 20\left[\frac{x^5}{5}\right]_0^2 \Rightarrow 4[x^5]_0^2 \Rightarrow 4[2^5 - 0^5]$$

$$= 4(32) \Rightarrow 128$$

(c) $I_{3a} = 0$

line $x = 0$ from (0,0) to (0,4)

$dx = 0$

$$I_{3b} = \int_C 3x^2(4)^2dx$$

line $y = 4$ from (0,4) to (2,4)

$dy = 0$

$$= \int_0^2 48x^2dx$$

$$= 48\left[\frac{x^3}{3}\right]_0^2 \Rightarrow 16[x^3]_0^2 \Rightarrow 16[2^3 - 0^3]$$

$$= 16(8) \Rightarrow 128 \quad \underline{\text{Ans.}}$$

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Lecture 33 Examples

Evaluate $I = \int_c \{3y\,dx + (3x+2y)dy\}$ from $A(1,2)$ to $B(3,5)$.

If no path is given, then integrand is an exact differential of some function $z = f(x,y)$. As

$$I = \int_c \{P\,dx + Q\,dy\}$$

here $P = 3y$, $z = \int 3y\,dx = 3xy + f(y)$

$Q = 3x+2y$, $z = \int (3x+2y)dy = 3xy + y^2 + f(x)$

in this case $f(y) = y^2$ & $f(x) = 0$

hence $z = 3xy + y^2$

$$I = \int_{(1,2)}^{(3,5)} d(3xy+y^2) \Rightarrow \left[3xy+y^2\right]_{(1,2)}^{(3,5)}$$

$$= \left[3(3)(5)+(5)^2\right] - \left[3(1)(2)+(2)^2\right]$$

$$= 3\left[45+25\right] - \left[6+4\right] \Rightarrow 70 - 10$$

$$= 60 \quad \text{Ans.}$$

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Example:-

Evaluate $I = \int_C \{(x^2+ye^x)dx + (e^x+y)dy\}$ b/w

A(0,1) & B(1,2).

$P = x^2+ye^x$, $z = \int (x^2+ye^x)dx$

$= \frac{x^3}{3} + ye^x + f(y)$

$Q = (e^x+y)$, $z = \int Q\,dy \Rightarrow \int (e^x+y)dy$

$= e^x y + \frac{y^2}{2} + f(x)$

here $f(y) = \frac{y^2}{2}$ & $f(x) = \frac{x^3}{3}$

So $z = \frac{x^3}{3} + ye^x + \frac{y^2}{2}$

Then $I = \left[\frac{x^3}{3} + ye^x + \frac{y^2}{2}\right]_{(0,1)}^{(1,2)}$

$= \left(\frac{1}{3} + (2)e^1 + \frac{2^2}{2}\right) - \left(0 + (1)e^0 + \frac{1^2}{2}\right)$

$= \left(\frac{1}{3} + 2e + 2\right) - \left(1 + \frac{1}{2}\right)$

$= \frac{1+6e+6}{3} - \frac{3}{2} \Rightarrow \frac{12e+14-9}{6}$

$= \frac{12e+5}{6} \Rightarrow 2e + \frac{5}{6}$ Ans.

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If $(Pdx + Qdy)$ is exact ~~function~~ differential

(a) $I = \int_c (Pdx + Qdy)$ is independent of path.

(b) $I = \oint_c (Pdx + Qdy)$ is zero.

Exact differentials in 3 independent variables:-

If $z = f(x, y, w)$ Then its exact differentia

will be $dz = Pdx + Qdy + R\,dw$

& $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$; $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$; $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

if test is successful Then

(a) $\int_c (Pdx + Qdy + R\,dw)$ is independent of path.

(b) $\oint_c (Pdx + Qdy + Rdw)$ is zero.

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Example:- Verify that

$dz = (3x^2yw + 6x)dx + (x^3w - 8y)dy + (x^3y + 1)dw$

is in exact differential & hence evaluate

$\int_c dz$ from $A(1,2,4)$ to $B(2,1,3)$.

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$P = 3x^2yw + 6x$, $Q = x^3w - 8y$, $R = x^3y + 1$

$\frac{\partial P}{\partial y} = 3x^2w$, $\frac{\partial Q}{\partial x} = 3x^2w$, $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

$\frac{\partial P}{\partial w} = 3x^2y$, $\frac{\partial R}{\partial x} = 3x^2y$, $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$

$\frac{\partial Q}{\partial w} = x^3$, $\frac{\partial R}{\partial y} = x^3$, $\frac{\partial Q}{\partial w} = \frac{\partial R}{\partial y}$

So dz is exact differential.

To find z,

$z = \int P\,dx = \int (3x^2yw + 6x)\,dx \Rightarrow \frac{3x^3}{3}yw + \frac{6x^2}{2}$

$= x^3yw + 3x^2 + f(y) + F(w)$

$z = \int Q\,dy = \int (x^3w - 8y)\,dy \Rightarrow x^3yw - \frac{8y^2}{2}$

$= x^3yw - 4y^2 + F(w) + g(x)$

$z = \int R\,dy = \int (x^3y + 1)\,dw \Rightarrow x^3yw + w + f(y) + g(x)$

Now

$z = x^3yw + 3x^2 - 4y^2 + w$

$I = [x^3yw + 3x^2 - 4y^2 + w]$

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$$I = \left[x^3yw + 3x^2 - 4y^2 + w\right]_{(1,2,4)}^{(2,1,3)}$$

$$= \left[(2)^3(1)(3) + 3(2)^2 - 4(1)^2 + 3\right] - \left[(1)^3(2)(4) + 3(1)^2 - 4(2)^2 + 4\right]$$

$$= (24 + 12 - 4 + 3) - (8 + 3 - 16 + 4)$$

$$= 36 \quad \underline{\text{Ans.}}$$

Green's Theorem:- Let P & Q are two functions of x & y that are finite & continuous. If first partial derivatives are continuous, Then

$$\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy = -\oint_C (P\,dx + Q\,dy)$$

Example:- Evaluate $I = \oint_C \{(2x - y)dx + (2y + x)dy\}$ around boundry C, the ellipse $x^2 + 9y^2 = 16$.

$P = 2x - y, \quad Q = 2y + x$

$\frac{\partial P}{\partial y} = 0 - 1 \Rightarrow -1, \quad \frac{\partial P}{\partial x} = 0 + 1 \Rightarrow 1$

$$I = -\iint_R (-1 - 1)\,dx\,dy = -\iint_R (-2)\,dx\,dy$$

$$= 2\iint_R dx\,dy$$ (Find area of ellipse & multip by 2 now)

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$x^2 + 9y^2 = 16$

divide by 16

$$\frac{x^2}{16} + \frac{9y^2}{16} = 1$$

comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{x^2}{(4)^2} + \frac{y^2}{\frac{(4)^2}{(3)^2}} = 1$$

$a = 4, \quad b = \frac{4}{3}$

Area of ellipse $= \pi(a.b)$

$= \pi\left(4 \times \frac{4}{3}\right) \Rightarrow \frac{16}{3}\pi$

Now put area in

$= 2\iint_R dx\,dy \Rightarrow 2A$

$= 2\left(\frac{16}{3}\pi\right) \Rightarrow \frac{32}{3}\pi$ Ans.

Example:- Evaluate $I = \oint_C [(2x+y)dx + (3x-2y)dy]$ taken in anticlockwise around the triangle with vertices O(0,0), A(1,0), B(1,

$$I = \oint_C [(2x+y)\,dx + (3x-2y)\,dy]$$

$P = 2x + y, \quad \frac{\partial P}{\partial y} = 1$

$Q = 3x - 2y, \quad \frac{\partial Q}{\partial x} = 3$

(not equal so apply green's Theorem)

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$$I = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$= -\iint_R (1 - 3)\, dx\,dy$$

$$= -\iint_R (-2)\, dx\,dy$$

$$= 2\iint_R dx\,dy \Rightarrow 2A$$

area of Triangle $= \frac{1}{2}$ (base x height)

$= \frac{1}{2}(1 \times 2)$

$= 1$

base = x value of A

height = y value of B.

put in

$2A = 2(1) = 2$ Ans.

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Lecture 34 Examples

Examples:- Evaluate the line integral

$I = \oint_C \{xy\,dx + (2x-y)\,dy\}$ round the region bounded by the curves $y = x^2$ & $\sqrt{x} = \sqrt{y}$ by use of Green's Theorem.

$$I = \oint_C \{xy\,dx + (2x-y)\,dy\}$$

Take $y = x^2$ & $x = y^2$

$x = (x^2)^2$

$x = x^4$

$x - x^4 = 0$

$x(1-x^3) = 0$

$\boxed{x = 0}$ | $1 - x^3 = 0$

$1 = x^3$

$\boxed{1 = x}$

put $x = 0$ in $y = x^2$

$\boxed{y = 0}$

$(0,0)$

put $x = 1$ in $y = x^2$

$y = (1)^2$

$\boxed{y = 1}$

$(1,1)$

points of intersection

Now $P = xy$ & $Q = 2x - y$

$\frac{\partial P}{\partial y} = x(1) \Rightarrow x$ & $\frac{\partial Q}{\partial x} = 2(1) - 0 \Rightarrow 2$

Both are not equal so use green's Theorem.

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$$I = \oint_C (P\,dx + Q\,dy) = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$= -\int_0^1 \int_{x^2}^{\sqrt{x}} (x-2)\,dy\,dx$$

$$= -\int_0^1 (x-2)\left[y\right]_{x^2}^{\sqrt{x}}\,dx$$

$$= -\int_0^1 (x-2)\left(\sqrt{x} - x^2\right) dx$$

$$= -\int_0^1 \left(x^{3/2} - x^3 - 2\sqrt{x} + 2x^2\right) dx$$

$$= -\left[\frac{x^{\frac{3}{2}+1}}{5/2} - \frac{x^4}{4} - 2\frac{x^{\frac{1}{2}+1}}{3/2} + \frac{2x^3}{3}\right]_0^1$$

$$= -\left[\frac{2}{5}x^{5/2} - \frac{x^4}{4} - 2\times\frac{2}{3}x^{3/2} + \frac{2x^3}{3}\right]_0^1$$

$$= -\left[\frac{2}{5}(1)^{5/2} - \frac{(1)^4}{4} - \frac{4(1)^{3/2}}{3} + \frac{2(1)^3}{3}\right] - 0$$

$$= -\left[\frac{2}{5} - \frac{1}{4} - \frac{4}{3} + \frac{2}{3}\right] \Rightarrow -\left[\frac{24-15-80+40}{60}\right]$$

$$= -\left[\frac{-31}{60}\right] \Rightarrow \frac{31}{60} \quad \text{Ans.}$$

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In special case when $P = y$, $Q = -x$, so

$$\frac{\partial P}{\partial y} = 1, \quad \frac{\partial Q}{\partial x} = -1$$

Green Theorem states that

$$+\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy = \oint_C (P\,dx + Q\,dy)$$

$$\iint_R \left(1 - (-1)\right) dx\,dy = -\oint_C (y\,dx - x\,dy)$$

$$2\iint_R dx\,dy = \oint_C (x\,dy - y\,dx)$$

$$\left(\text{area of closed region}\right) \iint_R dx\,dy = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

Example:- Determine the area of figure enclosed by $y = 3x^2$ & $y = 6x$. ~~Points of intersection~~

~~$3x^2 =$~~

$3x^2 = 6x$

$3x^2 - 6x = 0$

$3x(x-2) = 0$

$3x = 0$ | $x - 2 = 0$

$x = 0$ | $x = 2$

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For C_1 : $y = 3x^2$, $dy = 6x\,dx$

$$I_1 = \frac{1}{2}\int_0^2 (x\,dy - y\,dx)$$

$$= \frac{1}{2}\int_0^2 (x \cdot 6x\,dx - (3x^2)\,dx)$$

$$= \frac{1}{2}\int_0^2 (6x^2\,dx - 3x^2\,dx) \Rightarrow \frac{1}{2}\int_0^2 (6x^2 - 3x^2)\,dx$$

$$= \frac{1}{2}\int_0^2 3x^2\,dx \Rightarrow \frac{3}{2}\int_0^2 x^2\,dx \Rightarrow \frac{3}{2}\left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{2}\left[x^3\right]_0^2 \Rightarrow \frac{1}{2}(2^3 - 0^3) \Rightarrow \frac{1}{2}(8) = 4$$

For C_2 : $y = 6x$, $dy = 6\,dx$

$$I_2 = \frac{1}{2}\int_0^2 (x\,dy - y\,dx) \Rightarrow \frac{1}{2}\int_0^2 (x \cdot 6\,dx - 6x\,dx)$$

$$= \frac{1}{2}\int_0^2 (6x\,dx - 6x\,dx) = \frac{1}{2}\int_0^2 (6x - 6x)\,dx$$

$$= 0$$

Now

$$\text{Area} = I = I_1 + I_2 \Rightarrow 4 + 0 \Rightarrow 4 \quad \underline{\text{Ans.}}$$

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Example:- Determine the area bounded by the curves $y = 2x^3$, $y = x^3 + 1$ & the axis $x = 0$ for $x \geq 0$

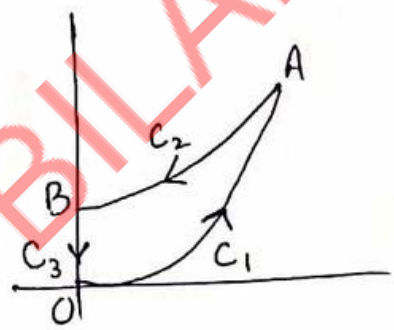
$y = 2x^3$, $y = x^3 + 1$, $x = 0$

put $2x^3 = x^3 + 1$

$2x^3 - x^3 = 1$

$x^3 = 1$

$\boxed{x = 1}$



I_1 = for C_1: $y = 2x^3$, $dy = 6x^2 dx$

$$I_1 = \frac{1}{2}\int_0^1 (x\,dy - y\,dx)$$

$$= \frac{1}{2}\int_0^1 \left[x(6x^2 dx) - (2x^3)\,dx\right]$$

$$= \frac{1}{2}\int_0^1 (6x^3\,dx - 2x^3 dx) \Rightarrow \frac{1}{2}\int_0^1 4x^3\,dx$$

$$= \frac{1}{2}\left[\frac{4x^4}{4}\right]_0^1 \Rightarrow \frac{1}{2}\left[x^4\right]_0^1 \Rightarrow \frac{1}{2}(1^4 - 0^4) \Rightarrow \frac{1}{2}$$

for C_2: $y = x^3 + 1$, $dy = 3x^2 dx$

$$I_2 = \frac{1}{2}\int_1^0 (x\,dy - y\,dx) \Rightarrow \frac{1}{2}\int_1^0 \left(x(3x^2 dx) - (x^3+1)dx\right]$$

$$= \frac{1}{2}\int_1^0 (3x^3 - x^3 - 1)\,dx \Rightarrow \frac{1}{2}\int_1^0 (2x^3 - 1)\,dx$$

$$= \frac{1}{2}\left[\frac{2x^4}{4} - x\right]_1^0 \Rightarrow \frac{1}{2}\left[0 - \left(\frac{(1)^4}{2} - 1\right)\right] \Rightarrow \frac{1}{2}\left[-\left(\frac{1-2}{2}\right)\right]$$

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$$= \frac{1}{2}\left[-\left(-\frac{1}{2}\right)\right] \Rightarrow \frac{1}{2}\left(\frac{1}{2}\right) \Rightarrow \frac{1}{4}$$

for C_3 : $x = 0$, $dx = 0$

$$I_3 = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

$$I_3 = 0$$

Now Area $= I = I_1 + I_2 + I_3$

$$= \frac{1}{2} + \frac{1}{4} + 0 \Rightarrow \frac{2+1}{4} = \frac{3}{4}$$ sq. units

Revision Summary:

Properties of line integral:-

1. Sign of integral is reversed when direction of integral is reversed.
2. Path of integration parallel to y-axis, $dx = 0$

   $\therefore I_c = \int_c Q\,dy$
3. Path of integration parallel to x-axis, $dy = 0$

   $\therefore I_c = \int_c P\,dx$
4. Path of integration must be continuous & single valued.
5. Dependence of line integral on Path of integration.

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6. The value of line integral depends on particular path of integration.

7. Exact differential

- If $Pdx + Qdy$ is an exact differential

(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

(b) $I = \int_C (Pdx + Qdy)$ is independent of path.

(c) $I = \oint_C (Pdx + Qdy)$ is zero.

8. Exact differential in three variables.

- If $Pdx + Qdy + Rdw$ is an exact differential

(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$, $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

(b) $\int_C (P(dx + Qdy + Rdw)$ is independent of path.

(c) $\oint_C (Pdx + Qdy + Rdw)$ is zero.

9. Green's Theorem.

$$\oint_C (Pdx + Qdy) = -\iint_R \left\{ \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right\} dx\,dy$$ & for simple closed curve $\oint_C (x\,dy - y\,dx) = 2\iint_R dx\,dy = 2A$

where A is area of closed figure.

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Gradient of Scalar function:-

Del operator is given by $\nabla = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right)$

$$\nabla\phi = \text{grad}\,\phi = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right)\phi$$

$$\nabla\phi = \text{grad}\,\phi = \frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j + \frac{\partial\phi}{\partial z} k$$

Div (Divergence of vector function).

If $A = a_1 i + a_2 j + a_3 k$, Then

$$\text{div } A = \nabla \cdot A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (a_1 i + a_2 j + a_3 k)$$

$$= \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$$

Note That

(a) grad operator ∇ acts on scalar & gives vector.

(b) div operator $\nabla\cdot$ acts on vector & gives scalar.

Example:- If $A = x^2 y i - xyz j + yz^2 k$. find div A.

$$\text{div } A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (x^2 y i - xyz j + yz^2 k)$$

$$= \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(-xyz) + \frac{\partial}{\partial z}(yz^2)$$

$$\nabla \cdot A = 2xy - xz + 2yz$$

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Example:- If $A = 2x^2y\,i - 2(xy^2 + y^3z)j + 3y^2z^2k$ determine $\nabla . A$ (div A).

$$\nabla . A = \frac{\partial}{\partial x}(2x^2y) + \frac{\partial}{\partial y}(-2(xy^2 + y^3z)) + \frac{\partial}{\partial z}(3y^2z^2)$$

$$= 4xy - 2(2xy + 3y^2z) + 6y^2z$$

$$= 4xy - 4xy - 6y^2z + 6y^2z \Rightarrow 0$$

vector for which $\nabla . A = 0$ called a Solenoid vector.

Curl (curl of a vector function) :-

Curl operator denoted by $\nabla \times$ acts on vector & also gives a vector.

If $A = a_1 i + a_2 j + a_3 k$ then curl $A = \nabla \times A$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \times (a_1 i + a_2 j + a_3 k)$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$\nabla \times A = i\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) - j\left(\frac{\partial a^3}{\partial x} - \frac{\partial a_1}{\partial z}\right) + k\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)$$

curl A is a vector function.

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Example:- If $A = (y^4 - x^2z^2)i + (x^2+y^2)j - x^2yz\,k$, determine curl A at (1, 3, −2)

$$\text{Curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^4 - x^2z^2 & x^2+y^2 & -x^2yz \end{vmatrix}$$

$$= i\left(\frac{\partial}{\partial y}(-x^2yz) - \frac{\partial}{\partial z}(x^2+y^2)\right) - j\left(\frac{\partial}{\partial x}(-x^2yz) - \frac{\partial}{\partial z}(y^4 - x^2z^2)\right)$$

$$+ k\left(\frac{\partial}{\partial x}(x^2+y^2) - \frac{\partial}{\partial y}(y^4 - x^2z^2)\right)$$

$$= i(-x^2z - 0) - j(-2xyz + 2x^2z) + k(2x - 4y^3)$$

$$= \text{put } (1, 3, -2)$$

$$= i(-(1)^2(-2)) - j[-2(1)(3)(-2) + 2(1)^2(-2)] + k(2(1) - 4(3)^3)$$

$$= i(2) - j(12-48) + k(2-108)$$

$$= 2i - 8j - 106k \quad \text{Ans.}$$

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Example: Determine curl F at point (2,0,3) given that $F = ze^{2xy}\,i + 2xz\cos y\,j + (x+2y)k$.

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ze^{2xy} & 2xz\cos y & x+2y \end{vmatrix}$$

$$= i\left[2 - 2x\cos y\right] - j\left[1 - e^{2xy}\right] + k\left[2z\cos y - ze^{2xy}\cdot 2x\right]$$

Put (2, 0, 3)

$$= i\left[2 - 2(2)\cos 0^\circ\right] - j\left[1 - e^{2(2)(0)}\right] + k\left[2(3)\cos 0^\circ - (3)e^{2(2)(0)}\cdot 2(2)\right]$$

$$= i(-2) - j(1-1) + k(6-12)$$

$$= -2i - 6k \Rightarrow -2(i + 3k).$$

Summary of grad, div, curl

(a) grad ∇ acts on scalar to give vector.

(b) Div $\nabla\cdot$ acts on vector to give scalar.

(c) Curl $\nabla\times$ acts on vector to give vector.

(d) with a scalar function. $\phi(x,y,z)$

$$\text{grad}\,\phi = \nabla\phi = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$$

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(e) With a vector function $A = a_1 i + a_2 j + a_3 k$

(i) $\text{div } A = \nabla . A = \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$

(ii) $\text{curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$

Multiple Operations:-

We can combine the operators grad, div & curl in multiple operations.

Example:- $A = x^2 y i + y z^3 j + z x^3 k$

div A = ?, grad (div A) = ?

$$\text{div } A = \nabla . A = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(y z^3) + \frac{\partial}{\partial z}(z x^3)$$

$$= 2xy + z^3 + x^3$$

Now

$$\text{grad (div } A) = \nabla(\nabla . A) = \frac{\partial}{\partial x}(\text{div } A) + \frac{\partial}{\partial y}(\text{div } A) + \frac{\partial (\text{div } A)}{\partial z}$$

$$= \frac{\partial}{\partial x}(2xy + z^3 + x^3) i + \frac{\partial}{\partial y}(2xy + z^3 + x^3) j + \frac{\partial}{\partial z}(2xy + z^3 + x^3) k$$

$$= (2y + 3x^2) i + 2x j + 3z^2 k \quad \underline{\text{Ans.}}$$

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Example:- If $\phi = xyz - 2y^2z + x^2z^2$.

find div grad ϕ at $(2,4,1)$

$$\text{grad}\,\phi = \nabla\phi = \frac{\partial}{\partial x}\phi i + \frac{\partial}{\partial y}\phi j + \frac{\partial}{\partial z}\phi k$$

$$= (yz - 2xz^2)i + (xz - 4yz)j + (xy - 2y^2 + 2x^2z)k$$

Now

$$\text{div grad}\,\phi = \nabla\cdot(\nabla\phi) = \frac{\partial}{\partial x}a_1 + \frac{\partial}{\partial y}a_2 + \frac{\partial}{\partial z}a_3$$

$$= 2z^2 - 4z + 2x^2$$

put $(2,4,1)$

$$= 2(1)^2 - 4(1) + 2(2)^2$$

$$= 2 - 4 + 8 \Rightarrow 6 \quad \text{Ans.}$$

$$\therefore \text{div grad}\,\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

Example:- If $F = x^2yz\,i + xyz^2\,j + y^2z\,k$

determine curl F at the point $(2,1,1)$. Determine an expression for curl F in the usual way, which will be a vector, & then curl of the result. Finally substitute values. (i.e $\nabla \times (\nabla \times F)$

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$$\text{curl } F = \nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & xyz^2 & y^2z \end{vmatrix}$$

$$= i(2yz - 2xyz) - j(0 - x^2y) + k(yz^2 - x^2z)$$

Now

$$\nabla \times (\nabla \times F) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2yz - 2xyz & +x^2y & yz^2 - x^2z \end{vmatrix}$$

$$= i[(z^2 - 0) - 0] - j[(0 - 2xz) - (2y - 2xy)] + k[2xy - (2y - 2xz)$$

$$= z^2 i - j(-2xz - 2y + 2xy) + k(2xy - 2y + 2xz)$$

Put (2, 1, 1)

$$= (1)^2 i - j[-2(2)(1) - 2(1) + 2(2)(1)] + k[2(2)(1) - 2(1) + 2(2)(1)$$

$$= i - j[-4 - 2 + 4] + k[4 - 2 + 4]$$

$$= i - j(-2) + k(6)$$

$$= i + 2j + 6k \quad \text{Ans.}$$

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Two intersecting general results-

(a) Curl grad ϕ where ϕ is scalar.

$$\text{Curl grad } \phi = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= i\left[\frac{\partial^2\phi}{\partial y \partial z} - \frac{\partial^2\phi}{\partial z \partial y}\right] - j\left[\frac{\partial^2\phi}{\partial x \partial z} - \frac{\partial^2\phi}{\partial z \partial x}\right] + k\left[\frac{\partial^2\phi}{\partial x \partial y} - \frac{\partial^2\phi}{\partial y \partial x}\right]$$

$$= 0$$

So curl grad ϕ

$$\nabla \times (\nabla \phi) = 0$$

(b) div Curl A where A is vector

$$\text{Curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$= i\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) - j\left(\frac{\partial a_3}{\partial x} - \frac{\partial a_1}{\partial z}\right) + k\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)$$

$$\text{div Curl } A = \nabla \cdot (\nabla \times A)$$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (\nabla \times A)$$

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$$= \frac{\partial^2 a_3}{\partial x \partial y} - \frac{\partial^2 a_2}{\partial z \partial x} - \frac{\partial^2 a_3}{\partial x \partial y} + \frac{\partial^2 a_1}{\partial y \partial z} - \frac{\partial^2 a_2}{\partial z \partial x} - \frac{\partial^2 a_1}{\partial y \partial z}$$

$$= 0$$

So $\boxed{\text{div curl } A = \nabla \cdot (\nabla \times A) = 0}$

(c) div grad ϕ where ϕ is scalar.

$$\nabla\phi = \text{grad } \phi = \frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k$$

Now

$$\text{div grad } \phi = \nabla \cdot (\nabla \phi)$$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k\right)$$

$$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

$$= \nabla^2 \phi$$

So

$$\boxed{\text{div grad } \phi = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi}$$

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Lecture 35

Definite Integrals

for $\sin^n x$ & $\cos^n x$, $\left(0 \le x \le \frac{\pi}{2}\right)$

$$\int_0^{\pi/2} \sin^2 x\,dx = \int_0^{\pi/2} \left(\frac{1-\cos 2x}{2}\right) dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)\,dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)\,dx = \frac{1}{2}\left[x - \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2}\left[\left\{\frac{\pi}{2} - \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right\} - \left\{0 - \frac{\sin 2(0)}{2}\right\}\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{2} - 0 - 0 + 0\right] \Rightarrow \frac{1}{2}\left(\frac{\pi}{2}\right)$$

$$= \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4} \quad \text{Ans.}$$

$$\sin\frac{x}{2} = \sqrt{\frac{1-\cos x}{2}}$$

$$\sin^2\frac{x}{2} = \frac{1-\cos x}{2}$$

$$\sin^2 x = 1-\cos 2x$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x\,dx = \int_0^{\frac{\pi}{2}} \left(\frac{1+\cos 2x}{2}\right) dx \Rightarrow \frac{1}{2}\int_0^{\frac{\pi}{2}} (1+\cos 2x)$$

$$= \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}} \Rightarrow \frac{1}{2}\left[\left\{\frac{\pi}{2} + \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right\} - \{0\}\right]$$

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$= \frac{1}{2}\left(\frac{\pi}{2}\right) \Rightarrow \frac{1}{2}\cdot\frac{\pi}{2} \Rightarrow \frac{\pi}{4}$ Ans.

$$\int_0^{\pi/2} \sin^3 x\,dx = \int_0^{\pi/2} \sin^2 x \sin x\,dx$$

$$= \int_0^{\pi/2} (1-\cos^2 x)\sin x\,dx \Rightarrow \int_0^{\pi/2} (\sin x - \cos^2 x \sin x)\,dx$$

$$= \int_0^{\pi/2} \sin x\,dx + \int_0^{\pi/2} \cos^2 x\,(-\sin x)\,dx$$

$$= \left[-\cos x\right]_0^{\frac{\pi}{2}} + \left[\frac{\cos^3 x}{3}\right]_0^{\frac{\pi}{2}}$$

$$= -\left[\cos\frac{\pi}{2} - \cos 0^\circ\right] + \frac{1}{3}\left[\cos^3\frac{\pi}{2} - \cos^3 0\right]$$

$$= -\left[0 - 1\right] + \frac{1}{3}\left[(0)^3 - (1)^3\right]$$

$$= 1 + \frac{1}{3}(-1) \Rightarrow 1 - \frac{1}{3} = \frac{3-1}{3} = \frac{2}{3}$$ Ans.

$$\int_0^{\pi/2} \cos^3 x\,dx = \int_0^{\pi/2} \cos^2 x \cos x\,dx = \int_0^{\pi/2} (1-\sin^2 x)\cos x\,dx$$

$$= \int_0^{\pi/2} (\cos x - \sin^2 x \cos x)\,dx \Rightarrow \left[\sin x\right]_0^{\pi/2} - \left[\frac{\sin^3 x}{3}\right]_0^{\frac{\pi}{2}}$$

$$= \left(\sin\frac{\pi}{2} - \sin 0\right) - \frac{1}{3}\left(\sin^3\frac{\pi}{2} - \sin^3 0\right)$$

$$= (1-0) - \frac{1}{3}\left((1)^3 - (0)^3\right) \Rightarrow 1 - \frac{1}{3} \Rightarrow \frac{2}{3}$$ Ans.

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$$\int_0^{\pi/2} \sin^4 x\,dx = \int_0^{\pi/2} (\sin^2 x)^2\,dx \Rightarrow \int_0^{\pi/2} \left[\frac{1-\cos 2x}{2}\right]^2$$

$$= \int_0^{\frac{\pi}{2}} \left[\frac{1 - 2\cos 2x + \cos^2 2x}{4}\right] dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 - 2\cos 2x + \frac{1+\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 - 2\cos 2x + \frac{1}{2} + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(\frac{3}{2} - 2\cos 2x + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\left[\frac{3}{2}x - 2\,\frac{\sin 2x}{2} + \frac{\sin 4x}{2\times 4}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4}\left[\frac{3}{2}\left(\frac{\pi}{2}\right) - \sin 2\left(\frac{\pi}{2}\right) + \frac{\sin 4\left(\frac{\pi}{2}\right)}{8}\right] - 0$$

$$= \frac{1}{4}\left(\frac{3}{2}\cdot\frac{\pi}{2} - 0 + 0\right) \Rightarrow \frac{1}{4}\cdot\frac{3}{2}\cdot\frac{\pi}{2}$$

$$= \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

$$\cos\frac{x}{2} = \sqrt{\frac{1+\cos x}{2}}$$

$$\cos^2\frac{x}{2} = \frac{1+\cos x}{2}$$

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

$$\cos^2 2x = \frac{1+\cos 4x}{2}$$

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$$\int_0^{\pi/2} \cos^4 x\,dx = \int_0^{\pi/2} (\cos^2 x)^2\,dx = \int_0^{\pi/2} \left[\frac{1+\cos 2x}{2}\right]^2 dx$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{1+2\cos 2x+\cos^2 2x}{4}\right) dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1+2\cos 2x+\frac{1+\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(1+2\cos 2x+\frac{1}{2}+\frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(\frac{3}{2}+2\cos 2x+\frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\left[\frac{3}{2}x+\frac{2\sin 2x}{2}+\frac{\sin 4x}{8}\right]_0^{\pi/2}$$

$$= \frac{1}{4}\left[\frac{3}{2}\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+\frac{\sin 4\left(\frac{\pi}{2}\right)}{8}\right]-0$$

$$= \frac{1}{4}\left[\frac{3}{2}\cdot\frac{\pi}{2}+0+0\right]$$

$$= \frac{1}{4}\cdot\frac{3}{2}\cdot\frac{\pi}{2}$$

$$= \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} \quad \text{Ans.}$$

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$$\Rightarrow \int_0^{\pi/2} \sin^5 x\,dx = \frac{4}{5}\cdot\frac{2}{3}, \quad \int_0^{\pi/2} \cos^5 x\,dx = \frac{4}{5}\cdot\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^6 x\,dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^6 dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\Rightarrow \int_0^{\pi/2} \sin^7 x\,dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}, \quad \int_0^{\pi/2} \cos^7 x\,dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^8 x\,dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^8 x\,dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\Rightarrow \int_0^{\pi/2} \sin^9 x\,dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}, \quad \int_0^{\pi/2} \cos^9 x\,dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{10} x\,dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^{10} x\,dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

Wallis sine formula:-

When n is even:

$$\int_0^{\pi/2} \sin^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

When n is odd:

$$\int_0^{\pi/2} \cos^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{11} x\,dx = \frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}, \quad \int_0^{\pi/2} \cos^{11} x\,dx = \frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{12} x\,dx = \frac{11}{12}\cdot\frac{9}{10}\cdot\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^{12} x\,dx = \frac{11}{12}\frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

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Integration by parts:-

$$\int UV\,dx = U\int V\,dx - \int\left(\int V\,dx \cdot \frac{dU}{dx}\right)dx$$

Example:- Evaluate $\int x \ln x \cdot dx$

$$\int \ln x \cdot x\,dx = \ln x \int x\,dx - \int\left(\int x\,dx \cdot \frac{d(\ln x)}{dx}\right)dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int\left(\frac{x^2}{2}\cdot\frac{1}{x}\right)dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int \frac{x}{2}\,dx \Rightarrow \ln x \cdot \frac{x^2}{2} - \frac{1}{2}\int x\,dx$$

$$= \ln x \cdot \frac{x^2}{2} - \frac{1}{2}\cdot\frac{x^2}{2} \Rightarrow \frac{x^2}{2}\left[\ln x - \frac{1}{2}\right] \text{ Ans}$$

Example:- Evaluate $\int x \sin x\,dx$

$$\int x \sin x\,dx = x\int \sin x\,dx - \int\left(\int \sin x\,dx \cdot \frac{d(x)}{dx}\right)dx$$

$$= x(-\cos x) - \int(-\cos x \cdot 1)\,dx$$

$$= -x\cos x + \int \cos x\,dx$$

$$= -x\cos x + \sin x \quad \text{Ans.}$$

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Line integrals

Length of curve $AB = \int_c dr$ when $dr \to 0$

Lecture 36 Scalar Field

If a scalar field $V(x)$ exist for all points on the curve, Then line integral $= \int_c V(r)\,dr$.

To evaluate line integral, integrand is expressed in terms of x, y, z with $dr = dx\,i + dy\,j + dz\,k$

Example:- If $V = xy^2z$, evaluate $\int_c V(r)\,dr$ along the curve c having parametric equations $x = 3u$, $y = 2u^2$, $z = u^3$ between $A(0,0,0)$ & $B = (3,2,1)$.

Solution:-

$$V = xy^2z = (3u)(2u^2)^2(u^3)$$
$$= (3u)(4u^4)(u^3)$$
$$= 12u^8$$

$x = 3u$, $dx = 3$

$y = 2u^2$, $dy = 4u$

$z = u^3$, $dz = 3u^2$

$$dr = dx\,i + dy\,j + dz\,k$$
$$= 3\,du\,i + 4u\,du\,j + 3u^2 k\,du$$
$$= (3i + 4u\,j + 3u^2 k)\,du$$

put $A(0,0,0)$ in

$0 = 3u$	,	$0 = 2u^2$	, $0 = u^3$
$0 = u$		$0 = u$	, $0 = u$

put $B(3,2,1)$,

$3 = 3u$	,	$2 = 2u^2$	, $1 = u^3$
$1 = u$		$1 = u$	$1 = u$

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$$\int_c V(r)\,dr = \int_0^1 12u^8(3i+4uj+3u^2k)\,du$$

$$= \int_0^1 (36u^8 i + 48u^9 j + 36u^{10} k)\,du$$

$$= \left[36\frac{u^9}{9}i + 48\frac{u^{10}}{10}j + 36\frac{u^{11}}{11}k\right]_0^1$$

$$= \left(4(1)^9 i + \frac{24}{5}(1)^{10} j + \frac{36}{11}(1)^{11} k\right) - 0$$

$$= 4i + \frac{24}{5}j + \frac{36}{11}k \quad \text{Ans.}$$

Example:- If $V = xy + y^2z$. Evaluate $\int_c V(r)\,dr$ along curve c defined by $x=t^2$, $y=2t$, $z=t+5$ between $(0,0,5)$ & $(4,4,7)$.

$$V = xy + y^2z = (t^2)(2t) + (2t)^2(t+5)$$

$$= 2t^3 + 4t^2(t+5)$$

$$= 2t^3 + 4t^3 + 20t^2$$

$$= 6t^3 + 20t^2$$

Now

$x = t^2$, $dx = 2t\,dt$

$y = 2t$, $dy = 2\,dt$

$z = t+5$, $dz = 1\,dt$

$$dr = dx\,i + dy\,j + dz\,k = 2t\,dt\,i + 2dt\,j + dt\,k$$

$$= (2t\,i + 2j + k)\,dt$$

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Put (0,0,5) in

$x = t^2$	$y = 2t$	$z = t+5$
$0 = t^2$	$0 = 2t$	$5 = t+5$
$0 = t$	$0 = t$	$0 = t$

Put (4,4,7) in

$4 = t^2$	$4 = 2t$	$7 = t+5$
$2 = t$	$\frac{4}{2} = t$	$7-5 = t$
	$2 = t$	$2 = t$

So

$$\int_C V(r)\,dr = \int_0^2 (6t^3 + 20t^2)(2ti + 2j + k)\,dt$$

$$= \int_0^2 (12t^4 i + 40t^3 i + 12t^3 j + 40t^2 j + 6t^3 k + 20t^2 k)\,dt$$

$$= 2\int_0^2 (6t^4 + 20t^3)i + (6t^3 + 20t^2)j + (3t^3 + 10t^2)k$$

$$= 2\left[\left(\frac{6t^5}{5} + \frac{20t^4}{4}\right)i + \left(\frac{6t^4}{4} + \frac{20t^3}{3}\right)j + \left(\frac{3t^4}{4} + \frac{10t^3}{3}\right)k\right]_0^2$$

$$= 2\left[\left(\frac{6(2)^5}{5} + \frac{20(2)^4}{4}\right)i + \left(\frac{6(2)^4}{4} + \frac{20(2)^3}{3}\right)j + \left(\frac{3(2)^4}{4} + \frac{10(2)^3}{3}\right)\right]$$

$$= 2\left[\left(\frac{6(32)}{5} + \frac{20(16)}{4}\right)i + \left(\frac{6(16)}{4} + \frac{20(8)}{3}\right)j + \left(\frac{3(16)}{4} + \frac{10(8)}{3}\right)\right]$$

$$= 2\left[\left(\frac{192}{5} + 80\right)i + \left(24 + \frac{160}{3}\right)j + \left(12 + \frac{80}{3}\right)k\right]$$

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$$= 2\left[\left(\frac{192+400}{5}\right)i + \left(\frac{72+160}{3}\right)j + \left(\frac{36+80}{3}\right)k \right]$$

$$= 2\left[\frac{592}{5}i + \frac{232}{3}j + \frac{116}{3}k \right]$$

$$= 2\times 4\left[\frac{148}{5}i + \frac{58}{3}j + \frac{29}{3}k \right]$$

$$= 8\left[\frac{444i + 290j + 145k}{15} \right]$$

$$= \frac{8}{15}\left[444i + 290j + 145k \right]$$ Ans.

Vector Field:-

If a vector field $F(r)$ exists for all points of the curve c, Then for each element of arc

The line integral $= \int_c F.dr$

here $F = F_1 i + F_2 j + F_3 k$ & $dr = dx\,i + dy\,j + dz\,k$

Then $\int_c F.dr = \int_c (F_1 i + F_2 j + F_3 k).(dx\,i + dy\,j + dz\,k)$

$$\int_c F.dr = \int_c F_1 dx + F_2 dy + F_3 dz$$

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Example:- If $F(r) = x^2yi + xzj + 2yzk$,

Evaluate $\int_c F.dr$ b/w $A(0,0,0)$ & $B(4,2,1)$ along curve c having parametric equations

$x = 4t, \quad y = 2t^2, \quad z = t^3$

$dx = 4dt, \quad dy = 4t\,dt, \quad dz = 3t^2dt$

$$F(r) = x^2yi + xzj + 2yzk$$

$$= (4t)^2(2t^2)\,i + (4t)(t^3)j + 2(2t^2)(t^3)k$$

$$= 32t^4 i + 4t^4 j + 4t^5 k$$

& Now $dr = (4i + 4tj + 3t^2k)\,dt$

$$\int_c F.dr = \int_c (32t^4 i + 4t^4 j + 4t^5 k).(4i + 4tj + 3t^2 k)\,dt$$

$$= \int_c (128t^4 + 16t^5 + 12t^7)\,dt$$

Limits:- put $(0,0,0)$ in

$x = 4t$	$y = 2t^2$	$z = t^3$
$0 = 4t$	$0 = 2t^2$	$0 = t^3$
$0 = t$	$0 = t$	$0 = t$

put $(4,2,1)$

$4 = 4t$	$2 = 2t^2$	$1 = t^3$
$1 = t$	$\frac{2}{2} = t^2$	$1 = t$
	$1 = t$	

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$$= \int_c (128t^4 + 16t^5 + 12t^7)\,dt$$

$$= \int_0^1 \left(128\frac{t^4}{5} + 16t^5 + 12t^7\right) dt$$

$$= \left[128\frac{t^5}{5} + 16\frac{t^6}{6} + 12\frac{t^8}{8}\right]_0^1$$

$$= \left(\frac{128(1)^5}{5} + \frac{16(1)^6}{6} + \frac{12(1)^8}{8}\right) - 0$$

$$= \frac{128}{5} + \frac{16}{6} + \frac{12}{8} \Rightarrow 25.6 + 2.67 + 1.5$$

$$\int_c F \cdot dr = 29.76 \text{ Ans.}$$

Example:- If $F(r) = x^2y\,i + 2yz\,j + 3z^2x\,k$.

Evaluate $\int_c F \cdot dr$ b/w $A(0,0,0)$ & $B(1,2,3)$

(a) along straight line

c_1 from $(0,0,0)$ to $(1,0,0)$

c_2 from $(1,0,0)$ to $(1,2,0)$

c_3 from $(1,2,0)$ to $(1,2,3)$

(b) along straight line c_4 joining $(0,0,0)$ to $(1,2,3)$.

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$$\int_C F \cdot dr = \int_C (x^2y\,i + 2yz\,j + 3z^2x\,k) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_C (x^2y\,dx + 2yz\,dy + 3z^2x\,dz)$$

(i) C_1, $y=0$, $z=0$, $dy=0$, $dz=0$

$$\int_{C_1} F \cdot dr = 0+0+0 \Rightarrow \underline{0}$$

(ii) C_2 $x=1$, $z=0$, $dx=0$, $dz=0$

$$\int_{C_2} F \cdot dr = 0+0+0 \Rightarrow \underline{0}$$

(iii) C_3

$x=1$, $y=2$, $dx=0$, $dy=0$

$$\int_{C_3} F \cdot dr = \int_{C_3} 0+0+3z^2(1)\,dz$$

$$= \int_0^3 3z^2\,dz \Rightarrow \left[\frac{3z^3}{3}\right]_0^3$$

$$= \left[z^3\right]_0^3 \Rightarrow (3)^3 - (0)^3 = \underline{27}$$

Add all

$$\int_{(0,0,0)}^{(1,2,3)} F \cdot dr = 0+0+27 \Rightarrow 27 \quad \underline{\text{Ans.}}$$

$$\therefore \int_{C_1+C_2+C_3} F \cdot dr = 27$$

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(b) parametric equation 'll be

$x = t,\ y = 2t,\ z = 3t$ according to $(1,2,3)$

$dx = dt,\ dy = 2dt,\ dz = 3dt$

$$F(t) = x^2 y\,i + 2yz\,j + 3z^2 x\,k$$

$$= (t)^2(2t)i + 2(2t)(3t)j + 3(3t)^2(t)k$$

$$F = 2t^3 i + 12t^2 j + 27t^3 k$$

$$dr = dx\,i + dy\,j + dz\,k$$

$$= dt\,i + 2dt\,j + 3dt\,k$$

$$dr = (i + 2j + 3k)\,dt$$

Now put $(0,0,0)$ in

$x = t$	$y = 2t$	$z = 3t$
$0 = t$	$0 = 2t$	$0 = 3t$
	$0 = t$	$0 = t$

put $(1,2,3)$ in

$1 = t$	$2 = 2t$	$3 = 3t$
	$1 = t$	$1 = t$

Now $$\int_C F \cdot dr = \int_0^1 (2t^3 i + 12t^2 j + 27t^3 k) \cdot (i + 2j + 3k)\,dt$$

$$= \int_0^1 (2t^3 + 24t^2 + 81t^3)\,dt$$

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$$= \int_0^1 (24t^2 + 83t^3)\,dt$$

$$= \left[\frac{24t^3}{3} + \frac{83t^4}{4}\right]_0^1$$

$$= \left(\frac{24(1)^3}{3} + \frac{83(1)^4}{4}\right) - 0$$

$$= \frac{24}{3} + \frac{83}{4} \Rightarrow$$

$$= 8 + 20.75 \Rightarrow 28.75 \quad \underline{\text{Ans.}}$$

So **(a)** $\int F \cdot dr$ via $C_1, C_2, C_3 = 27$

(b) $\int F \cdot dr$ via $C_4 = 28.4$

value of line integral depends on path b/w two end points A & B.

Example:- Evaluate $\int_V F\,dv$ where v is region bounded by planes $x=0, y=0, z=0$ & $2x+y+z=2$ & $F = 2zi + yk$.

In $2x+y+z=2$

when $z=0$ $\quad 2x+y=2$, $y=2-2x$ $\quad \boxed{y=2(1-x)}$

when $y=0$ $\quad 2x+z=2$, $z=2-2x$

when $x=0$ $\quad y+z=2$, $z=2-y$

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as $2x+y+z=2$

$z = 2-2x-y$

$\boxed{z = 2(1-x)-y}$

$2x = 2-y-z$

$2x = x-x+2x-2+2x$

$2x = 4x-2$

$2 = 4x-2x$

$2 = 2x$

$\boxed{1 = x}$

Now

$$\int_V F\,dV = \int_0^1\int_0^{2(1-x)}\int_0^{2(1-x)-y} (2z\,i + y\,k)\,dz\,dy\,dx$$

$$= \int_0^1\int_0^{2(1-x)} \left[\frac{2z^2}{2}i + yz\,k\right]_0^{2(1-x)-y}$$

$$= \int_0^1\int_0^{2(1-x)} \left[[2(1-x)-y]^2 i + y[2(1-x)-y]k\right] - 0$$

$$= \int_0^1\int_0^{2(1-x)} \left\{\left[4(1-x)^2 + y^2 - 4(1-x)y\right]i + \left[2(1-x)y - y^2\right]k\right\}$$

$$= \int_0^1 \left\{\left[4(1-x)^2 y + \frac{y^3}{3} - 4(1-x)\frac{y^2}{2}\right]i + \left[2(1-x)\frac{y^2}{2} - \frac{y^3}{3}\right]k\right\}_0^{2(1-x)}$$

$$= \int_0^1 \left\{\left(4(1-x)^2\cdot 2(1-x) + \frac{[2(1-x)]^3}{3} - 4(1-x)\frac{[2(1-x)]^2}{2}\right)i + \left(\frac{2(1-x)[2(1-x)]^2}{2} - \frac{[2(1-x)]^3}{3}\right)k\right\}$$

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$$= \int_0^1 \left\{ \left(8(1-x)^3 + \frac{8(1-x)^3}{3} - \frac{18(1-x)^3}{2} \right) i + \left(\frac{8(1-x)^3}{2} - \frac{8(1-x^3)}{3} \right) \right.$$

$$= \int_0^1 \left\{ \left(\frac{48(1-x)^3 + 16(1-x)^3 - 48(1-x)^3}{6} \right) i + \left(\frac{24(1-x)^3 - 16(1-x)}{6} \right) \right.$$

$$= \int_0^1 \left[\frac{16(1-x)^3}{6} i + \frac{8}{6}(1-x)^3 k \right]$$

$$= \int_0^1 \left[\frac{8}{3}(1-x)^3 i + \frac{4}{3}(1-x)^3 k \right]$$

$$= \left[\frac{8}{3} \frac{(1-x)^4}{4} i + -\frac{4}{3} \frac{(1-x)^4}{4} k \right]_0^1$$

$$= \frac{1}{3} \left[-2(1-x)^4 i + -(1-x)^4 k \right]_0^1$$

$$= \frac{1}{3} \left[0 - \left\{ -2(1)^4 i \right\} - \left\{ (-(1)^4) k \right\} \right]$$

$$= \frac{1}{3} (2i + k) \quad \text{Ans.}$$

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Lecture 37:- Examples

Evaluate $\int_V F\,dV$ where $F = 2i + 2zj + yk$ & V is region bounded by planes $z=0$, $z=4$ & surface $x^2+y^2=9$. we'll use polar coordinates (cylindrical) (r,θ,z). bcz limits for x & y are not available

$x = r\cos\theta$, $y = r\sin\theta$, $z = z$ (comparing with (x,y,z))

$$\int_V F\,dV = \iiint_V (2i + 2zj + yk)\,dx\,dy\,dz$$

$$= \iiint_V (2i + 2zj + r\sin\theta\, k)\, r\,dr\,d\theta\,dz$$

$$= \int_0^{2\pi}\int_0^3\int_0^4 (2ri + 2rzj + r^2\sin\theta\, k)\, dz\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 \left[2rzi + \frac{2rz^2}{2}j + r^2\sin\theta\, z\, k\right]_0^4 dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 [8ri + 16rj + 4r^2\sin\theta\, k]\,dr\,d\theta$$

$$= 4\int_0^{2\pi}\int_0^3 [2ri + 4rj + r^2\sin\theta\, k]\,dr\,d\theta$$

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$$= 4\int_0^{2\pi} \left[\frac{2r^2}{2}i + \frac{4r^2}{2}j + \frac{r^3}{3}\sin\theta k\right]_0^3 d\theta$$

$$= 4\int_0^{2\pi} \left[9i + 18j + 9\sin\theta k\right] d\theta$$

$$= 4\times 9\int_0^{2\pi} \left[i + 2j + \sin\theta k\right] d\theta$$

$$= 36\left[\theta i + 2\theta j + (-\cos\theta)\right]_0^{2\pi}$$

$$= 36\left\{\left[2\pi i + 2(2\pi)j - \cos 2\pi k\right] - \left[0 + 0 - \cos 0 k\right]\right\}$$

$$= 36\left\{\left[2\pi i + 4\pi j - (1)k\right] - \left[-(1)k\right]\right\}$$

$$= 36\left[2\pi i + 4\pi j - k + k\right]$$

$$= 36\times 2(\pi i + 2\pi j) \Rightarrow 72(\pi i + 2\pi j) \text{ Ans.}$$

$$= 72\pi(i + 2j)$$

Scalar Field:- A Scalar field $F = xyz$ exists over curved Surface S defined by $x^2+y^2=4$ b/w The planes $z=0$ & $z=3$ in first octant.

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Evaluate $\int_S F\,ds$ over this surface.

As $F = xyz$, $S = x^2+y^2 = 4$, $z = 0$ to $z = 3$

$x^2+y^2-4=0$

$\mathbf{dS} = \hat{n}\,ds$ where $\hat{n} = \dfrac{\nabla S}{|\nabla S|}$

Now $\nabla S = \dfrac{\partial S}{\partial x}i + \dfrac{\partial S}{\partial y}j + \dfrac{\partial S}{\partial z}k$

$= 2xi + 2yj + 0k \Rightarrow 2xi + 2yj$

$|\nabla S| = \sqrt{(2x)^2+(2y)^2+(0)^2} \Rightarrow \sqrt{(4x^2+4y^2)} \Rightarrow \sqrt{4(x^2+y^2)}$

$= 2\sqrt{x^2+y^2} \Rightarrow 2\sqrt{4} \Rightarrow 2(2) \Rightarrow 4$

$\therefore \hat{n} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{2xi+2yj}{4} = \dfrac{2(xi+yj)}{4}$

$\hat{n} = \dfrac{xi+yj}{2}$

$dS = \hat{n}\,ds = \dfrac{xi+yj}{2}\,ds$

$\int_S F\,ds = \int_S F\,\hat{n}\,ds \Rightarrow \int_S xyz\cdot\left(\dfrac{xi+yj}{2}\right)ds$

$= \dfrac{1}{2}\int_S (x^2yzi + xy^2zj)\,ds$

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As $x = r\cos\theta$, $y = r\sin\theta$, $z = z$

$x = 2\cos\theta$, $y = 2\sin\theta$

Now

$$\int_S F\,ds = \frac{1}{2}\int_S (x^2yz\,i + xy^2z\,j)\,ds$$

$$= \frac{1}{2}\int_S \left[(2\cos\theta)^2(2\sin\theta)z\,i + (2\cos\theta)(2\sin\theta)^2 z\,j\right] ds$$

$$= \frac{1}{2}\int_S \left[8\sin\theta\cos^2\theta\, z\,i + 8\sin^2\theta\cos\theta\, z\,j\right] r\,d\theta\,dz$$

$$= \frac{1}{2}\cdot 8\int_0^{\pi/2}\int_0^3 \left[\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j\right] z\,dz\,d\theta$$

here $0 \le \theta \le \frac{\pi}{2}$ (bcz first octant).

$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j)\left[\frac{z^2}{2}\right]_0^3 d\theta$$

$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j)\, 9\,d\theta$$

$$= 36\left[\left(-\frac{\cos^3\theta}{3}\, i + \frac{\sin^3\theta}{3}\, j\right)\right]_0^{\pi/2}$$

$$= \frac{36}{3}\left[\left(-\cos^3\frac{\pi}{2}\, i + \sin^3\frac{\pi}{2}\, j\right) - \left(-\frac{\cos^3 0}{}\, i + \sin^3(0)\, j\right)\right]$$

$$= \frac{36}{3}\left[j - (-i)\right] \Rightarrow 12(i + j)$$ Ans.

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Vector Field:-

A vector field $F = yi + 2j + k$ exists over a surface S defined by $x^2 + y^2 + z^2 = 9$ bounded by $x = 0, y = 0$ $z = 0$ in first octant.

Evaluate $\int_S F \cdot dS$ over surface indicated.

$$dS = \hat{n}\, ds \quad , \quad \hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$S = x^2 + y^2 + z^2 = 9$$

$$x^2 + y^2 + z^2 - 9 = 0$$

$$\nabla S = \frac{\partial S}{\partial x} i + \frac{\partial S}{\partial y} j + \frac{\partial S}{\partial z} k \Rightarrow 2xi + 2yj + 2zk$$

$$|\nabla S| = \sqrt{(2x)^2 + (2y)^2 + (2z)^2} \Rightarrow \sqrt{4x^2 + 4y^2 + 4z^2}$$

$$= \sqrt{4(x^2 + y^2 + z^2)} \Rightarrow \sqrt{4(9)} \Rightarrow \sqrt{36} \Rightarrow 6$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|} = \frac{2xi + 2yj + 2zk}{6} \Rightarrow \frac{2(xi + yj + zk)}{6}$$

$$= \frac{xi + yj + zk}{3}$$

$$\hat{n}\, ds = \frac{1}{3}(xi + yj + zk)\, ds$$

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$$\int_S F \cdot ds = \int_S F \cdot \hat{n}\, ds = \int_S (yi + 2j + k) \cdot \frac{1}{3}(xi + yj + zk)\, ds$$

$$= \frac{1}{3}\int_S (xy + 2y + z)\, ds$$

converting into spherical coordinates

$x = r\sin\varphi\cos\theta$, $y = r\sin\varphi\sin\theta$, $z = r\cos\varphi$

$x = 3\sin\varphi\cos\theta$, $y = 3\sin\varphi\sin\theta$, $z = 3\cos\varphi$

$ds = r^2\sin\varphi\, d\varphi\, d\theta \Rightarrow 9\sin\varphi\, d\varphi d\theta$

Now

$$= \frac{1}{3}\int_S (xy + 2y + z)\, ds$$

$$= \frac{1}{3}\int_S \left[(3\sin\varphi\cos\theta)(3\sin\varphi\sin\theta) + 2(3\sin\varphi\sin\theta) + 3\cos\varphi\right] 9\sin\varphi\, d\varphi d\theta$$

$$= \frac{1}{3}\int_S \left[9\sin^2\varphi\sin\theta\cos\theta + 6\sin\varphi\sin\theta + 3\cos\varphi\right] 9\sin\varphi\, d\varphi d\theta$$

$$= 9.\frac{3}{3}\int_0^{\pi/2}\int_0^{\pi/2}\left[3\sin^3\varphi\sin\theta\cos\theta + 2\sin^2\varphi\sin\theta + \sin\varphi\cos\varphi\right] d\varphi\, d\theta$$

As we know, by Wallis formula.

$$\int_0^{\pi/2} \sin^3\varphi\, d\varphi = \frac{n-1}{n} = \frac{2}{3}$$

$$\int_0^{\pi/2} \sin^2\varphi\, d\varphi = \frac{n-1}{n}\cdot\frac{\pi}{2} = \frac{1}{2}\frac{\pi}{2}$$

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So we get

$$= 9\int_0^{\pi/2}\left(3\cdot\frac{2}{3}\sin\theta\cos\theta + 2\cdot\frac{1}{2}\cdot\frac{\pi}{2}\sin\theta + \int_0^{\pi/2}\frac{2\sin\varphi\cos\varphi}{2}d\varphi\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\int_0^{\pi/2}\sin 2\varphi\ d\varphi\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\left[\frac{\cos 2\varphi}{2}\right]_0^{\pi/2}\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{4}\left[-\cos\pi-(-\cos 0^\circ)\right]\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{4}\left(-(-1)-(-1)\right)\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\right)d\theta$$

$$= 9\left[\frac{-\cos 2\theta}{2} + \frac{\pi}{2}(-\cos\theta) + \frac{1}{2}\theta\right]_0^{\pi/2}$$

$$= 9\left[\frac{-\cos\pi}{2} - \frac{\pi}{2}\cos\frac{\pi}{2} + \frac{1}{2}\left(\frac{\pi}{2}\right)\right] - \left[\frac{-\cos 0^\circ}{2} - \frac{\pi}{2}\cos 0^\circ + \frac{1}{2}(0)\right]$$

$$= 9\left[\frac{1}{2} - 0 + \frac{\pi}{4}\right] - \left[-\frac{1}{2} - \frac{\pi}{2} + 0\right]$$

$$= 9\left[\frac{1}{2} + \frac{\pi}{4} + \frac{1}{2} + \frac{\pi}{2}\right] \Rightarrow 9\left[\frac{2+\pi+2+2\pi}{4}\right] \to 9\left(\frac{4+3\pi}{4}\right)$$

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Example:- Evaluate $\int_S F.ds$ where

$F = 2yj + zk$ & S is surface $x^2+y^2=4$ in first two octants bounded by the planes $z=0$, $z=5$ & $y=0$

$$S = x^2+y^2=4$$
$$x^2+y^2-4=0$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial S}{\partial x} i + \frac{\partial S}{\partial y} j + \frac{\partial S}{\partial z} k$$

$$= 2xi + 2yj$$

$$|\nabla S| = \sqrt{(2x)^2+(2y)^2} = \sqrt{4x^2+4y^2} = \sqrt{4(x^2+y^2)}$$

$$= \sqrt{4(4)} = \sqrt{16} = 4$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|} = \frac{2xi+2yj}{4} = \frac{2(xi+yj)}{4} \Rightarrow \frac{xi+yj}{2}$$

$$\int_S F.\hat{n}\,ds = \int_S (2yj+zk).\left(\frac{xi+yj}{2}\right)$$

$$= \frac{1}{2}\int_S (2y^2)\,ds$$

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$$= \frac{2}{2}\int_S y^2\,dS = \int_S y^2\,dS$$

Now $x = r\cos\theta$ & $y = r\sin\theta$, $z = z$, $dS = r\,d\theta\,dz$

$x = 2\cos\theta$ & $y = 2\sin\theta$ $= 2\,d\theta\,dz$

$$\int_S F\cdot dS = \int_S (2\sin\theta)^2\, 2\,d\theta\,dz$$

$$= \int_S 8\sin^2\theta\,d\theta\,dz \Rightarrow 8\int_S \frac{1-\cos 2\theta}{2}\,d\theta\,dz$$

$$= 4\int_0^5\int_0^{\pi}(1-\cos 2\theta)\,d\theta\,dz$$

$$= 4\int_0^5\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} dz$$

$$= 4\int_0^5\left[\left(\pi - \frac{\sin 2\pi}{2}\right) - \left(0 - \frac{\sin 2(0)}{2}\right)\right] dz$$

$$= 4\int_0^5 \pi\,dz \Rightarrow 4\pi\,[z]_0^5$$

$$= 4\pi[5-0) \Rightarrow 20\pi \quad \underline{\text{Ans.}}$$

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Conservative Vector Field:-

For a closed path in a conservative field

$$\oint_C F \cdot dr = 0$$

if field is conservative

$$\int_{C_1(AB)} F \cdot dr = \int_{C_2(AB)} F \cdot dr$$

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Lecture 38

Conservative Vector Fields

For a closed path in conservative field $\oint F \cdot dr = 0$

Example:- If $F = 2xyz\,i + x^2z\,j + x^2y\,k$, evaluate the line integral $\int F \cdot dr$ b/w $A(0,0,0)$ & $B(2,4,6)$

(a) along the curve C whose parametric equations are $x = u,\ y = u^2,\ z = 3u$.

(b) along the three straight lines

C_1 $(0,0,0)$ to $(2,0,0)$

C_2 $(2,0,0)$ to $(2,4,0)$

C_3 $(2,4,0)$ to $(2,4,6)$

(a) $F = 2xyz\,i + x^2z\,j + x^2y\,k$

$x = u,\ y = u^2,\ z = 3u$

$dx = du,\ dy = 2u\,du,\ dz = 3\,du$

$F \cdot dr = (2xyz\,i + x^2z\,j + x^2y\,k) \cdot (dx\,i + dy\,j + dz\,k)$

$= 2xyz\,dx + x^2z\,dy + x^2y\,dz$

put $x = u,\ y = u^2$ & $z = 3u$

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$$F.dr = 2(u)(u^2)(3u)(du) + (u)^2(3u^2)\,2u\,du + (u)^2(u^2)\,3du$$

$$= \left(6u^4 + 6u^4 + 3u^4\right) du$$

$$= 15u^4 du$$

$$\int_0^2 F.dr = \int_0^2 15u^4 du =$$

$$= 15\int_0^2 u^4 du$$

$$= 15\left[\frac{u^5}{5}\right]_0^2$$

$$= \frac{15}{5}\left[2^5 - 0^5\right]$$

$$= 3(32)$$

$$= 96 \text{ Ans.}$$

$x = u$, $y = u^2$, $z = 3u$

put $A(0,0,0)$ for lower limit

$0 = u$, $0 = u^2$, $0 = 3u$

$0 = u$, $0 = u$

put $B(2,4,6)$ for upper limit

$2 = u$, $4 = u^2$, $6 = 3u$

$2 = u$, $2 = u$

(b) C_1 : along x-axis $(0,0,0)$ to $(2,0,0)$

put $y=0$, $z=0$, $dy=0$, $dz=0$

$$\int F.dr = \int (2xyz\,dx + x^2z\,dy + x^2y\,dz)$$

$$= 0 + 0 + 0 \Rightarrow 0$$

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C_2: along y-axis $(2,0,0)$ to $(2,4,0)$

put $x=2$, $z=0$, $dx=0$, $dz=0$

$$\int_{C_2} F \cdot dr = \int_{C_2} (2xyz\,dx + x^2 z\,dy + x^2 y\,dz)$$

$$= 0+0+0 \Rightarrow 0$$

C_3: along z-axis $(2,4,0)$ to $(2,4,6)$

$x=2$, $y=4$, $dx=0$, $dy=0$

$$\int_{C_3} F \cdot dr = \int_{C_3} (2xyz\,dx + x^2 z\,dy + x^2 y\,dz)$$

$$= \int_{C_3} [2(2)(4)z(0) + (2)^2(z)(0) + (2)^2(4)dz)$$

$$= \int_0^6 16\,dz \Rightarrow 16[z]_0^6 \Rightarrow 16(6-0) \Rightarrow 96$$

Now

$$\int_{C_1+C_2+C_3} F \cdot dr = 0+0+96 \Rightarrow 96$$

here the value of line integral is independent of two paths. So F is a conservative field.

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$$\text{Curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix}$$

$$= i(x^2 - x^2) - j(2xy - 2xy) + k(2xz - 2xz)$$

$$= 0 - 0 + 0 \Rightarrow 0$$

So There are three test to check whether a vector field is conservative or not.

(a) $\oint F.dr = 0$

(b) curl $F = 0$

(c) $F = \text{grad } V$

Divergence Theorem:- (Gauss' Theorem).

For a closed surface S, with region V & vector field F

$$\int_V \text{div } F dV = \int_S F.dS$$

generally, volume integral (Tripple integral) can be expressed as a surface integral (double integral).

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Example:- Verify The divergence Theorem for The vector field $F = x^2 i + zj + yk$ taken over The region bounded by planes $z=0$, $z=2$, $x=0$, $x=1$, $y=0$, $y=3$.

$dV = dx\,dy\,dz$.

We have to show That

$$\int_V \text{div } F\, dV = \int_S F \cdot dS$$

So $\text{div } F = \nabla \cdot F = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (x^2 i + zj + yk)$

$= 2x + 0 + 0 \Rightarrow 2x$

$$\int_V \text{div } F\, dV = \int_V 2x\, dV = \int_0^1 \int_0^3 \int_0^2 2x\, dz\, dy\, dx$$

$$= \int_0^1 \int_0^3 [2xz]_0^2 dy\, dx \Rightarrow \int_0^1 \int_0^3 2x(2-0) \Rightarrow \int_0^1 \int_0^3 4x\, dy\, dx$$

$$= \int_0^1 4x [y]_0^3 dx \Rightarrow \int_0^1 4x(3-0)dx = \int_0^1 12x\, dx$$

$$= 12 \int_0^1 x\, dx \Rightarrow 12\left[\frac{x^2}{2}\right]_0^1 \Rightarrow 6[x^2]_0^1 \Rightarrow 6[1^2 - 0^2]$$

$= 6$ Ans.

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$$\int_S F \cdot ds = \int_S F \cdot \hat{n}\, ds$$

Surface consists of 6 separate plane faces $S_1, S_2, S_3, S_4, S_5, S_6$. & $F = x^2 i + z j + y k$

S_1 (for z) $z = 0$, $\hat{n} = -k$

$F = x^2 i + y k$

$$\int_{S_1} F \cdot \hat{n}\, ds_1 = \int_0^1 \int_0^3 (x^2 i + y k)(-k)\, dy\, dx$$

$$= \int_0^1 \int_0^3 (-y)\, dy\, dx \Rightarrow \int_0^1 -\left[\frac{y^2}{2}\right]_0^3 dx$$

$$= \int_0^1 -\left(\frac{3^2}{2} - \frac{0^2}{2}\right) dx = \int_0^1 -\frac{9}{2}\, dx$$

$$= -\frac{9}{2}[(x)]_0^1 \Rightarrow -\frac{9}{2}(1-0) \Rightarrow \boxed{-\frac{9}{2}}$$

S_2 (for z) $z = 2$, $\hat{n} = k$

$F = x^2 i + 2j + y k$

$$\int_{S_2} F \cdot \hat{n}\, ds_2 = \int_0^1 \int_0^3 (x^2 i + 2j + y k)(k)\, dy\, dx$$

$$= \int_0^1 \int_0^3 y\, dy\, dx \Rightarrow \int_0^1 \left[\frac{y^2}{2}\right]_0^3 dx$$

$$= \int_0^1 \frac{9}{2}\, dx \Rightarrow \frac{9}{2}[x]_0^1 \Rightarrow \frac{9}{2}(1-0) \Rightarrow \boxed{\frac{9}{2}}$$

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S_3 (for y) $\quad y = 0, \quad \hat{n} = -j$

$F = x^2 i + z j$

$$\int_{S_3} F \cdot \hat{n}\, ds_3 = \iint_{S_3} (x^2 i + z j) \cdot (-j)\, ds_3$$

$$= \int_0^1 \int_0^2 -z \, dz\, dx \Rightarrow \int_0^1 -\left[\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 -\left(\frac{2^2}{2} - \frac{0^2}{2}\right) dx = \int_0^1 -2 \, dx \Rightarrow -2[x]_0^1$$

$$= -2(1-0) \Rightarrow -2(1) \Rightarrow \boxed{-2}$$

S_4 (for y) $\quad y = 3, \quad \hat{n} = j$

$F = x^2 i + z k + 3k$

$$\int_{S_4} F \cdot \hat{n}\, ds_4 = \int_{S_4} (x^2 i + z k + 3k) \cdot (j)\, ds_4$$

$$= \int_0^1 \int_0^2 z \, dz\, dx \Rightarrow \int_0^1 \left[\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 \left(\frac{4}{2} - 0\right) dx = \int_0^1 2 \, dx$$

$$= 2[x]_0^1 \Rightarrow 2(1-0) \Rightarrow \boxed{2}$$

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S_5 (for x) $x = 0$, $\hat{n} = -i$

$F = x^2 i + z j + y k = z j + y k$

$$\int_{S_5} F \cdot \hat{n}\, ds_5 = \int_{S_5} (z j + y k)(-i)\, ds_5$$

$$= \int_0^2 \int_0^3 0 \, dy\, dz \Rightarrow \boxed{0}$$

S_6 (for x) $x = 1$, $\hat{n} = i$

$F = i + z j + y k$

$$\int_{S_6} F \cdot \hat{n}\, ds_6 = \int_{S_6} (i + z j + y k) \cdot (i)\, ds_6$$

$$= \int_0^2 \int_0^3 1 \, dy\, dz \Rightarrow \int_0^2 [y]_0^3 \, dz$$

$$= \int_0^2 (3 - 0)\, dz \Rightarrow 3 \int_0^2 1 \, dz$$

$$= 3 [z]_0^2 \Rightarrow 3(2 - 0) \Rightarrow 3(2) = \boxed{6}$$

Now for whole surface

$$\int_S F \cdot dS = \frac{-9}{2} + \frac{9}{2} - 2 + 2 + 0 + 6 = 6$$

So proved that

$$\int_V \text{div } F \, dV = \int_S F \cdot dS$$

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Stokes Theorem:- If F is a vector field existing over an open surface S & around its boundary closed curve C, Then $\int_S \text{curl } F \cdot dS = \oint_C F \cdot dr$

We can express a surface integral in terms of a line integral.

Example:- A hemisphere S is defined by $x^2+y^2+z^2=+4$ $(z \geq 0)$. A vector field $F = 2yi - xj + xzk$ exist over the surface & around its boundary C. verify Stoke's Theorem.

$S = x^2+y^2+z^2 = 4$, $F = 2yi - xj + xzk$

$x^2+y^2+z^2-4=0$

if we ignore z ↑ then put $z=0$ $x^2+y^2=4$ (circle)

$$\oint_C F \cdot dr = \int_C (2yi - xj + xzk) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_C [2y\,dx - x\,dy + xz\,dz]$$

converting to polar coordinates

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$x = r\cos\theta$, $y = r\sin\theta$, $z = 0$

$x = 2\cos\theta$, $y = 2\sin\theta$, $dz = 0$

$dx = -2\sin\theta\, d\theta$, $dy = 2\cos\theta\, d\theta$

$$\oint_C F \cdot dr = \int_0^{2\pi} \left[2(2\sin\theta)(-2\sin\theta\, d\theta) - (2\cos\theta)(2\cos\theta\, d\theta) \right]$$

$$= \int_0^{2\pi} \left(-8\sin^2\theta - 4\cos^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(2\sin^2\theta + \cos^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(2\sin^2\theta + 1 - \sin^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(1 + \sin^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(1 + \frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(\frac{2 + 1 - \cos 2\theta}{2} \right) d\theta$$

$$= -2 \int_0^{\pi/2} \left(3 - \cos 2\theta \right) d\theta$$

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$$= -2\left[3\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi} \Rightarrow$$

$$= -2\left[\left\{3(2\pi) - \frac{\sin 2(2\pi)}{2}\right\} - \left\{3(0) - \frac{\sin 2(0)}{2}\right\}\right]$$

$$= -2\left[6\pi - 0\right] \Rightarrow -2(6\pi) \Rightarrow -12\pi$$

Now $\int \text{curl } F \cdot dS = \int \text{curl } F \cdot \hat{n}\, dS.$

$$F = 2yi - xj + xzk$$

$$\text{curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & -x & +xz \end{vmatrix}$$

$$= i(0-0) + (-j)(z-0) + k(-1-2)$$

$$= -zj - 3k$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial}{\partial x} Si + \frac{\partial}{\partial y} Sj + \frac{\partial}{\partial z} Sk$$

$$= 2xi + 2yj + 2zk$$

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$$|\nabla S| = \sqrt{(2x)^2+(2y)^2+(2z)^2}$$

$$= \sqrt{4x^2+4y^2+4z^2}$$

$$= \sqrt{4(x^2+y^2+z^2)}$$

$$= 2\sqrt{4(4)} = \sqrt{16} = 4$$

So $\hat{n} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{2xi + 2yj + 2zk}{4}$

$$= \frac{2(xi+yj+zk)}{4}$$

$$= \frac{xi+yj+zk}{2}$$

Now

$$\int_S \text{curl } F \cdot \hat{n}\, ds = \int_S (-3j-3k)\cdot\left(\frac{xi+yj+zk}{2}\right) ds$$

$$= \frac{1}{2}\int_S (-yz-3z)\, ds$$

$x = r\sin\phi\cos\theta$, $y = r\sin\phi\sin\theta$, $z = r\cos\phi$

$x = 2\sin\phi\cos\theta$, $y = 2\sin\phi\sin\theta$, $z = 2\cos\phi$

& $dS = r^2\sin\phi\, d\phi\, d\theta$

$= 4\sin\phi\, d\phi\, d\theta$

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$$\int_S \text{curl } F \cdot \hat{n}\, dS = \frac{1}{2}\int_S \left[-2\sin\varphi\sin\theta\, 2\cos\varphi - 3.\ 2\cos\varphi\right] ds$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\pi/2}\left[-4\sin\theta\sin\varphi\cos\varphi - 6\cos\varphi\right] 4\sin\varphi\, d\varphi\, d\theta$$

$$= 4\left(\frac{-2}{2}\right)\int_0^{2\pi}\int_0^{\pi/2}\left[2\sin^2\varphi\sin\theta\cos\varphi + 3\sin\varphi\cos\varphi\right] d\varphi\, d\theta$$

$$= -4\int_0^{2\pi}\left[\frac{2\sin\theta\,\sin^3\varphi}{3} + 3\,\frac{\sin^2\varphi}{2}\right]_0^{\frac{\pi}{2}} d\theta$$

$$= -4\int_0^{2\pi}\left[\frac{2\sin\theta\,\sin^3\frac{\pi}{2}}{3} + \frac{3\sin^2\frac{\pi}{2}}{2}\right] - 0 \quad d\theta$$

$$= -4\int_0^{2\pi}\left(\frac{2}{3}\sin\theta + \frac{3}{2}\right) d\theta$$

$$= -4\left[-\frac{2}{3}\cos\theta + \frac{3}{2}\theta\right]_0^{2\pi}$$

$$= -4\left[-2\,\frac{\cos(2\pi)}{3} + \frac{3}{2}(2\pi)\right] - \left[\frac{-2\cos 0}{3} + \frac{3}{2}(0)\right]$$

$$= -4\left[-\frac{2}{3} + 3\pi + \frac{2}{3}\right] \Rightarrow -4(3\pi)$$

$$= -12\pi \quad \text{So} \int_S \text{curl } F \cdot dS = \oint_C F \cdot dr$$

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Lecture 39. Periodic Functions

A function $f(x)$ is said to be periodic if its function values repeat at regular intervals of the independent variable. The regular interval b/w repetitions is the period of oscillations.

$$f(x+p) = f(x)$$

$y = A \sin nx$

The period of sine is 360° or 2π.
The period of cos is also 360° or 2π.

In $y = 5 \sin 2x$ (amplitude is 5 & period is 180°)

$$y = 5 \sin(2x + 360^\circ)$$

$$= 5 \sin 2\left(\frac{2x}{2} + \frac{360^\circ}{2}\right)$$

$$= 5 \sin 2(x + 180^\circ)$$

Example:-

(i) $y = 3 \sin 5x$ amplitude = 3

$$y = 3 \sin(5x + 360)$$

$$= 3 \sin 5\left(\frac{5x}{5} + \frac{360}{5}\right)$$

$$= 3 \sin 5(x + 72^\circ)$$ So period $= 72^\circ$

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(ii) $y = 2\cos 3x$

amplitude $= 2$

$y = 2\cos(3x + 2\pi)$

$= 2\cos 3\left(\frac{3x}{3} + \frac{360}{3}\right)$

$= 2\cos 3(x + 120^\circ)$

So period $= 120^\circ$

(iii) $y = \sin\frac{x}{2}$

Amplitude $= 1$

$y = \sin\left(\frac{x}{2} + 360\right)$

$y = \sin\frac{1}{2}\left(\frac{2x}{2} + 360 \times 2\right)$

$= \sin\frac{1}{2}(x + 720^\circ)$

So period is 720°

(iv) $y = 4\sin 2x$ amplitude $= 4$

$y = 4\sin(2x + 360)$

$= 4\sin 2\left(\frac{2x}{2} + \frac{360}{2}\right)$

$= 4\sin 2(x + 180^\circ)$ so period $= 180^\circ$

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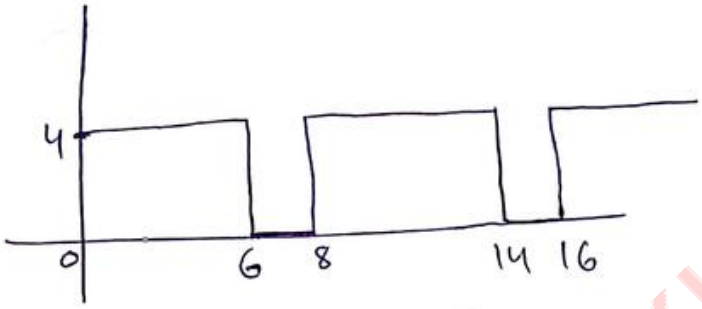
For $y = A \sin nx$

Amplitude $= A$

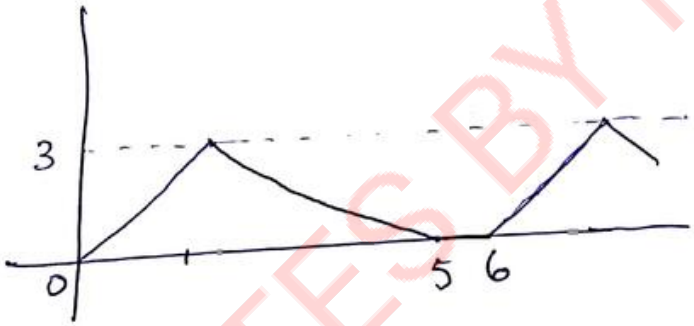
Period $= \frac{360}{n} = \frac{2\pi}{n}$

Same goes for $y = A \cos nx$

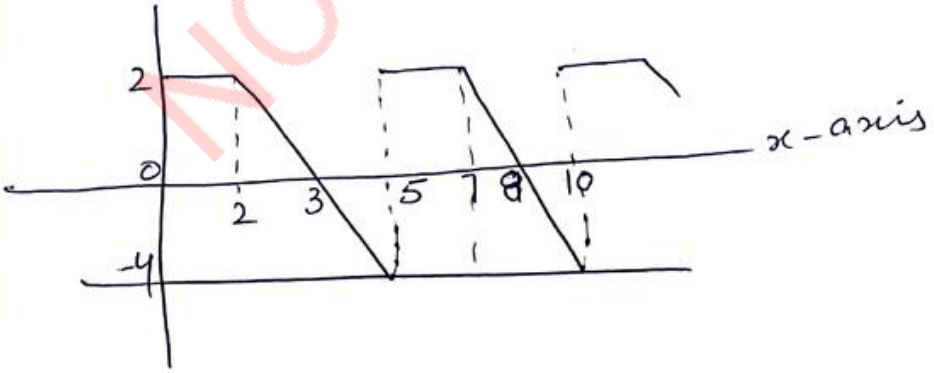
Examples:-



period = 8



period = 6



Period = 5

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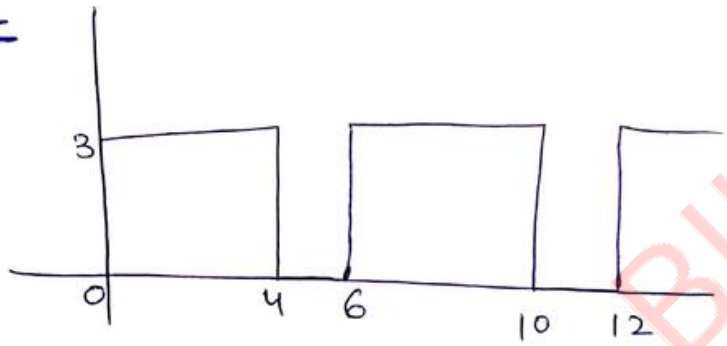
Analytical description of a periodic func

A periodic function can be defined analytically in many cases.

Example:-



If $y=3$, when $0 \leq x \leq 4$

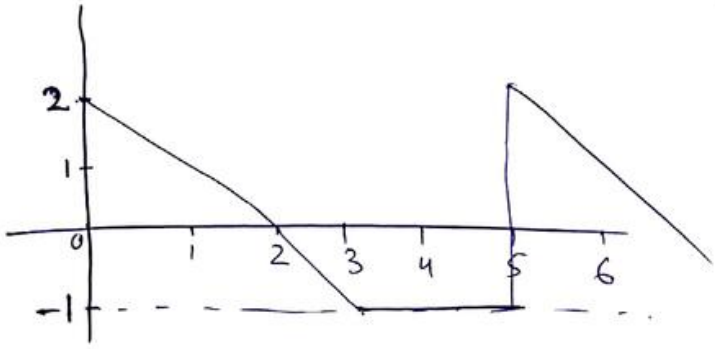
$f(x)=3$

& $y=0$ when $4 \leq x \leq 6$

$f(x)=0$

$f(x) = f(x+6)$ (function is periodic with period 6)

Example:-



when $0 < x < 3$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-0}{3-2} \Rightarrow -1$

$y - y_0 = m(x - x_0)$

$y - 2 = -1(x - 0)$

$y - 2 = -x$

$y = 2 - x$

$f(x) = 2 - x$ when $0 < x < 3$

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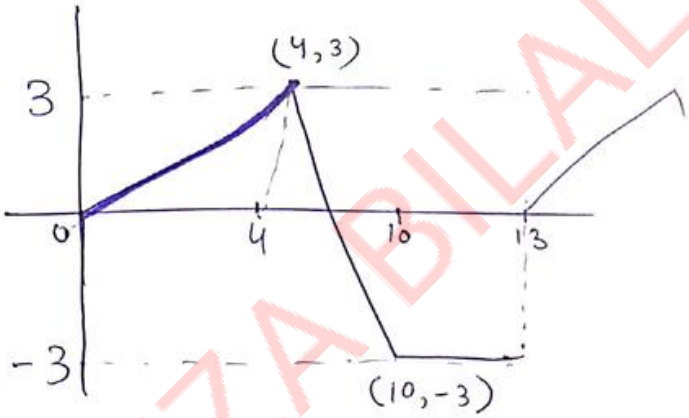
$y = -1$ when $3 < x < 5$

$f(x) = -1$

& $f(x) = f(x+5)$

function is periodic with period 5.

Example:-



when $0 < x < 4$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-0}{4-0} \Rightarrow \frac{3}{4}$$

$$y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{3}{4}(x - 0)$$

$$y = \frac{3}{4}x \quad \text{or} \quad f(x) = \frac{3}{4}x \quad \text{when} \quad 0 < x < 4$$

when $4 < x < 10$

$$m = \frac{-3-3}{10-4} \Rightarrow \frac{-6}{6} \Rightarrow -1$$

$$y - y_0 = m(x - x_0)$$

$$y - 3 = -1(x - 4)$$

$$y - 3 = -x + 4$$

$$y = -x + 4 + 3$$

$$y = 7 - x$$

$f(x) = 7 - x$ where $4 < x < 10$

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when $10 < x < 13$

$f(x) = -3$

& $f(x) = f(x+13)$

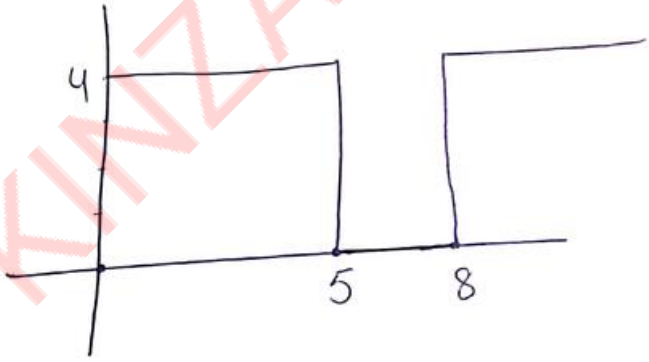
function is periodic with period 13.

Example:- Sketch the graph

1. $f(x) = 4$ $\quad 0 < x < 5$

$f(x) = 0$ $\quad 5 < x < 8$

$f(x) = f(x+8)$



2. $f(x) = 3x - x^2$ $\quad 0 < x < 3$

$f(x) = f(x+3)$

$f(x) = -x^2 + 3x + 0$ (parabola)

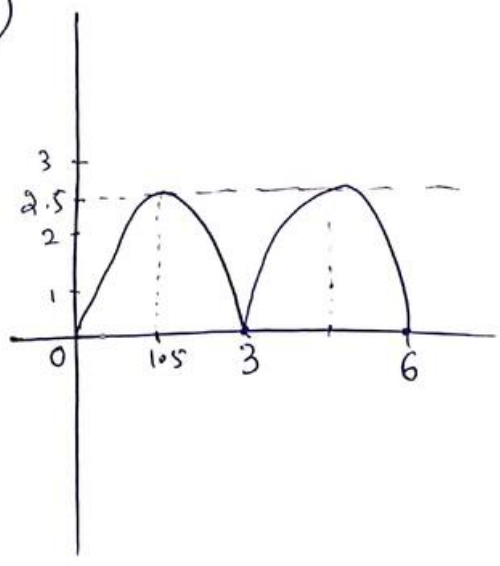
$x = \text{Vertex} = -\frac{b}{2a} = \frac{-3}{2(-1)} \Rightarrow \frac{3}{2} \Rightarrow 1.5$

put $x = 1.5$ in

$f(x) = 3(1.5) - (1.5)^2$

$f(x) = 2.5$

opening downward bcz $a < 0$ (-ve)



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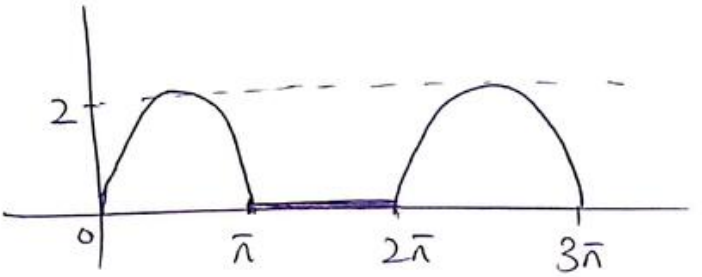
Example:-

$f(x) = 2\sin 2x \qquad 0 < x < \pi$

$f(x) = 0 \qquad \pi < x < 2\pi$

$f(x) = f(x + 2\pi)$

amplitude = 2



Example:-

$f(x) = \frac{x^2}{4} \qquad 0 < x < 4$ (portion of parabola)
a is +ve so opens upward

$f(x) = 4 \qquad 4 < x < 6$

$f(x) = 0 \qquad 6 < x < 8$

$f(x) = f(x + 8)$

Useful integrals

Here are some integrals that appear frequently in our work on Fourier Series.

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$$\int_{-\pi}^{\pi} \sin nx\,dx = \left[\frac{-\cos nx}{n}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{n}\left[-\cos n\pi - (-\cos n(-\pi))\right]$$

$$= \frac{1}{n}\left[-\cos n\pi + \cos n\pi\right] = 0$$

$$\int_{-\pi}^{\pi} \cos nx\,dx = \left[\frac{\sin nx}{n}\right]_{-\pi}^{\pi} \Rightarrow \frac{1}{n}\left[\sin n\pi - (-\sin n\pi)\right]$$

$$= \frac{1}{n}\left[\sin n\pi + \sin n\pi\right] = \frac{1}{n}\left[0+0\right] \Rightarrow 0$$

$$\int_{-\pi}^{\pi} \sin^2 nx\,dx = \int_{-\pi}^{\pi} \frac{1-\cos 2nx}{2}dx \Rightarrow \frac{1}{2}\int_{-\pi}^{\pi}\left[1-\cos 2nx\right]dx$$

$$= \frac{1}{2}\left[x - \frac{\sin 2nx}{2n}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2}\left[\left\{\pi - \frac{\sin 2n\pi}{2n}\right\} - \left\{-\pi + \frac{\sin 2n\pi}{2n}\right\}\right]$$

$$= \frac{1}{2}\left[\pi - 0 + \pi + 0\right] = \frac{1}{2}\left[2\pi\right] = \pi$$

$$\int_{-\pi}^{\pi} \cos^2 nx\,dx = \int_{-\pi}^{\pi} \frac{1+\cos 2nx}{2}\,dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi}(1+\cos 2nx)\,dx = \frac{1}{2}\left[x + \frac{\sin 2nx}{2n}\right]_{-\pi}^{\pi}$$

$$= \pi \quad \text{Ans.}$$

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$$\frac{1}{2}\int_{-\pi}^{\pi} 2\sin nx\cos mx\,dx = \frac{1}{2}\int_{-\pi}^{\pi}\left[\sin(n+m)x + \sin(n-m)x\right]dx$$

Product to sum

$$= \frac{1}{2}\{0+0\} = 0$$

$$\int_{-\pi}^{\pi}\cos nx\cos mx\,dx = \frac{1}{2}\int_{-\pi}^{\pi}\left[\cos(n+m)x + \cos(n-m)x\right]dx$$

$$= \frac{1}{2}\{0+0\} = 0$$

Note:- If $n=m$

$\int_{-\pi}^{\pi}\cos nx\cos mx\,dx$ becomes $\int_{-\pi}^{\pi}\cos^2 nx\,dx = \pi$

$$\int_{-\pi}^{\pi}\sin nx\sin mx\,dx = -\frac{1}{2}\int_{-\pi}^{\pi}\left(\cos(n+m)x - \cos(n-m)x\right)dx$$

$$= -\frac{1}{2}\{0-0\} = 0$$

Note:-
If $n=m$ it becomes $\int_{-\pi}^{\pi}\sin^2 mx = \pi$

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Summary of Integrals :-

$$\int_{-\pi}^{\pi} \sin nx\, dx = 0 \quad , \quad \int_{-\pi}^{\pi} \cos nx\, dx = 0$$

$$\int_{-\pi}^{\pi} \cos^2 nx\, dx = \pi \quad , \quad \int_{-\pi}^{\pi} \sin^2 nx\, dx = \pi$$

$$\int_{-\pi}^{\pi} \sin mx \cos mx\, dx = 0 \quad , \quad \int_{-\pi}^{\pi} \cos nx \cos mx\, dx = 0 \ (n \neq m)$$

$$\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = 0 \ (n \neq m) \quad , \quad \int_{-\pi}^{\pi} \cos nx \cos mx\, dx = \pi \ (n = m)$$

$$\int_{-\pi}^{\pi} \sin nx \sin mx\, dx = \pi \ (n = m)$$

limits can be changed, but the condition is to keep the interval of 2π. like $-\pi$ to π, $\frac{-\pi}{2}$ to $\frac{3\pi}{2}$, 0 to 2π etc.

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Lecture 40

Fourier Series

$$f(x) = A_0 + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$

can be written in expanded form

$$f(x) = A_0 + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x) + \ldots + (a_n \cos nx + b_n \sin nx)$$

Fourier Coefficients

To find A_0, integrate $f(x)$ from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi}^{\pi} A_0\,dx + \sum_{n=1}^{\infty} \left\{ \int_{-\pi}^{\pi} a_n \cos nx\,dx + \int_{-\pi}^{\pi} b_n \sin nx \right\}$$

$$= [A_0 x]_{-\pi}^{\pi} + \sum_{n=1}^{\infty} \{0 + 0\}$$

$$= A_0 (\pi - (-\pi)) = A_0 (2\pi) = 2\pi A_0$$

So

$$\int_{-\pi}^{\pi} f(x)\,dx = 2A_0\pi$$

$$\boxed{\frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\,dx = A_0}$$

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If $\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = a_0$

Then $\boxed{A_0 = \frac{1}{2}a_0}$

To find a_n we multiply $f(x)$ by $\cos mx$ & integrate from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)\cos mx\,dx = \int_{-\pi}^{\pi} A_0 \cos mx\,dx + \sum_{n=1}^{\infty}\left\{\int_{-\pi}^{\pi} a_n \cos nx \cos mx\,dx + \int_{-\pi}^{\pi} b_n \sin nx \cos mx\,dx\right\}$$

$$= A_0\{0\} + \sum_{n=1}^{\infty}\left(a_n(0) + b_n(0)\right)$$

$$= 0 \quad \text{for } n \neq m$$

$$\int_{-\pi}^{\pi} f(x)\cos mx\,dx = 0 + a_n\pi + 0 \Rightarrow a_n\pi \quad \text{for } n = m$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

To find b_n we multiply $f(x)$ by $\sin mx$ & integrate from $-\pi$ to π.

$$\int_{-\pi}^{\pi} f(x)\sin mx\,dx = \int_{-\pi}^{\pi} A_0 \sin mx + \sum_{n=1}^{\infty}\left\{\int_{-\pi}^{\pi} a_n \cos nx \sin mx\,dx + \int_{-\pi}^{\pi} b_n \sin nx \sin mx\,dx\right\}$$

$$= 0 \text{ if } n \neq m$$

$$= 0 + 0 + b_n\pi \quad \text{if} \quad n = m$$

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$$\int_{-\pi}^{\pi} f(x) \sin nx\, dx = b_n \pi$$

$$\frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx = b_n$$

Result for Fourier Series:-

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\{a_n \cos nx + b_n \sin nx\}$$

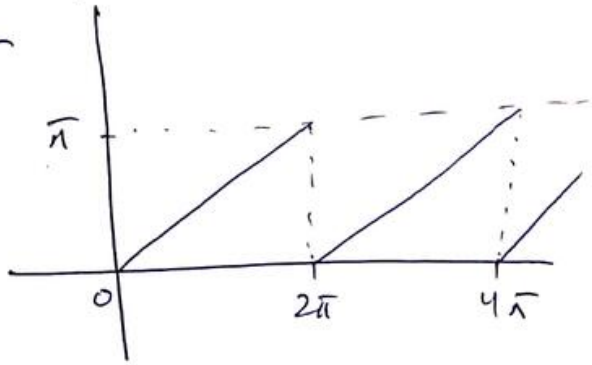
(a) $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = 2$ (mean value of $f(x)$ over a period)

(b) $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = 2$ (mean value of $f(x)\cos nx$ over a period)

(c) $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = 2$ (mean value of $f(x)\sin nx$ over a period)

In each case $n = 1, 2, 3, 4, \ldots$

Example:- Determine the Fourier Series to represent the periodic function shown.



here limits are $0 < x < 2\pi$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\pi - 0}{2\pi - 0}$$

$$m = \frac{\pi}{2\pi} = \frac{1}{2}$$

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$$y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{1}{2}(x - 0)$$

$$y = \frac{x}{2}$$

$$f(x) = \frac{x}{2}, \quad 0 < x < 2\pi$$

$$f(x) = f(x + 2\pi) \quad \text{period} = 2\pi$$

Now we'll find Fourier Series

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\{a_n \cos nx + b_n \sin nx\}$$

First of all

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\, dx$$

$$= \frac{1}{\pi}\int_0^{2\pi}\left(\frac{x}{2}\right)\cos nx\, dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi} x\cos nx\, dx$$

Integrating by parts

$$= \frac{1}{2\pi}\left\{ x\int_0^{2\pi}\cos nx\, dx + \int_0^{2\pi}\left[\int_0^{2\pi}\cos nx\, dx \cdot \frac{d}{dx}(x)\right]dx\right\}$$

$$= \frac{1}{2\pi}\left\{ x\left[\frac{\sin nx}{n}\right]_0^{2\pi} + \int_0^{2\pi}\frac{\sin nx}{n}\, dx\right\}$$

$$= \frac{1}{2\pi}\{x(0) + 0\}\Big|_0^{2\pi} \Rightarrow \boxed{0 = a_n}$$

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$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{2\pi}\left(\frac{x}{2}\right)dx$$

$$= \frac{1}{\pi}\left[\frac{x^2}{4}\right]_0^{2\pi} = \frac{1}{4\pi}\left[x^2\right]_0^{2\pi}$$

$$= \frac{1}{4\pi}\left[(2\pi)^2 - 0^2\right] \Rightarrow \frac{1}{4\pi}(4\pi^2)$$

$$\boxed{a_0 = \pi}$$

Now

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi}\left(\frac{x}{2}\right)\sin nx\,dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi} x\sin nx\,dx$$

$$= \frac{1}{2\pi}\left\{x\int_0^{2\pi}\sin nx\,dx - \int_0^{2\pi}\left(\int_0^{2\pi}\sin nx\cdot\frac{dx}{dx}\right)dx\right\}$$

$$= \frac{1}{2\pi}\left\{-x\left[\frac{\cos nx}{n}\right]_0^{2\pi} + \int_0^{2\pi}\left[\frac{\cos nx}{n}\right]_0^{2\pi}dx\right\}$$

$$= \frac{1}{2\pi}\left\{-\frac{2\pi\cos n2\pi}{n} + 0\right\}$$

$$= \frac{1}{2\pi}\left(-\frac{2\pi}{n}\right) = -\frac{1}{n}$$

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$$a_0 = \pi, \quad a_n = 0, \quad b_n = -\frac{1}{n}$$

Now put values in Fourier series

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \{ a_n \cos nx + b_n \sin nx \}$$

$$= \frac{1}{2}\pi + \sum_{n=1}^{\infty} \left\{ -\frac{1}{n} \sin nx \right\}$$

$$= \frac{1}{2}\pi + \left\{ \frac{-1}{1} \sin x - \frac{1}{2} \sin 2x - \frac{1}{3} \sin 3x - \cdots \right\}$$

$$= \frac{1}{2}\pi + \left(-\sin x - \frac{1}{2} \sin 2x - \frac{1}{3} \sin 3x \cdots \right)$$

$$= \frac{\pi}{2} - \left(\sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \cdots \right)$$

Dirichlet Conditions:-

The $f(x)$ must be defined & single-valued

$f(x)$ must be continuous or have a finite number of finite discontinuities within a periodic interval.

$f(x)$ & $f'(x)$ must be piecewise continuous in the periodic interval.

If these dirichlet conditions are satisfied fourier series converges to $f(x_1)$ if $x = x_1$ is a point of continuity.

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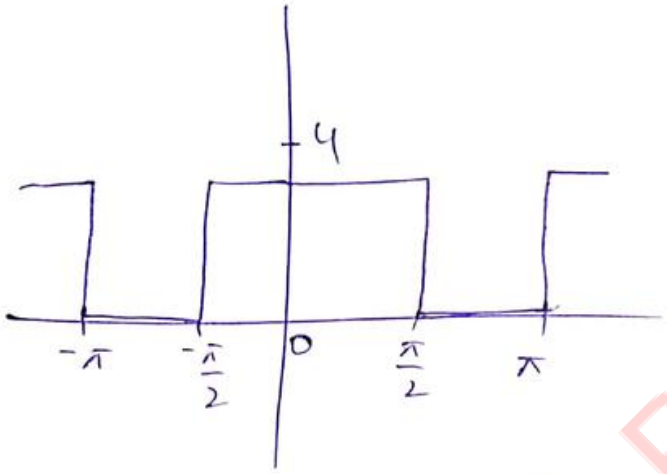
Example:- Find fourier series for function shown.



$f(x) = 0 \qquad -\pi < x < -\frac{\pi}{2}$

$f(x) = 4 \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$

$f(x) = 0 \qquad \frac{\pi}{2} < x < \pi$

$f(x) = f(x + 2\pi)$

As

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\{a_n \cos nx + b_n \sin nx\}$$

Now

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} 0\,dx + \int_{-\pi/2}^{\pi/2} 4\,dx + \int_{\pi/2}^{\pi} 0\,dx\right\}$$

$$= \frac{1}{\pi}\,4\,[x]_{-\pi/2}^{\pi/2} \Rightarrow \frac{1}{\pi}\,4\left[\frac{\pi}{2} + \frac{\pi}{2}\right]$$

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$$= \frac{1}{\pi} 4\left(\frac{2\pi}{2}\right) = 4$$

(b) $$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} 0\cos nx\,dx + \int_{-\pi/2}^{\pi/2} 4\cos nx\,dx + \int_{\pi/2}^{\pi} 0\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\,4\left[\left(\frac{\sin nx}{n}\right)\right]_{\frac{-\pi}{2}}^{\pi/2} \Rightarrow \frac{4}{n\pi}\left[\sin n\frac{\pi}{2} + \sin n\frac{\pi}{2}\right]$$

$$= \frac{4}{n\pi}\,2\sin n\frac{\pi}{2} \Rightarrow \frac{8}{n\pi}\sin\frac{n\pi}{2}$$

considering different integer values of n

if n is even $a_n = 0$

if $n = 1, 5, 9, \ldots$ $a_n = \frac{8}{n\pi}$

if $n = 3, 7, 11, \ldots$ $a_n = \frac{-8}{n\pi}$

(c) To find b_n

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

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$$= \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} 0 \sin nx\,dx + \int_{-\pi/2}^{\pi/2} 4 \sin nx\,dx + \int_{\pi/2}^{\pi} 0 \sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{4\int_{-\pi/2}^{\pi/2} \sin nx\,dx\right\}$$

$$= \frac{4}{\pi}(0) \Rightarrow 0$$

Fourier series is

$$f(x) = 2 + \frac{8}{\pi}\left\{\cos x - \frac{1}{3}\cos 3x + \frac{1}{5}\cos 5x - \cdots\right\}$$

Effect of Harmonics :-

If we include just one cosine term, we can see effect to including subsequent harmonics. As number of terms is increased, the graph gradually approaches to original wave form. ripples increases in number & decreases in amplitude. A perfect wave is impossible to find, so usually first few terms give accuracy of acceptable level.

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Lecture 41 Examples

Example:- Find fourier series for function defined by

$$f(x) = -x \qquad -\pi < x < 0$$

$$f(x) = 0 \qquad 0 < x < \pi$$

$$f(x) = f(x + 2\pi)$$

first we'll find a_0, a_n & b_n.

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \Rightarrow \frac{1}{\pi}\left\{\int_{-\pi}^{0} -x\,dx + \int_{0}^{\pi} 0\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[-\frac{x^2}{2}\right]_{-\pi}^{0}\right\} + 0 \Rightarrow \frac{1}{\pi}\left[-\left(\frac{0^2}{2} - \frac{(-\pi)^2}{2}\right)\right]$$

$$= \frac{1}{\pi}\left[-\left(0 - \frac{\pi^2}{2}\right)\right] \Rightarrow \frac{1}{\pi}\left(\frac{\pi^2}{2}\right) \Rightarrow \frac{\pi}{2}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left\{\int_{-\pi}^{0} (-x)\cos nx\,dx + \int_{0}^{\pi} 0\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} -x\cos nx\,dx + 0\right\}$$

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$$-\frac{1}{\pi}\int_{-\pi}^{0} x\cos nx\,dx$$

$$= -\frac{1}{\pi}\left\{\left[x\frac{\sin nx}{n}\right]_{-\pi}^{0} - \frac{1}{n}\int_{-\pi}^{0}(\sin nx)\,dx\right\}$$

$$= -\frac{1}{\pi}\left\{0-0 - \frac{1}{n}\left[\frac{-\cos nx}{n}\right]_{-\pi}^{0}\right\}$$

$$= -\frac{1}{\pi}\left\{\frac{1}{n^2}\left[\cos 0 - \cos n(-\pi)\right]\right\}$$

$$= -\frac{1}{\pi}\left\{\frac{1}{n^2}\left[1-\cos n\pi\right]\right\}$$

$$= \frac{-1}{\pi n^2}\left[1-\cos n\pi\right]$$

now if n is even, $\cos n\pi = 1$

$$a_n = \frac{-1}{\pi n^2}(1-1) = 0$$

if n is odd, $\cos n\pi = -1$

$$a_n = \frac{-1}{\pi n^2}\left[1-(-1)\right] \Rightarrow \frac{-2}{\pi n^2}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin x\,dx$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0}(-x)\sin x\,dx + \int_{0}^{\pi} 0\,dx\right\}$$

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$$= \frac{1}{\pi}\left\{ \int_{-\pi}^{0} -x \sin nx \, dx \right\}$$

$$= \frac{1}{\pi}\left\{ \left[-x \frac{\cos nx}{n}\right]_{-\pi}^{0} + \frac{1}{n}\int_{-\pi}^{0} \cos nx \, dx \right\}$$

$$= \frac{1}{\pi}\left\{ \left(\frac{\pi \cos n\pi}{n}\right) + \frac{1}{n}\left[\frac{\sin nx}{n}\right]_{-\pi}^{0} \right\}$$

$$= \frac{1}{\pi}\left(\frac{\pi \cos n\pi}{n} \right) \Rightarrow \frac{\cos n\pi}{n}$$

if n is even

$$b_n = \frac{1}{n}$$

if n is odd

$$b_n = -\frac{1}{n}$$

So we have $a_0 = \frac{\pi}{2}$

$a_n = 0$ (n even) , $a_n = -\frac{2}{\pi n^2}$ (n odd)

$b_n = \frac{1}{n}$ (n even) , $b_n = -\frac{1}{n}$ (n odd)

As $f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \{ a_n \cos nx + b_n \sin nx \}$

$$= \frac{1}{2} \cdot \frac{\pi}{2} + \left\{ \left(1 \cdot \cos x + 0 + \frac{-2}{\pi 3^2} \cos 3x + 0 + \left(\frac{-2}{\pi (5)^2} \cos 5x\right) + \cdots\right) + \left(\frac{-1}{\pi} \sin x + \frac{1}{2} \sin 2x - \frac{1}{3} \sin 3x + \frac{1}{4} \sin 4x - \cdots\right)\right\}$$

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$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left(\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \cdots\right) +$$

$$\left(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \frac{1}{4}\sin 4x + \cdots\right)$$

Odd & Even functions :-

Even functions:

A function is said to be even if $f(x) = f(-x)$. Also even function is symmetrical about y-axis.

$y = f(x) = x^2$

put $x = -x$

$$f(-x) = (-x)^2$$

$$f(-x) = x^2 = f(x)$$

So it is even function.

put $x = 2$ & $x = -2$

$$f(x) = f(2) = x^2 = (2)^2 = 4$$

$$f(-2) = (-2)^2 = 4$$

$y = \cos x$

$$f(x) = \cos x$$

$$f(-x) = \cos(-x) = \cos(x)$$

$$f(-x) = f(x)$$

So it is an even function.

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Odd function:-

A function $f(x)$ is said to be odd if $f(-x) = -f(x)$. its graph is symmetrical about origin.

$$y = f(x) = x^3$$

put $x = -x$

$$f(-x) = (-x)^3 = -x^3$$

$$f(-x) = -f(x)$$

So it is odd function

put $x = 2$ & -2

$$f(2) = (2)^3 = 8$$

$$f(-2) = (-2)^3 = -8 = -f(2)$$

So $\boxed{f(-2) = -f(2)}$

$$y = f(x) = \sin x$$

put $x = -x$

$$f(-x) = \sin(-x) = -\sin x$$

$$f(-x) = -f(x)$$

So it is odd function.

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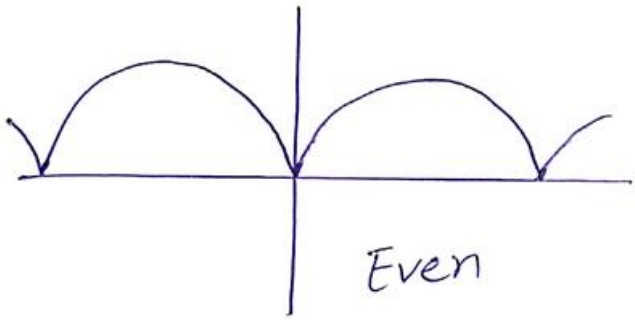
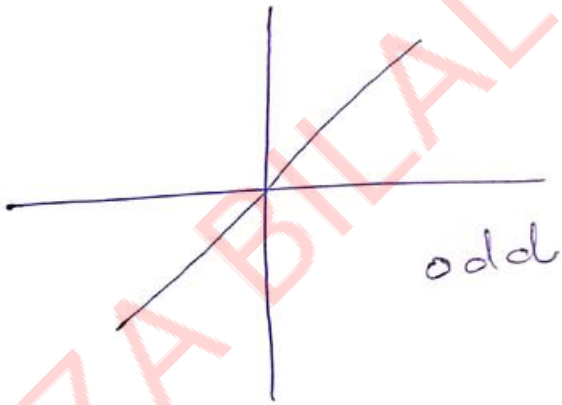
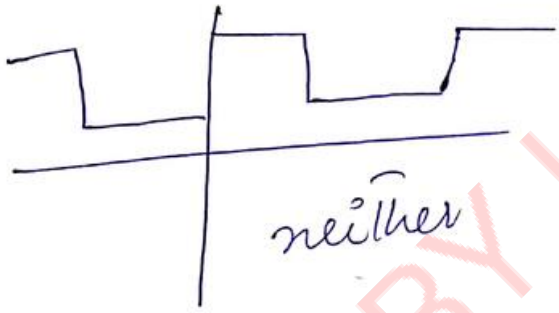
If a function is not symmetric about y-axis & not-symmetric about origin

Then it is neither even nor odd.







Products of Odd & Even functions:-

even x even = even like (+)(+) = (+)

odd x odd = even , (−)(−) = (+)

odd x even = odd , (−)(+) = (−)

even x odd = odd , (+)(−) = (−)

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Two even functions:-

$$F(x) = f(x)\,g(x)$$

$$F(-x) = f(-x)\,g(-x)$$

$\because f(-x) = f(x)$, $g(-x) = g(x)$

$$F(-x) = (f(x))(g(x))$$

$$F(-x) = F(x)$$

Two odd functions:

$$F(x) = f(x)g(x)$$

$$F(-x) = f(-x)\,g(-x)$$

$$= [-f(x)][-g(x)]$$

$\because f(-x) = -f(x)$, $g(-x) = -g(x)$

$$F(-x) = f(x)\,g(x)$$

$$F(-x) = F(x)$$

One odd & one even func:-

$$F(x) = f(x)g(x)$$

when $f(x)$ is odd & $g(x)$ is even

$$F(-x) = f(-x)\,g(-x)$$

$$= [-f(x)]\,g(x)$$

$$= -f(x)g(x)$$

$$F(-x) = -F(x)$$

Examples:

1. $x^2 \sin 2x$ odd $(E)(O) = O$

2. $x^3 \cos x$ odd $(O)(E) = O$

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3. $\cos 2x \cos 3x$ — even — $(E)(E) = E$

4. $x \sin nx$ — even — $(O)(O) = E$

5. $3 \sin x \cos 4x$ — odd — $(O)(E) = O$

6. $(2x+3) \sin 4x$ — neither — $(N)(O) = N$

7. $\sin^2 x \cos 3x$ — Even — $(E)(E) = E$

8. $x^3 e^x$ — neither — $(O)(N) = N$

9. $(x^4+4) \sin 2x$ — Odd — $(E)(O) = O$

Two useful facts from odd & even func:-

For an even function

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

For an odd function

$$\int_{-a}^{a} f(x)\,dx = 0$$

Theorem:- If $f(x)$ is defined over the interval $-\pi < x < \pi$ & $f(x)$ is even, then fourier series for $f(x)$ contains cosine terms only.

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Proof:-

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx \Rightarrow \frac{1}{\pi}\, 2\int_{0}^{\pi} f(x)dx \quad (\text{bcz } f(x) \text{ is even})$$

$$a_0 = \frac{2}{\pi}\int_{0}^{\pi} f(x)dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\, dx \quad (\text{product of even func. is even})$$

$$a_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\, dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\, dx \quad (\text{Product of even \& odal func is odal.})$$

$$= \frac{1}{\pi}(0)$$

$b_n = 0$

So proved that there will not be any sine term in fourier series.

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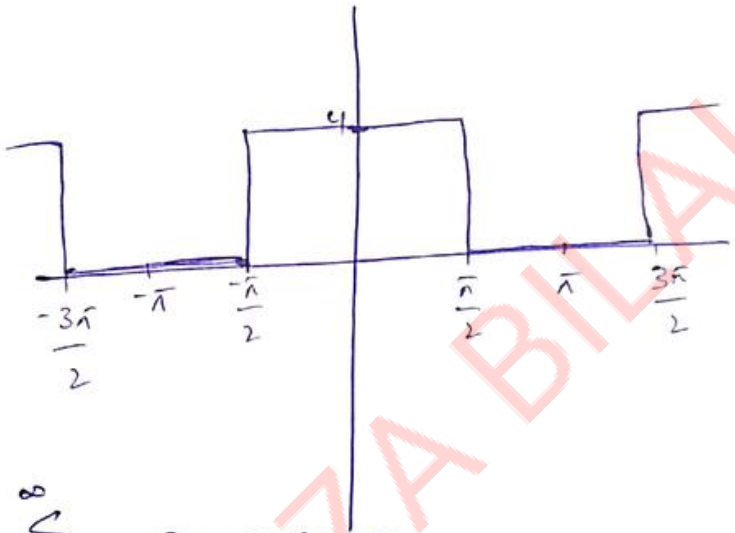
Example:- The waveform shown is symmetric about y-axis. The func. is therefore even & there'll be no sine terms in series.



$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \Rightarrow \frac{2}{\pi}\int_{0}^{\pi} f(x)\,dx$$

$$= \frac{2}{\pi}\left\{\int_{0}^{\pi/2} f(x)\,dx + \int_{\pi/2}^{\pi} f(x)\,dx\right\}$$

$$= \frac{2}{\pi}\left\{\int_{0}^{\pi/2} 4\,dx + 0\right\}$$

$$= \frac{2}{\pi}\left\{4[x]_0^{\pi/2}\right\} \Rightarrow \frac{2}{\pi}\left(4\frac{\pi}{2}\right) \Rightarrow \boxed{4}$$

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$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$= \frac{2}{\pi}\left\{\int_0^{\pi} f(x)\cos nx\,dx\right\}$$

$$= \frac{2}{\pi}\left\{\int_0^{\pi/2} 4\cos nx\,dx + \int_{\pi/2}^{\pi} 0\cos nx\,dx\right\}$$

$$= \frac{2}{\pi}\left\{4\left[\frac{\sin nx}{n}\right]_0^{\pi/2} + 0\right\}$$

$$= \frac{2}{\pi}\cdot\frac{4}{n}\cdot\left[\sin n\cdot\frac{\pi}{2} - \sin n(0)\right]$$

$$\boxed{a_n = \frac{8}{n\pi}\sin\frac{n\pi}{2}}$$

But $\sin\frac{n\pi}{2} = 0$ (for even values of n)

$= 1$ (for $n = 1, 5, 9, \ldots$

$= -1$ (for $n = 3, 7, 11, \ldots$

$\boxed{a_n = \frac{8}{n\pi}}$ for $n = 1, 5, 9$

$\boxed{a_n = \frac{-8}{n\pi}}$ for $n = 3, 7, 11$

$$f(x) = \frac{1}{2}(4) + \left[\frac{8}{\pi}\cos x + 0 - \frac{8}{3\pi}\cos 3x + 0 + \frac{8}{5\pi}\cos 5x + \cdots\right]$$

$$= 2 + \frac{8}{\pi}\left[\cos x - \frac{\cos 3x}{3} + \frac{\cos 5x}{5} - \frac{\cos 7x}{7} + \frac{\cos 9x}{9} + \cdots\right]$$

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Theorem 2:-

If $f(x)$ is an odd function over $[-\pi, \pi]$ Then Fourier Series for $f(x)$ contains sine terms only.

Since $f(x)$ is odd, so

$$\int_{-\pi}^{\pi} f(x)\,dx = 0$$

$$\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx = 0$$

$$\int_{-\pi}^{0} f(x)\,dx = -\int_{0}^{\pi} f(x)\,dx$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \Rightarrow 0 \qquad (f(x) \text{ is odd})$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = 0 \qquad ([O][E]=[O])$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \qquad ((O)(O)=(E))$$

$$= \frac{1}{\pi}\cdot 2\int_{0}^{\pi} f(x)\sin nx\,dx$$

So fourier series contain sine terms only.

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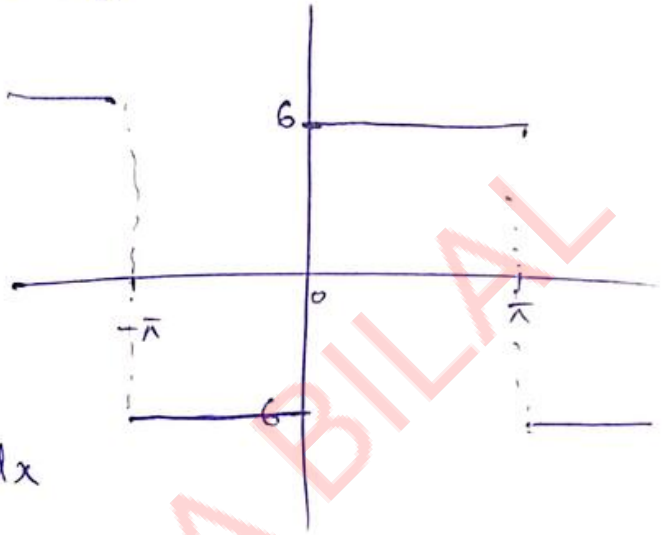
Example:- $f(x) = -6$ $\quad -\pi < x < 0$

$f(x) = 6$ $\quad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



function is odd, so

$a_0 = 0$, $a_n = 0$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} f(x) \sin nx\, dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} 6 \sin nx\, dx \Rightarrow \frac{2}{\pi} \cdot 6 \left[\frac{-\cos nx}{n}\right]_0^{\pi}$$

$$= \frac{12}{n\pi}\left[-\cos n\pi - (-\cos 0)\right]$$

$$= \frac{12}{n\pi}\left[-\cos n\pi + 1\right] \Rightarrow \frac{12}{n\pi}\left(1 - \cos n\pi\right)$$

if n is even

$$b_n = \frac{12}{n\pi}(1 - 1) \Rightarrow 0$$

if n is odd

$$b_n = \frac{12}{n\pi}(1 - (-1)) \Rightarrow \frac{24}{n\pi}$$

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Fourier Series will be

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$= \frac{24}{\pi}\sin x + 0 + \frac{24}{3\pi}\sin 3x + 0 + \frac{24}{5\pi}\sin 5x + \cdots$$

$$= \frac{24}{\pi}\left[\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\right].$$

If $f(x)$ is neither even nor odd, we'll must find expressions for a_0, a_n, b_n in full.

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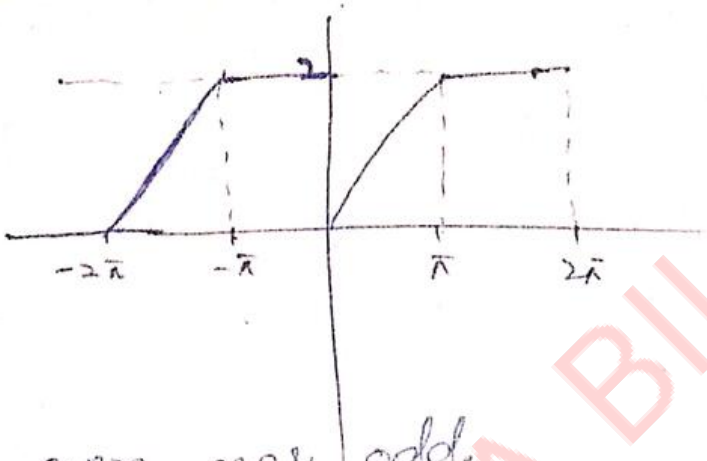
Examples: Lecture 42.

Example: Determine the Fourier Series for the function shown.



This is neither even nor odd.

Therefore we'll find a_0, a_n, b_n.

As Fourier Series are

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\{a_n \cos nx + b_n \sin nx\}$$

$$m = \frac{2-0}{\pi-0} = \frac{2}{\pi}$$

$$y - y_0 = m(x - x_0)$$

$$y - 0 = \frac{2}{\pi}(x - 0)$$

$$y = \frac{2}{\pi}x$$

$$f(x) = \frac{2}{\pi}x \quad \text{when} \quad 0 < x < \pi$$

$$f(x) = 2 \quad \text{when} \quad \pi < x < 2\pi$$

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$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{\pi}\left\{\int_0^{\pi} f(x)\,dx + \int_{\pi}^{2\pi} f(x)\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\int_0^{\pi} \frac{2}{\pi}x\,dx + \int_{\pi}^{2\pi} 2\,dx\right.$$

$$= \frac{2}{\pi}\left\{\int_0^{\pi} \frac{1}{\pi}x\,dx + \int_{\pi}^{2\pi} dx\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{\pi}\cdot\left[\frac{x^2}{2}\right]_0^{\pi} + [x]_{\pi}^{2\pi}\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{\pi}\left(\frac{\pi^2}{2}\right) + (2\pi - \pi)\right\}$$

$$= \frac{2}{\pi}\left\{\frac{\pi}{2} + \pi\right\} \Rightarrow \frac{2}{\pi}\left\{\frac{\pi + 2\pi}{2}\right\} \Rightarrow \frac{2}{\pi}\left\{\frac{3\pi}{2}\right\}$$

$$\boxed{a_0 = 3}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx \Rightarrow \frac{1}{\pi}\left\{\int_0^{\pi} f(x)\cos nx\,dx + \int_{\pi}^{2\pi} f(x)\cos nx\,dx\right.$$

$$= \frac{1}{\pi}\left\{\int_0^{\pi} \frac{2}{\pi}x\cos nx\,dx + \int_{\pi}^{2\pi} 2\cos nx\,dx\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{\pi}\left(\left[x\,\frac{\sin nx}{n}\right]_0^{\pi} - \int_0^{\pi}\frac{\sin nx}{n}\,dx\right) + \left[\frac{\sin nx}{n}\right]_{\pi}^{2\pi}\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{n\pi}\left((0) - \left(-\frac{\cos nx}{n}\right)\right)_0^{\pi} + 0\right\}$$

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$$= \frac{2}{\pi}\left\{\frac{1}{n\pi}\left[\left(\frac{\cos nx}{n}\right)\right]_0^{\pi}\right.$$

$$= \frac{2\left[\cos nx\right]_0^{\pi}}{n^2\pi^2} \Rightarrow \frac{2}{n^2\pi^2}\left(\cos n\pi - \cos 0^\circ\right)$$

$$= \frac{2}{n^2\pi^2}\left(\cos n\pi - 1\right)$$

$$= 0 \quad \text{if} \quad n \text{ is even}$$

$$= \frac{-4}{\pi^2 n^2} \quad \text{if} \quad n \text{ is odd.}$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left\{\int_0^{\pi} f(x)\sin nx\,dx + \int_{\pi}^{2\pi} f(x)\sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\int_0^{\pi}\left(\frac{2}{\pi}x\sin nx\right)dx + \int_{\pi}^{2\pi} 2\sin nx\,dx\right\}$$

$$= \frac{2}{\pi}\left\{\frac{1}{\pi}\left(\left[x\left(\frac{-\cos nx}{n}\right)\right]_0^{\pi} - \int_{\pi}^{2\pi}\left(\frac{-\cos nx}{n}\right)dx\right) + 2\left[\frac{\cos nx}{n}\right]_{\pi}^{2\pi}\right.$$

$$= \frac{2}{\pi}\left\{\frac{1}{n\pi}\left[\left(-\pi\cos n\pi\right) + \frac{1}{n}\left[\sin nx\right]_{\pi}^{2\pi}\right] - \left(\frac{\cos 2\pi n - \cos n\pi}{n}\right)\right.$$

$$= \frac{2}{\pi}\left\{-\frac{\cos n\pi}{n} - \frac{\cos 2\pi n}{n} + \frac{\cos n\pi}{n}\right\}$$

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$$= \frac{2}{\pi}\left(-\frac{\cos 2\pi n}{n}\right)$$

$$= \frac{-2}{n\pi}\cos 2\pi n$$

$$= \frac{-2}{n\pi}(1)$$

$$\boxed{b_n = \frac{-2}{n\pi}}$$

So fourier series is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\{a_n \cos nx + b_n \sin nx\}$$

$$= \frac{1}{2}(3) + \left\{\frac{-4}{\pi^2}\cos x + 0 - \frac{4}{9\pi^2}\cos 3x + 0 \mp \frac{4}{25\pi^2}\cos 5x + \cdots\right\}$$

$$+ \left\{-\frac{2}{\pi}\sin x - \frac{2}{2\pi}\sin 2x - \frac{2}{3\pi}\sin 3x - \frac{2}{4\pi}\sin 4x \cdots\right\}$$

$$= \frac{3}{2} - \frac{4}{\pi^2}\left\{\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \cdots\right\} +$$

$$- \frac{2}{\pi}\left\{\sin x + \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x + \frac{1}{4}\sin 4x + \cdots\right\}$$

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Sum of a fourier series at a point of discontinuity:-

At $x = x_1$ series converges to $f(x_1)$

A particular point of interest occurs at a point of finite discontinuity or jump. of $f(x)$.

At $x = x_1$, the function appears to have two distinct values y_1 & y_2.

Fourier series that results is now converging to the average of y_1 & y_2

(i-e) $= \frac{1}{2}(y_1 + y_2)$.

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Example:-

Consider the function

$$f(x) = 0 \qquad -\pi < x < 0$$

$$f(x) = a \qquad 0 < x < \pi$$

$$f(x) = f(x+2\pi)$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} a\,dx\right\}$$

$$= \frac{1}{\pi}\left\{a\,[x]_0^{\pi}\right\} \Rightarrow \frac{1}{\pi}\,a(\pi - 0) = \frac{1}{\pi}\,a(\pi)$$

$$\boxed{a_0 = a}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$\frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\cdot\cos nx\,dx + \int_{0}^{\pi} a\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{0 + a\left[\frac{\sin x}{n}\right]_0^{\pi}\right\}$$

$$\Rightarrow \frac{1}{\pi}\{a(0)\} \Rightarrow \boxed{0 = a_n}$$

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$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\cdot\sin nx\,dx + \int_{0}^{\pi} a\sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{0 + a\left[\left(-\frac{\cos nx}{n}\right)\right]_0^{\pi}\right\}$$

$$= \frac{1}{\pi}\left\{\frac{a}{n}\left[-(\cos n\pi - \cos 0^\circ)\right]\right\}$$

$$= \frac{a}{n\pi}\left[-(\cos n\pi - 1)\right] \Rightarrow \frac{a}{n\pi}\left[(1-\cos nx)\right]$$

$b_n = 0$ if n is even.

$b_n = \frac{2a}{n\pi}$ if n is odd.

So $f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)$

$$= \frac{1}{2}(a) + 0 + \frac{2a}{\pi}\sin x + 0 + \frac{2a}{3\pi}\sin 3x + 0 + \frac{2a}{5\pi}\sin 5x + \cdots$$

$$= \frac{a}{2} + \frac{2a}{\pi}\left\{\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\right\}$$

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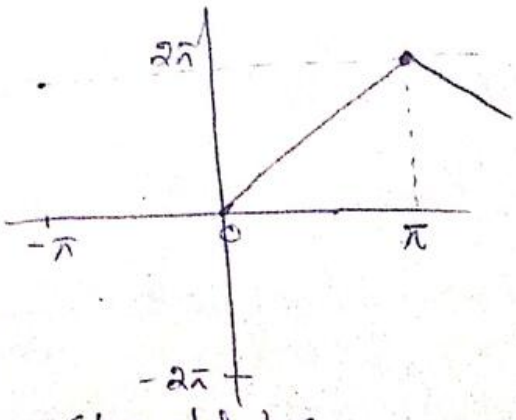
A finite discontinuity or jump occurs at $x=0$. If we substitute $x=0$, all sine terms vanishes. & we get $f(x)=\frac{a}{2}$ which is average of two func. values at $x=0$.

$y=0$ & $y=a$

$= \frac{1}{2}(0+a) \Rightarrow \frac{a}{2}$.

Half Range Series-

Sometimes a ~~series~~ function of period 2π is defined over range 0 to π. instead of $-\pi$ to π or 0 to 2π.



if it is even function it should be symmetrical about y-axis having only cosine values.
If it is odd func. it should be Symmetric about origin. having only sine terms.

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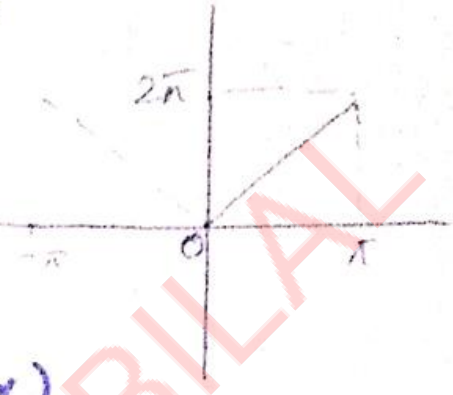
Example:-

A function $f(x)$ is defined by

$f(x) = 2x \quad 0 < x < \pi$

$f(x) = f(x + 2\pi)$



Obtain a cosine series of half Range to represent $f(x)$.

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} 2x\,dx \Rightarrow \frac{2}{\pi}\,2\left[\left(\frac{x^2}{2}\right)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\pi^2 - 0^2\right] = \frac{2}{\pi}(\pi^2) \Rightarrow \boxed{2\pi = a_0}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \Rightarrow \frac{2}{\pi}\int_0^{\pi} 2x\cos nx\,dx$$

$$= \frac{4}{\pi}\left\{\left[\frac{x\sin nx}{n}\right]_0^{\pi} - \int_0^{\pi}\frac{\sin nx}{n}\,dx\right\}$$

$$= \frac{4}{\pi}\left\{\frac{\pi\sin n\pi}{n} - 0 - \frac{1}{n}\left[-\frac{\cos nx}{n}\right]_0^{\pi}\right\}$$

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$$= \frac{4}{\pi}\left(0 - 0 - \frac{1}{n^2}\left[-(\cos n\pi - \cos 0^\circ) \right] \right)$$

$$= \frac{4}{\pi}\left(-\frac{1}{n^2}(1 - \cos nx) \right)$$

$$= \frac{-4}{n^2\pi}(1 - \cos nx)$$

$$= \frac{4}{n^2\pi}(\cos nx - 1)$$

$\boxed{a_n = 0}$ if n is even

$\boxed{a_n = \frac{-8}{n^2\pi}}$ if n is odd.

$\boxed{b_n = 0}$ bcz $f(x)$ is even & contain only cosine terms.

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

$$= \frac{1}{2}(2\pi) + \frac{-8}{\pi}\cos x + 0 - \frac{8}{9\pi}\cos 3x + 0 - \frac{8}{25\pi}\cos 5x + \cdots$$

$$= \pi - \frac{8}{\pi}\left(\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \cdots \right)$$

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Example:- Determine a half range sine series to represent the function $f(x)$ defined by $f(x) = 1+x \quad 0 < x < \pi$

$$f(x) = f(x+2\pi)$$

it is odd function, so $a_0 = 0$ & $a_n = 0$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} f(x)\sin nx\,dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} (1+x)\sin nx\,dx$$

$$= \frac{2}{\pi}\left\{\left[(1+x)\frac{-\cos nx}{n}\right]_0^{\pi} - \int_0^{\pi} \frac{-\cos nx}{n}\,dx\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{(1+\pi)\cos n\pi}{n} + \frac{+1}{n} + \frac{1}{n}\left[\frac{\sin nx}{n}\right]_0^{\pi}\right\}$$

$$= \frac{2}{\pi}\left\{-\frac{(1+\pi)\cos n\pi}{n} + \frac{1}{n} + 0\right\}$$

$$= \frac{2}{n\pi}\left(1 - (1+\pi)\cos n\pi\right)$$

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if n is even

$$= \frac{2}{n\pi}\left(1 - (1+\pi)(1)\right) \Rightarrow \frac{2}{n\pi}\left(1-1+\pi\right)$$

$$= \frac{2}{n\pi}(\pi) \Rightarrow \boxed{\frac{2}{n}}$$

if n is odd

$$= \frac{2}{n\pi}\left(1 - (1+\pi)(-1)\right) \Rightarrow \frac{2}{n\pi}(1+1+\pi)$$

$$= \boxed{\frac{4+2\pi}{n\pi}}$$

putting values in

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

$$= 0 + 0 + \frac{4+2\pi}{\pi}\sin x + \frac{2}{2}\sin 2x + \frac{4+2\pi}{3\pi}\sin 3x$$

$$+ \frac{2}{4}\sin 4x + \cdots$$

$$f(x) = \frac{4+2\pi}{\pi}\sin x + \sin 2x + \frac{4+2\pi}{3\pi}\sin 3x + \frac{1}{2}\sin 4x + \cdots$$

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Lecture 43

Function with Periods over Than 2π

If $y = f(x)$ is defined in the range $-\frac{T}{2}$ to $\frac{T}{2}$ i.e has a period T, we can convert this to an interval of 2π by changing units of independent variable. usually independent variable is time(t) in physical oscillations. Periodic interval is usually T.

$$f(t) = f(t+T)$$

Each cycle is completed in T seconds & frequency is $f = \frac{1}{T}$. If angular velocity, ω radians per second, defined by $\omega = 2\pi f$, then $\omega = \frac{2\pi}{T}$ & $T = \frac{2\pi}{\omega}$

also $x = \omega t$ so fourier series will be

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$$f(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega t + b_n \sin n\omega t)$$

can also be written as

$$f(t) = \frac{1}{2}A_0 + \sum_{n=1}^{\infty} B_n(\sin(n\omega t + \phi_n))$$

$n = 1, 2, 3, \ldots\ldots$

Fourier Co-efficients:-

$$f(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega t + b_n \sin n\omega t)$$

where

$$a_0 = \frac{2}{T}\int_0^T f(t)dt = \frac{\omega}{\pi}\int_0^{\frac{2\pi}{\omega}} f(t)dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega t\, dt = \frac{\omega}{\pi}\int_0^{\frac{2\pi}{\omega}} f(t)\cos n\omega t\, dt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t\, dt = \frac{\omega}{\pi}\int_0^{2\pi/\omega} f(t)\sin n\omega t\, dt$$

limits can be 0 to T, $-\frac{T}{2}$ to $\frac{T}{2}$, $-\frac{\pi}{\omega}$ to $\frac{\pi}{\omega}$, 0 to $\frac{2\pi}{\omega}$ etc.

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Example:- Determine the fourier series for a periodic function defined by

$$f(t) = 2(1+t) \qquad -1 < t < 0$$

$$f(t) = 0 \qquad 0 < t < 1$$

$$f(t) = f(t+2)$$

As $f(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\{a_n \cos n\omega t + b_n \sin n\omega t\}$

$$a_0 = \frac{2}{T}\int_{-1}^{1} f(t)\,dt \Rightarrow \frac{2}{2}\int_{-1}^{1} f(t)\,dt$$

$$= \int_{-1}^{0} 2(1+t)\,dt + \int_{0}^{1} 0\,dt$$

$$= 2\left[t + \frac{t^2}{2}\right]_{-1}^{0} + 0 \Rightarrow 2\left[\left(0 + \frac{0^2}{2}\right) - \left(-1 + \frac{(-1)^2}{2}\right)\right]$$

$$= 2\left(-\left(-1 + \frac{1}{2}\right)\right) \Rightarrow$$

$$= 2\left(1 - \frac{1}{2}\right) = 2\left(\frac{1}{2}\right) \Rightarrow \boxed{1}$$

$$a_n = \frac{2}{T}\int_{-1}^{0} f(t)\cos n\omega t\,dt$$

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$$= \frac{2}{2}\int_{-1}^{0} 2(1+t)\cos n\omega t\,dt + \int_{0}^{1} 0\cos n\omega t\,dt$$

$$= 2\left[(1+t)\frac{\sin n\omega t}{n\omega}\right]_{-1}^{0} - \int_{-1}^{0}\frac{\sin n\omega t}{n\omega}\,dt\Bigg] + 0$$

$$= 2\left[0 - 0 - \frac{1}{n\omega}\left[\frac{-\cos n\omega t}{n\omega}\right]_{-1}^{0}\right]$$

$$= 2\left[\frac{1}{n^2\omega^2}\left[1 - \cos n\omega\right]\right]$$

$$a_n = \frac{2}{n^2\omega^2}\left(1 - \cos n\omega\right)$$

as $\omega = \frac{2\pi}{T} \Rightarrow \frac{2\pi}{2} \Rightarrow \pi$

So $a_n = \frac{2}{n^2\omega^2}\left(1 - \cos n\pi\right)$

$$a_n = 0 \quad (n \text{ even})$$

$$a_n = \frac{4}{n^2\omega^2} \quad (n \text{ odd})$$

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$$b_n = \frac{2}{T}\int_{-1}^{1} f(x)\sin n\omega t\,dt$$

$$= \frac{2}{2}\left[\int_{-1}^{0} 2(1+t)\sin n\omega t\,dt + \int_{0}^{1} 0\sin n\omega t\,dt\right]$$

$$= 2\left\{\left[(1+t)\left(\frac{-\cos n\omega t}{n\omega}\right)\right]_{-1}^{0} - \int_{-1}^{0}\frac{-\cos n\omega t}{n\omega}\,dt\right\}$$

$$= 2\left\{\frac{-1}{n\omega} + \frac{1}{n\omega}\left[\left(\frac{\sin n\omega t}{n\omega}\right)\right]_{-1}^{0}\right\}$$

$$= 2\left\{\frac{-1}{n\omega} + \frac{1}{n^2\omega^2}\left(0 - (-\sin n\omega)\right)\right\}$$

$$= 2\left(\frac{-1}{n\omega} + \frac{1}{n^2\omega^2}\sin n\omega\right)$$

As we deeived for a_n, $\omega = \pi$

$$= 2\left(\frac{-1}{n\omega}\right) \Rightarrow \boxed{\frac{-2}{n\omega} = b_n}$$

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So fouriee series'll be

$$f(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos n\omega t + b_n \sin n\omega t)$$

$$= \frac{1}{2}(1) + \left\{\frac{4}{\omega^2}\cos\omega t + 0 + \frac{4}{9\omega^2}\cos 3\omega t + 0 + \right.$$

$$\left.\frac{4}{25\omega^2}\cos 5\omega t + \cdots\right\} + \left\{\frac{-2}{\omega}\sin\omega t - \frac{2}{2\omega}\sin 2\omega t\right.$$

$$\left. - \frac{2}{3\omega}\sin 3\omega t - \frac{2}{4\omega}\sin 4\omega t + \cdots\right\}$$

$$= \frac{1}{2} + \frac{4}{\omega^2}\left\{\cos\omega t + \frac{1}{9}\cos 3\omega t + \frac{1}{25}\cos 5\omega t + \cdots\right\} +$$

$$-\frac{2}{\omega}\left\{\sin\omega t + \frac{1}{2}\sin 2\omega t + \frac{1}{3}\sin 3\omega t + \frac{1}{4}\sin 4\omega t + \right.$$

Half Range Series:-

Even function:- Half Range cosine series

$$y = f(t), \quad 0 < t < \frac{T}{2}$$

$y = f(t+T)$. Symmetrical about y-axis

In even func. $b_n = 0$ So

$$f(t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t$$

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where

$$a_0 = \frac{4}{T}\int_0^{T/2} f(t)\,dt$$

$$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega t\,dt$$

Odd functions:-

Half Range sine series

$y = f(t)$, $0 < t < \frac{T}{2}$

$f(t) = f(t+T)$, symmetrical about origin

$$a_0 = 0,\quad a_n = 0,\quad b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega t\,dt$$

$$f(t) = \sum_{n=1}^{\infty} b_n \sin n\omega t.$$

Example:- A function $f(t)$ is defined by $f(t) = 4-t$ $\quad 0 < t < 4$

Form a half range cosine series to represent the function.

$$a_0 = \frac{4}{T}\int_0^4 f(t)\,dt \Rightarrow \frac{4}{8}\int_0^4 (4-t)\,dt$$

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$$a_0 = \frac{1}{2}\left[4t - \frac{t^2}{2}\right]_0^4 \Rightarrow \frac{1}{2}\left[\left(4(4) - \frac{(4)^2}{2}\right) - (0)\right]$$

$$= \frac{1}{2}\left(16 - \frac{16}{2}\right) \Rightarrow \frac{1}{2}(16-8) \Rightarrow \frac{8}{2} \Rightarrow \boxed{4 = a_0}$$

$$a_n = \frac{4}{T}\int_0^4 f(t)\cos n\omega t\, dt$$

$$= \frac{4}{8}\int_0^4 (4-t)\cos n\omega t\, dt$$

$$= \frac{1}{2}\left[(4-t)\frac{\sin n\omega t}{n\omega}\right]_0^4 - \int_0^4 \frac{\sin n\omega t}{n\omega}(-1)\, dt$$

$$= \frac{1}{2}\left[0 - 0 + \frac{1}{n\omega}\left[\frac{-\cos n\omega t}{n\omega}\right]_0^4\right.$$

$$= \frac{1}{2}\left(-\frac{1}{n^2\omega^2}(\cos 4n\omega - 1)\right)$$

$$= -\frac{1}{2n^2\omega^2}(\cos 4n\omega - 1)$$

As $\omega = \frac{2\pi}{T} \Rightarrow \frac{2\pi}{8} \Rightarrow \frac{\pi}{4}$

$$= -\frac{1}{2n^2\omega^2}\left(\cos 4n\left(\frac{\pi}{4}\right) - 1\right) \Rightarrow \frac{-1}{2n^2\omega^2}(\cos n\pi - 1)$$

$a_n = 0$ (n even)

$a_n = \frac{1}{n^2\omega^2}$ (n odd)

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$$f(t) = \frac{1}{2}(4) + \frac{1}{\pi w^2}\cos wt + 0 + \frac{1}{9w^2}\cos 3wt + 0 + \frac{1}{25w^2}\cos 5wt + \dots$$

$$= 2 + \frac{1}{w^2}\left(\cos wt + \frac{1}{9}\cos 3wt + \frac{1}{25}\cos 5wt + \dots\right)$$

Example:- A function $f(t)$ is defined by

$$f(t) = 3 + t \qquad 0 < t < 2$$

$$f(t) = f(t+4)$$

obtain half range sine series.

function is odd so $a_n = 0$, $a_0 = 0$

$$b_n = \frac{4}{T}\int_0^2 f(t)\sin nwt\, dt$$

$$= \frac{4}{4}\int_0^2 (3+t)\sin nwt\, dt$$

$$= \left[(3+t)\frac{-\cos nwt}{nw}\right]_0^2 - \int_0^2 -\frac{\cos nwt}{nw}\, dt$$

$$= -\frac{5\cos 2nw}{nw} + \frac{3}{nw} + \frac{1}{nw}\left[\frac{\sin nwt}{nw}\right]_0^2$$

$$= -\frac{5\cos 2nw}{nw} + \frac{3}{nw} + \frac{1}{n^2w^2}\sin 2nw$$

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$$= \frac{1}{n\omega}\left[3 - 5\cos 2n\omega\right] + \frac{1}{n^2\omega^2}\sin 2n\omega$$

But $\omega = \frac{2\pi}{T} = \frac{2\pi}{4} \Rightarrow \frac{\pi}{2}$

$$= \frac{1}{n\omega}\left[3 - 5\cos n\pi\right] + \frac{1}{n^2\omega^2}\sin n\pi$$

$$b_n = \frac{-2}{n\omega} \quad (n \text{ even})$$

$$b_n = \frac{8}{n\omega} \quad (n \text{ odd})$$

$$f(t) = \sum_{n=1}^{\infty} b_n \sin n\omega t$$

$$= \frac{8}{\omega}\sin\omega t + \left(\frac{-2}{2\omega}\right)\sin 2\omega t + \frac{8}{3\omega}\sin 3\omega t +$$

$$\left(\frac{-2}{4\omega}\right)\sin 4\omega t + \frac{8}{5\omega}\sin 5\omega t + \cdots$$

$$= \frac{2}{\omega}\left(4\sin\omega t + \frac{1}{2}\sin 2\omega t + \frac{4}{3}\sin 3\omega t + \right.$$

$$\left. \frac{1}{4}\sin 4\omega t + \frac{4}{5}\sin 5\omega t + \cdots\right)$$

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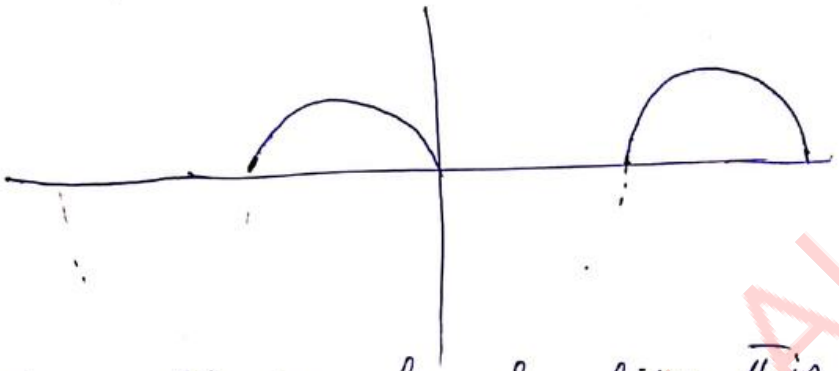
Half Wave Rectifier:-

Half wave rectifier clips the -ve portion of wave.



Fourier series will be found like this.

$$u(t) = 0 \qquad -\frac{T}{2} < t < 0$$

$$u(t) = E \sin \omega t \qquad 0 < t < \frac{T}{2}$$

here $T = \frac{2\pi}{\omega}$

$$a_0 = \frac{2}{T} \int_{-T/2}^{T/2} u(t)\, dt$$

$$= \frac{2}{T} \left\{ \int_{-T/2}^{0} 0\, dt + \int_{0}^{T/2} E \sin \omega t\, dt \right\}$$

$$= \frac{2}{2\pi/\omega} \left\{ 0 + \int_{0}^{\frac{2\pi}{\omega 2}} E \sin \omega t\, dt \right\}$$

$$= \frac{\omega}{\pi} \int_{0}^{\pi/\omega} E \sin \omega t\, dt \Rightarrow \frac{\omega}{\pi} E \left| \frac{-\cos \omega t}{\omega} \right|_{0}^{\pi/\omega}$$

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$$= -\frac{\omega}{\pi}\frac{E}{\omega}\left(\cos\omega\left(\frac{\pi}{\omega}\right) - \cos 0\right)$$

$$= \frac{-E}{\pi}\left(-1-1\right) \Rightarrow -\frac{E}{\pi}(-2)$$

$$\boxed{a_0 = \frac{2E}{\pi}}$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} u(t)\cos n\omega t\, dt$$

$$= \frac{2}{2\pi/\omega}\int_{-T/2}^{0} 0\cos n\omega t\, dt + \int_{0}^{T/2} E\sin\omega t \cos n\omega t\, dt$$

$$= \frac{\omega}{2\pi}E\int_{0}^{\pi/\omega} 2\sin\omega t\cos n\omega t\, dt$$

$$= \frac{\omega}{2\pi}E\int_{0}^{\pi/\omega}\left[\sin(1+n)\omega t + \sin(1-n)\omega t\right]dt$$

for $n=1$, $a_n = 0$

if $n \neq 1$

$$= \frac{\omega E}{2\pi}\left[\frac{-\cos(1+n)\omega t}{(1+n)\omega} + \frac{-\cos(1-n)\omega t}{(1-n)\omega}\right]_0^{\pi/\omega}$$

$$= +\frac{\omega E}{2\pi}\left[\frac{-\cos(1+n)\pi+1}{(1+n)\omega} + \frac{-\cos(1-n)\pi+1}{(1-n)\omega}\right] -$$

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If n is odd

$$\boxed{a_n = 0}$$

If n is even

$$a_n = \frac{E}{2\pi}\left(\frac{2}{1+n} + \frac{2}{1-n}\right)$$

$$= \frac{E}{2\pi}\left(\frac{2-2n+2+2n}{(1+n)(1-n)}\right) \Rightarrow \frac{E}{2\pi}\left(\frac{4}{1^2-n^2}\right)$$

$$a_n = \frac{4E}{2\pi(1-n^2)} \Rightarrow \boxed{\frac{2E}{\pi(1-n^2)} = a_n}$$

For $b_n = \frac{2}{T}\int_{-T/2}^{T/2} y(t)\sin n\omega t\, dt$

$$= \frac{2}{2\pi/\omega}\left[\int_{-T/2}^{0} 0\cdot \sin n\omega t + \int_{0}^{T/2} E\sin\omega t \sin n\omega t\, dt\right]$$

$$= \frac{\omega E}{-2\pi}\int_0^{\pi/\omega} -2\sin\omega t \sin n\omega t\, dt$$

$$= -\frac{\omega E}{2\pi}\int_0^{\pi/\omega}\left(\cos(1+n)\omega t - \cos(1-n)\omega t\right)$$

if $n \neq 1$

$$= -\frac{\omega E}{2\pi}\left[\frac{\sin(1+n)\omega t}{(1+n)\omega} - \frac{\sin(1-n)\omega t}{(1-n)\omega}\right]_0^{\pi/\omega}$$

for $n=1$

$$b_n = \frac{E}{2}$$

$$b_n = -\frac{\omega E}{2\pi}\int_0^{\pi/\omega}(\cos 2\omega t - 1)$$

$$= \frac{-\omega E}{2\pi}\left[\frac{\sin 2\omega t}{2\omega} - t\right]_0^{\frac{\pi}{\omega}}$$

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$$= -\frac{\omega E}{2\pi}\left[\frac{\sin(1+n)\pi}{(1+n)\omega} - \frac{\sin(1-n)\pi}{(1-n)\omega}\right]$$

$b_n = 0$

$$u(t) = \frac{1}{2}a_0 + \sum_{n=2}^{\infty}\left(a_n \cos n\omega t + b_n \cos n\omega t\right)$$

$$= \frac{1}{2}\left(\frac{2E}{\pi}\right) + \left\{0 + \frac{2E}{\pi(-3)}\cos 2\omega t + 0 + \frac{2E}{\pi(-15)}\cos 4\omega t + \text{-----}\right\} + \frac{E}{2}\sin \omega t$$

$$= \frac{E}{\pi} - \frac{2E}{\pi}\left\{\frac{1}{3}\cos 2\omega t + \frac{1}{15}\cos 4\omega t + \text{---}\right\} + \frac{E}{2}\sin \omega t$$

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Lecture 44

Laplace Transforms

Laplace Transform of a function $F(t)$ is denoted by $L\{F(t)\}$ & is defined as the integral of $F(t)e^{-st}$ b/w the limits $t=0$ & $t=\infty$

$$L\{F(t)\} = \int_0^\infty e^{-st} F(t)\,dt$$

s is always assumed to be +ve & $F(t)e^{-st}$ converges to zero at $t \to \infty$. whatever the function $F(t)$ is.

for example for $F(t) = a$

$$L(a) = \int_0^\infty a\, e^{-st}\, dt \Rightarrow a \int_0^\infty e^{-st}\, dt$$

$$= a\left[\frac{e^{-st}}{-s}\right]_0^\infty = -\frac{a}{s}\left[e^{-st}\right]_0^\infty \Rightarrow -\frac{a}{s}\left[\frac{1}{e^{st}}\right]_0^\infty$$

$$= -\frac{a}{s}(0-1) \Rightarrow \frac{a}{s} \quad \text{Ans.}$$

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Example:- Find Laplace Transform of the form e^{at} that $F(t) = e^{at}$

$$L(e^{at}) = \int_0^\infty e^{at} e^{-st}\, dt$$

$$= \int_0^\infty e^{at-st}\, dt$$

$$= \int_0^\infty e^{-t(s-a)}\, dt \Rightarrow \left[\frac{e^{-t(s-a)}}{-(s-a)}\right]_0^\infty$$

$$= -\frac{1}{s-a}\left[\frac{1}{e^{t(s-a)}}\right]_0^\infty \Rightarrow \frac{-1}{s-a}\left[0-1\right]$$

$$L(e^{at}) = \frac{1}{s-a}$$

So we have two standard Transforms.

$$L\{a\} = \frac{a}{s}, \quad L\{e^{at}\} = \frac{1}{s-a}$$

$$L\{4\} = \frac{4}{s}, \quad L\{e^{4t}\} = \frac{1}{s-4}$$

$$L\{-5\} = \frac{-5}{s}, \quad L\{e^{-5t}\} = \frac{1}{s+5}$$

Laplace Transform is always a function of s.

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Show That $L\{t^3\} = \frac{3!}{s^4}$

$$L\{t^3\} = \int_0^\infty t^3 e^{-st}\, dt =$$

$$= \left[t^3 \frac{e^{-st}}{-s}\right]_0^\infty - \int_0^\infty \frac{e^{-st}}{-s} \cdot 3t^2\, dt$$

$$= 0 - 0 + \frac{1}{s}\int_0^\infty 3t^2 e^{-st}\, dt$$

$$= \frac{1}{s}\left[\left[3t^2 \frac{e^{-st}}{-s}\right]_0^\infty - \int_0^\infty \frac{e^{-st}}{-s} \cdot 6t\right]$$

$$= \frac{1}{s}\left[0 - 0 + \frac{6}{s}\int_0^\infty t e^{-st}\, dt\right]$$

$$= \frac{1}{s}\left[\frac{6}{s}\int_0^\infty t e^{-st}\, dt\right] \Rightarrow \frac{6}{s^2}\int_0^\infty t e^{-st}\, dt$$

$$= \frac{6}{s^2}\left\{\left[t \frac{e^{-st}}{-s}\right]_0^\infty - \int_0^\infty \frac{e^{-st}}{-s} \cdot 1\, dt\right\}$$

$$= \frac{6}{s^2}\left\{0 - 0 + \frac{1}{s}\int_0^\infty e^{-st}\, dt\right\}$$

$$= \frac{6}{s^3}\int_0^\infty e^{-st}\, dt \Rightarrow \frac{6}{s^3}\left\{\left[\frac{e^{-st}}{-s}\right]_0^\infty\right\}$$

$$\Rightarrow \frac{-6}{s^4}\left[e^{-st}\right]_0^\infty \Rightarrow \frac{-6}{s^4}[0 - 1] \Rightarrow \frac{3!\cdot 1}{s^4} = \frac{3!}{s^4}$$

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Complex Numbers power of i

$i = \sqrt{-1}$, $i^2 = -1$, $i^4 = 1$

Solve

$$i^9 = i^8 \cdot i = (i^4)^2 \cdot i = i$$

$$i^{20} = (i^4)^5 = (1)^5 = 1$$

$$i^{30} = (i^2)^{15} = (-1)^{15} = -1$$

$$i^{15} = i^{14} \cdot i = (i^2)^7 \cdot i = (-1)^7 \cdot i = -i$$

Complex numbers:-

$z = 3 + 5i$ is called a complex number where 3 is real part & 5 is imaginary part.

In general, $a + bi$ is a complex number where a is real & b is imaginary part.

So

Complex no. = real part + (imaginary part) i

Conjugate Complex numbers:-

For a complex no. $a + bi$, the complex no. $a - bi$ is conjugate of $a + bi$. Conjugate complex no. are identical except the sign in the middle of brackets.

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$(4+5i)$ & $(4-5i)$ are conjugate Complex no.

$(6+2i)$ & $(2+6i)$ are not conjugate.

$(5-3i)$ & $(-5+3i)$ are not conjugate.

Remember:-

Product of a complex no. & its conjugate is a real no. entirely.

$(3+4i)(3-4i) \Rightarrow (3)^2-(4i)^2 \Rightarrow 9-16i^2 \Rightarrow 9-16(-1)$

$= 9+16 = \underline{25}$

$(a+bi)(a-bi) \Rightarrow (a)^2-(bi)^2 \Rightarrow a^2-b^2i^2$

$= a^2-b^2(-1) \Rightarrow \underline{a^2+b^2}$

Euler Formula:-

As we know that series expansion of e^x, $\cos x$ & $\sin x$ are

$$e^x = 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\cdots\cdots$$

$$\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\cdots\cdots$$

$$\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots\cdots$$

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$$e^{it} = 1 + it + \frac{(it)^2}{2!} + \frac{(it)^3}{3!} + \frac{(it)^4}{4!} + \cdots\cdots$$

$$= 1 + it - \frac{t^2}{2!} - \frac{it^3}{3!} + \frac{t^4}{4!} + \frac{it^5}{5!} - \frac{t^6}{6!} - \frac{it^7}{7!} + \cdots\cdots$$

$$= \left\{1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \cdots\cdots\right\} + i\left\{t - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \cdots\cdots\right\}$$

$$\cos t = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \frac{t^6}{6!} + \cdots\cdots$$

$$\sin t = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \cdots\cdots$$

So

$$e^{it} = \cos t + i \sin t$$

$$R(e^{it}) = \cos t \quad \& \quad I(e^{it}) = \sin t$$

The Laplace Transformation of $F(t) = \sin at$

$$L(\sin at) = L\{I(e^{iat})\} = I\int_0^\infty e^{iat} e^{-st}\, dt \Rightarrow I\int_0^\infty e^{iat - st}\, dt$$

$$= I\int_0^\infty e^{-t(s-ia)}\, dt = I\left[\frac{e^{-t(s-ia)}}{-(s-ia)}\right]_0^\infty$$

$$= I\left\{\frac{-1}{(s-ia)}[0-1]\right\} = I\left\{\frac{1}{(s-ia)}\right\}$$

Rationalizing it

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$$= I\left\{\frac{1}{s-ia} \times \frac{s+ia}{s+ia}\right\} \Rightarrow I\left\{\frac{s+ia}{s^2 - i^2a^2}\right\}$$

$$= I\left\{\frac{s+ia}{s^2+a^2}\right\} \Rightarrow I\left[\frac{s}{s^2+a^2} + \frac{ia}{s^2+a^2}\right]$$

$$\boxed{L\{\sin at\} = \frac{a}{s^2+a^2}}$$

Similarly for cos

$$L\{\cos at\} = R\left[\frac{s}{s^2+a^2} + \frac{ia}{s^2+a^2}\right]$$

$$\boxed{L\{\cos at\} = \frac{s}{s^2+a^2}}$$

The Transform of $F(t) = t^n$ where n is +ve

As $L\{t^n\} = \int_0^\infty t^n e^{-st}\, dt$

$$= \left[\frac{t^n e^{-st}}{-s}\right]_0^\infty - \int_0^\infty \frac{e^{-st}}{-s} \cdot n t^{n-1}\, dt$$

$$= 0 - 0 + \frac{n}{s}\int_0^\infty t^{n-1} e^{-st}\, dt$$

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$$L\{t^n\} = \frac{n}{s}\int_0^\infty t^{n-1}e^{-st}\,dt$$

If $I_n = \int_0^\infty t^n e^{-st}\,dt$ then $I_{n-1} = \int_0^\infty t^{n-1}e^{-st}\,dt$

So $I_n = \frac{n}{s} I_{n-1}$ (replaced n by $n-1$)

again replacing n by $n-1$

$$I_{n-1} = \frac{n-1}{s} I_{n-2}$$

again replacing n by $n-1$

$$I_{n-2} = \frac{n-2}{s} I_{n-3}$$

So

$$I_n = \int_0^\infty t^n e^{-st}dt = \frac{n}{s} I_{n-1} = \frac{n}{s}\cdot\frac{n-1}{s} I_{n-2} = \frac{n}{s}\cdot\frac{n-1}{s}\cdot\frac{n-2}{s}\cdot I_{n-3}$$

So finally

$$I_n = \frac{n}{s}\cdot\frac{n-1}{s}\cdot\frac{n-2}{s}\cdot\frac{n-3}{s}\cdot\frac{n-4}{s}\cdots\cdots\frac{2}{s}\cdot\frac{1}{s}\cdot I_0$$

But $I_0 = L\{t^0\} = L\{1\} = \int_0^\infty e^{-st}\,dt$

$$= \left[\frac{e^{-st}}{-s}\right]_0^\infty \Rightarrow \frac{-1}{s}\left[e^{-st}\right]_0^\infty \Rightarrow \frac{-1}{s}(0-1) \Rightarrow \frac{1}{s}$$

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$$I_n = \frac{n}{s}\cdot\frac{n-1}{s}\cdot\frac{n-2}{s}\cdot\frac{n-3}{s}\cdot\frac{n-4}{s}\cdots\cdots\frac{2}{s}\cdot\frac{1}{s}\cdot\frac{1}{s}$$

$$I_n = \frac{n!}{s^n \cdot s} \Rightarrow \frac{n!}{s^{n+1}}$$

$$\boxed{L\{t^n\} = \frac{n!}{s^{n+1}}}$$

(e.g)

$$L\{t\} = \frac{n!}{s^{n+1}} = \frac{1!}{s^{1+1}} \Rightarrow \frac{1}{s^2}$$

$$L\{t^2\} = \frac{2!}{s^{2+1}} = \frac{2!}{s^3} \Rightarrow \frac{2}{s^3}$$

$$L\{t^3\} = \frac{3!}{s^{3+1}} = \frac{3!}{s^4} \Rightarrow \frac{6}{s^4}$$

Laplace Transform of $F(t) = \sinh at$ & $F(t) = \cosh at$

$$\sinh at = \frac{1}{2}\left(e^{at} - e^{-at}\right) \quad \& \quad \cosh at = \frac{1}{2}\left(e^{at} + e^{-at}\right)$$

$F(t) = \sinh at$

$$L\{\sinh at\} = \int_0^\infty \sinh at\, e^{-st}\, dt \Rightarrow \int_0^\infty \frac{1}{2}\left(e^{at} - e^{-at}\right) e^{-st}\, dt$$

$$= \frac{1}{2}\int_0^\infty \left(e^{at}\cdot e^{-st} - e^{-at}e^{-st}\right) dt \Rightarrow \frac{1}{2}\int_0^\infty \left(e^{at-st} - e^{-at-st}\right) dt$$

$$= \frac{1}{2}\int_0^\infty \left(e^{-t(s-a)} - e^{-t(s+a)}\right) dt$$

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$$= \frac{1}{2}\left\{\left[\frac{e^{-t(s-a)}}{-(s-a)}\right]_0^\infty - \left[\frac{e^{-t(s+a)}}{-(s+a)}\right]_0^\infty\right\}$$

$$= \frac{1}{2}\left\{\frac{1}{-(s-a)}[0-1] - \frac{1}{-(s+a)}[0-1]\right\}$$

$$= \frac{1}{2}\left[\frac{1}{s-a} - \frac{1}{s+a}\right] \Rightarrow \frac{1}{2}\left[\frac{s+a-s+a}{(s-a)(s+a)}\right]$$

$$= \frac{1}{2}\left[\frac{2a}{s^2-a^2}\right] \Rightarrow \boxed{\frac{a}{s^2-a^2} = L\{\sin h\, at\}}$$

$F(t) = \cosh at$

$$L\{\cos h\, at\} = \int_0^\infty \cos h\, at\, e^{-st}\, dt$$

$$= \int_0^\infty \frac{1}{2}\left(e^{at} + e^{-at}\right)e^{-st}\, dt$$

$$= \frac{1}{2}\int_0^\infty \left(e^{at}\cdot e^{-st} + e^{-at}\cdot s^{-st}\right) dt$$

$$= \frac{1}{2}\int_0^\infty \left(e^{at-st} + e^{-at-st}\right) dt$$

$$= \frac{1}{2}\int_0^\infty \left(e^{-t(s-a)} + e^{-t(s+a)}\right) dt$$

$$= \frac{1}{2}\left\{\left[\frac{e^{-t(s-a)}}{-(s-a)}\right]_0^\infty + \left[\frac{e^{-t(s+a)}}{-(s+a)}\right]_0^\infty\right\} dt$$

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$$= \frac{1}{2}\left\{\frac{1}{-(s-a)}[0-1] + \frac{1}{-(s+a)}(0-1)\right\}$$

$$= \frac{1}{2}\left\{\frac{1}{s-a} + \frac{1}{s+a}\right\} \Rightarrow \frac{1}{2}\left[\frac{s+a+s-a}{(s-a)(s+a)}\right]$$

$$= \frac{1}{2}\left\{\frac{2s}{s^2-a^2}\right\} \Rightarrow \boxed{\frac{s}{s^2-a^2} = L\{\cosh at\}}$$

Several Standard Results:-

- $L\{a\} = \frac{a}{s}$
- $L\{e^{at}\} = \frac{1}{s-a}$
- $L\{t^n\} = \frac{n!}{s^{n+1}}$
- $L\{\sin at\} = \frac{a}{s^2+a^2}$
- $L\{\cos at\} = \frac{s}{s^2+a^2}$
- $L\{\sinh at\} = \frac{a}{s^2-a^2}$
- $L\{\cosh at\} = \frac{s}{s^2-a^2}$

Example:-

a) $L\{2\sin 3t + \cos 3t\}$

$$= 2L\{\sin 3t\} + L\{\cos 3t\}$$

$$= 2\left[\frac{3}{s^2+3^2}\right] + \left[\frac{s}{s^2+3^2}\right] \Rightarrow \frac{6}{s^2+9} + \frac{s}{s^2+9}$$

$$= \frac{6+s}{s^2+9^2}$$ Ans.

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$$L\{4e^{2t} + 3\cosh 4t\}$$

$$= 4L\{e^{2t}\} + 3L\{\cosh 4t\}$$

$$= 4 \cdot \frac{1}{s-2} + 3 \cdot \frac{s}{s^2-4^2}$$

$$= \frac{4}{s-2} + \frac{3s}{s^2-16}$$

$$= \frac{4(s^2-16) + 3s(s-2)}{(s-2)(s^2-16)}$$

$$= \frac{4s^2 - 64 + 3s^2 - 6s}{(s-2)(s^2-16)}$$

$$= \frac{7s^2 - 6s - 64}{(s-2)(s^2-16)}$$

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Lecture 45

Theorems

Theorem 1:

The first Shift Theorem:-

The first shift theorem states that if $L\{F(t)\} = f(s)$ Then $L\{e^{-at} F(t)\} = f(s+a)$.

The Transform $L\{e^{-at} F(t)\}$ is Thus the same as $L\{F(t)\}$ with s everywhere in the result replaced by $(s+a)$.

Example:- $L\{\sin 2t\} = \frac{2}{s^2+4}$, Then

$$L\{e^{-3t} \sin 2t\} = \frac{2}{(s+3)^2+4} \Rightarrow \frac{2}{s^2+9+6s+4}$$

$$= \frac{2}{s^2+6s+13}$$

Example:-

$L\{t^2\} = \frac{2}{s^3}$; $L\{t^2 e^{4t}\}$ is same, s replaced by $(s-4)$, So

$$L\{t^2 e^{4t}\} = \frac{2}{(s-4)^3}$$

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Theorem 2:-

Multiplying by t

If $L\{F(t)\} = f(s)$ Then $L\{t\,F(t)\} = -\frac{d}{ds}\{f(s)\}$

Example:- $L\{\sin 2t\} = \frac{2}{s^2+4}$

$$L\{t\sin 2t\} = -\frac{d}{ds}\left[\frac{2}{s^2+4}\right] \Rightarrow -2\frac{d}{ds}(s^2+4)^{-1}$$

$$= -2\left[-1(s^2+4)^{-1-1}(2s)\right] \Rightarrow 2(s^2+4)^{-2}(2s)$$

$$= \frac{2(2s)}{(s^2+4)^2} \Rightarrow \frac{4s}{(s^2+4)^2}$$

Example:- $L\{t\cos 3t\} = \frac{s}{s^2-9}$

$$L\{t\cos 3t\} = -\frac{d}{dx}\left[\frac{s}{s^2-9}\right]$$

$$= -\left[\frac{(s^2-9)(1) - s(2s)}{(s^2-9)^2}\right] \Rightarrow \frac{-(s^2-9-2s^2)}{(s^2-9)^2}$$

$$= \frac{-(-s^2-9)}{(s^2-9)^2} \Rightarrow \frac{s^2+9}{(s^2-9)^2}$$

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if $$L\{t^2 \cos 3t\} = \frac{-d}{ds}\left[\frac{s^2+9}{(s^2-9)^2}\right]$$

$$= -\left[\frac{(s^2-9)^2(2s) - (s^2+9)2(s^2-9)(2s)}{(s^2-9)^4}\right]$$

$$= -\left[\frac{(s^2-9)[(s^2-9)(2s) - 4s(s^2+9)]}{(s^2-9)^4}\right]$$

$$= -\left[\frac{2s^3-18s-4s^3-36s}{(s^2-9)^3}\right] \Rightarrow -\left[\frac{-2s^3-54s}{(s^2-9)^3}\right]$$

$$= \frac{2s^3+54s}{(s^2-9)^3} \Rightarrow \frac{2s(s^2+54)}{(s^2-9)^3}$$

So if

$L\{F(t)\} = f(s)$ Then

$$L\{t^n F(t)\} = (-1)^n \frac{d^n}{ds^n}\{f(s)\}$$

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Theorem:-

Dividing by t

If $L\{F(t)\} = f(s)$, Then

$$L\left\{\frac{F(t)}{t}\right\} = \int_s^\infty f(s)\,ds$$

Example:- Determine $L\left\{\frac{\sin at}{t}\right\}$

As $L\{\sin at\} = \frac{a}{s^2+a^2}$

$$L\left\{\frac{\sin at}{t}\right\} = \int_s^\infty \frac{a}{s^2+a^2}\,ds \qquad \because \operatorname{Tan}^{-1}\left(\frac{x}{a}\right) = \frac{a}{x^2+a^2}$$

$$= \left[\operatorname{Tan}^{-1}\left(\frac{s}{a}\right)\right]_s^\infty \Rightarrow \frac{\pi}{2} - \operatorname{Tan}^{-1}\left(\frac{s}{a}\right)$$

=

Example:- Determine $L\left\{\frac{1-\cos 2t}{t}\right\}$

As $L\{1-\cos 2t\} = \frac{1}{s} - \frac{s}{s^2+4}$

$$L\left\{\frac{1-\cos 2t}{t}\right\} = \int_s^\infty \left(\frac{1}{s} - \frac{s}{s^2+4}\right) ds = \int_s^\infty \left(\frac{1}{s} - \frac{2s}{2(s^2+4)}\right) ds$$

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$$= \left[\ln s - \frac{1}{2}\ln(s^2+4)\right]_s^\infty$$

$$= \left[\frac{1}{2}\cdot 2\ln s - \frac{1}{2}\ln(s^2+4)\right]_s^\infty$$

$$= \frac{1}{2}\left[\ln s^2 - \ln(s^2+4)\right]_s^\infty$$

$$= \frac{1}{2}\left\{\ln \frac{s^2}{s^2+4}\right\}_s^\infty$$

$$= \frac{1}{2}\left[0 - \ln \frac{s^2}{s^2+4}\right]$$

$$= -\frac{1}{2}\ln \frac{s^2}{s^2+4} \Rightarrow \ln\left[\frac{s^2}{s^2+4}\right]^{-\frac{1}{2}}$$

$$= \ln\left[\frac{s^2+4}{s^2}\right]^{\frac{1}{2}} \Rightarrow \ln\sqrt{\frac{s^2+4}{s^2}}$$

Standard forms:-

F(t)	$L\{f(t)\}=f(s)$
a	$\frac{a}{s}$
e^{at}	$\frac{1}{s-a}$
sin at	$\frac{a}{s^2+a^2}$
cos at	$\frac{s}{s^2+a^2}$

F(t)	$L\{F(t)\}=f(s)$
sinh at	$\frac{a}{s^2-a^2}$
cosh at	$\frac{s}{s^2-a^2}$
$t^n =$	$\frac{n!}{s^{n+1}}$

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Inverse Transforms:-

Here is reverse process, i-e given a laplace Transform & we have to find the function of t to which it belongs. We know that

$$\frac{a}{s^2+a^2} = L\{\sin at\}$$

$$L^{-1}\left[\frac{a}{s^2+a^2}\right] = \sin at$$, here L^{-1} is indicating inverse Transform, not a reciprocal.

$$\Rightarrow L^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2t}$$

$$\Rightarrow L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 5t$$

$$\Rightarrow L^{-1}\left\{\frac{4}{s}\right\} = 4$$

$$\Rightarrow L^{-1}\left\{\frac{12}{s^2-9}\right\} = 4\sinh 3t$$

$$\Rightarrow L^{-1}\left\{\frac{3s}{s^2-s-6}\right\} = ?$$

We'll Solve it using partial fraction rules.

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Rules of Partial Fractions:-

⇒ Numerator's degree must be lower than denominator, otherwise divide it first.

⇒ Factorize denominator into prime factors.

⇒ A linear factor $(s+a)$ gives partial fraction

⇒ $\frac{A}{s+a}$ is a constant to be determined.

⇒ A repeated factor $(s+a)^2$ gives $\frac{A}{s+a} + \frac{B}{(s+a)^2}$

⇒ Similarly $(s+a)^3$ gives $\frac{A}{s+a} + \frac{B}{(s+a)^2} + \frac{C}{(s+a)^3}$

⇒ A quadratic factor (s^2+px+q) gives $\frac{Ps+Q}{s^2+ps+q}$

⇒ Similarly $(s^2+px+q)^2$ gives $\frac{Ps+Q}{s^2+ps+q} + \frac{Rs+T}{s^2+ps+q}$

⇒ So $\frac{s-19}{(s-2)(s-5)}$ has partial fractions $\frac{A}{s-2} + \frac{B}{s-5}$

⇒ Eg $\frac{3s^2-4s+11}{(s+3)(s-2)^2} = \frac{A}{s+3} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$

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Example:- $L^{-1}\left\{\frac{5s+1}{s^2-s-12}\right\}$

$$\frac{5s+1}{s^2-(4-3)s-12} = \frac{5s+1}{s^2-4s+3s-12} = \frac{5s+1}{s(s-4)+3(s-4)}$$

let

$$\frac{5s+1}{(s-4)(s+3)} = \frac{A}{s-4} + \frac{B}{s+3}$$

multiply b/s by $(s-4)(s+3)$

$$5s+1 = A(s+3) + B(s-4)$$

put $s=4$

$$5(4)+1 = A(4+3)+0$$

$$21 = 7A$$

$$\boxed{3 = A}$$

put $s=-3$

$$5(-3)+1 = A(0) + B(-3-4)$$

$$-15+1 = -7B$$

$$-14 = -7B$$

$$\frac{-14}{-7} = B$$

$$\boxed{2 = B}$$

So $$\frac{5s+1}{s^2-s-12} = \frac{3}{s-4} + \frac{2}{s+3}$$

$$L^{-1}\left\{\frac{5s+1}{s^2-s-12}\right\} = 3e^{4t} + 2e^{-3t}$$

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Determine $L^{-1}\left\{\frac{9s-8}{s^2-2s}\right\}$

$$\frac{9s-8}{s^2-2s} \Rightarrow \frac{9s-8}{s(s-2)} = \frac{A}{s} + \frac{B}{s-2}$$

multiply b/s by $s(s-2)$

$$9s-8 = A(s-2) + Bs$$

put $s-2=0$, $s=2$

$$9(2)-8 = 0 + B(2)$$

$$10 = 2B$$

$$\boxed{5 = B}$$

put $s=0$

$$9(0)-8 = A(0-2) + 0$$

$$-8 = -2A$$

$$8 = 2A$$

$$\boxed{4 = A}$$

So $\frac{9s-8}{s^2-2s} = \frac{4}{s} + \frac{5}{s-2}$

$$L^{-1}\left\{\frac{9s-8}{s^2-2s}\right\} = 4 + 5e^{2t}$$

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Table of inverse Transforms:-

$f(s)$	$F(t)$
$\frac{a}{s}$	a
$\frac{1}{s+a}$	e^{-at}
$\frac{n!}{s^{n+1}}$	t^n
$\frac{1}{s^n}$	$\frac{t^{n-1}}{(n-1)!}$
$\frac{a}{s^2+a^2}$	$\sin at$
$\frac{s}{s^2+a^2}$	$\cos at$
$\frac{a}{s^2-a^2}$	$\sinh at$
$\frac{s}{s^2-a^2}$	$\cosh at$

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Transforms of derivatives:-

$F'(t)$ denotes first derivative of $F(t)$ w.r.t t

$F''(t)$ denotes second derivative of $F(t)$ w.r.t t

Then

$$L\{F'(t)\} = \int_0^\infty e^{-st} F'(t)\,dt$$

Integrating by parts

$$= \left[e^{-st} F(t) \right]_0^\infty - \int_0^\infty F(t) \left(e^{-st} \cdot (-s) \right) dt$$

$$= 0 - (1)(F(0)) + s \int_0^\infty e^{-st} F(t)\,dt$$

$$L\{F'(t)\} = -F(0) + s\,L\{F(t)\}$$

Similarly

$$L\{F''(t)\} = -F'(0) + s\,L\{F'(t)\}$$

$$= -F'(0) + s\left[-F(0) + sL\{F(t)\}\right]$$

$$= -F'(0) - sF(0) + s^2 L\{F(t)\}$$

$$= s^2 L\{F(t)\} - sF(0) - F'(0)$$

$$L\{F'''(t)\} = s^3 L\{F(t)\} - s^2 F(0) - sF'(0) - F''(0)$$

$$L\{F^{IV}(t)\} = s^4 L\{F(t)\} - s^3 F(0) - s^2 F'(0) - sF''(0) - F'''(0$$

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Differential Equ. & Its solution:-

Solve $\frac{dx}{dt} - 2x = 4$ given that at $t=0$ & $x=1$

$$L\left\{\frac{d\,x(t)}{dt} - 2x(t)\right\} = L\{4\}$$

$$L\left[\frac{d}{dt}x(t)\right] - 2L\{x(t)\} = L\{4\}$$

$$-x(0) + sL[x(t)] - 2L[x(t)] = \frac{4}{s}$$

$$-x(0) + L[x(t)](s-2) = \frac{4}{s}$$

$$-1 + L[x(t)](s-2) = \frac{4}{s}$$

$\because$ at $t=0$, $x=1$
So $x(0)=1$

$$L[x(t)](s-2) = \frac{4}{s} + 1$$

$$L[x(t)](s-2) = \frac{4+s}{s}$$

$$L[x(t)] = \frac{4+s}{s(s-2)}$$

$$x(t) = L^{-1}\left[\frac{4+s}{s(s-2)}\right]$$

Now by partial fractions.

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$$\frac{4+S}{S(S-2)} = \frac{A}{S} + \frac{B}{S-2}$$

multiply b/s by $S(S-2)$

$$4+S = A(S-2) + B(S)$$

put $S=0$

$$4 = A(-2) + 0$$

$$\frac{4}{-2} = A$$

$$\boxed{-2 = A}$$

put $S-2=0$, $S=2$

$$4+2 = A(0) + B(2)$$

$$6 = 2B$$

$$\boxed{3 = B}$$

So $$\frac{4+S}{S(S-2)} = \frac{-2}{S} + \frac{3}{S-2}$$

$$L^{-1}\left[\frac{4+S}{S(S-2)}\right] = -2 + 3e^{2t}$$

Solution of differential equ by laplace

(a) Rewrite equ in term of Laplace Transform.

(b) Insert given initial conditions.

(c) Re-arrange equ algebrically to give Transform of solution.

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(d) Determine inverse Transform to obtain the particular Solution.

Solve the equ.

$$\frac{d^2x}{dt^2} - 3\frac{dx}{dt} + 2x = 2e^{3t}$$ given that at

$t=0$ means $x(0)=5$, $x=5$ & $\frac{dx}{dt}=7$ $x'(0)=7$

$$x''(t) - 3x'(t) + 2x(t) = 2e^{3t}$$

$$L[x''(t)] - 3L[x'(t)] + 2L[x(t)] = 2L[e^{3t}]$$

$$s^2L\{x(t)\} - sx(0) - x'(0) - 3[sL(x(t)) - x(0)] + 2Lx(t) = \frac{2}{s-3}$$

$$s^2L\{x(t)\} - s(5) - 7 - 3[sL(x(t))] + 3(5) + 2L(x(t)) = \frac{2}{s-3}$$

$$s^2L\{x(t)\} - 3sL(x(t)) + 2L(x(t)) = \frac{2}{s-3} + 7 + 5s - 15$$

$$L\{x(t)\}[s^2 - 3s + 2] = \frac{2}{s-3} + 5s - 8$$

$$L\{x(t)\}(s-1)(s-2) = \frac{2 + 5s^2 - 15s - 8s + 24}{(s-3)}$$

$$L\{x(t)\} = \frac{5s^2 - 23s + 26}{(s-1)(s-2)(s-3)}$$

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Now partial fractions

$$L\{x(t)\} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$$

multiply b/s by $(s-1)(s-2)(s-3)$

$$5s^2 - 23s + 26 = A(s-2)(s-3) + B(s-1)(s-3) + C(s-1)(s-2)$$

put $s-1=0$, $s=1$

$$5 - 23 + 26 = A(-1)(-2) + 0 + 0$$

$$31 - 23 = 2A$$

$$8 = 2A$$

$$\boxed{4 = A}$$

put $s=2$

$$20 - 46 + 26 = 0 + B(2-1)(2-3) + 0$$

$$46 - 46 = -B$$

$$\boxed{0 = B}$$

put $s=3$

$$45 - 69 + 26 = 0 + 0 + C(3-1)(3-2)$$

$$71 - 69 = 2C$$

$$2 = 2C$$

$$\boxed{1 = C}$$

$$L\{x(t)\} = \frac{4}{s-1} + \frac{0}{s-2} + \frac{1}{s-3}$$

$$x(t) = L^{-1}\left[\frac{4}{s-1} + \frac{1}{s-3}\right] \Rightarrow 4e^{t} + e^{3t}$$ Ans.

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